Econometrics Guide

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Preface

The book covers the core estimation and inference methods used in applied econometrics. It begins with OLS — matrix derivation, the conditional expectation function, and the best linear predictor — then moves to maximum likelihood, quasi-MLE and M-estimation, and generalized least squares. Endogeneity, instrumental variables, and GMM follow. The second half treats models for specific data structures: censored and truncated data, discrete choice, count data, panel data, dynamic panels, survival analysis, and missing data. Two closing chapters develop the modern interpretation of OLS under heterogeneous treatment effects: effect weights and outcome weights.

Examples are in R or Stata, whichever has the cleaner implementation for the procedure at hand. For applied causal-inference material — difference-in-differences, matching, doubly robust methods, mediation, and causal discovery — see the companion books Introduction to Causal Econometrics and Topics on Econometrics and Causal Inference.

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1 OLS

A linear model with n observations and k regressors can be written as (in vector forms)

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{u} \tag{1.1}$$

The idea of least-squares is to choose β to minimize the residual sum of squares (SSR),

$$SSR(\beta) = (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})'(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) = \mathbf{y}'\mathbf{y} - \mathbf{2}\boldsymbol{\beta}'\mathbf{X}'\mathbf{y} + \boldsymbol{\beta}'\mathbf{X}'\mathbf{X}\boldsymbol{\beta} \tag{1.2}$$

The first-order condition to minimize SSR are:

$$\frac{\partial(SSR)}{\partial\beta} = -\mathbf{2X}'\mathbf{y} + \mathbf{2X}'\mathbf{X}\boldsymbol{\beta} = \mathbf{0} \tag{1.3}$$

This generates the so-called normal equations

$$(\mathbf{X}'\mathbf{X})\boldsymbol{\beta} = \mathbf{X}'\mathbf{y} \tag{1.4}$$

Therefore, Ordinary Least Squares (OLS) estimator of β is

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-\mathbf{1}}\mathbf{X}'\mathbf{y} \tag{1.5}$$

1.1 Fitness of OLS

We can see y vector as the part explained by the regression and the unexplained (residual) part,

$$\mathbf{y} = \hat{\mathbf{y}} + \hat{\mathbf{u}} = \mathbf{X}\hat{\boldsymbol{\beta}} + \hat{\mathbf{u}} \tag{1.6}$$

where $\hat{u}$ is the OLS residual. Because the residual is orthogonal to the fitted values, $\hat{\mathbf{y}}'\hat{\mathbf{u}} = \mathbf{0}$, so the cross term drops:

$$\mathbf{y}'\mathbf{y} = (\hat{\mathbf{y}} + \hat{\mathbf{u}})'(\hat{\mathbf{y}} + \hat{\mathbf{u}}) = \hat{\mathbf{y}}'\hat{\mathbf{y}} + \hat{\mathbf{u}}'\hat{\mathbf{u}} = \hat{\boldsymbol{\beta}}'\mathbf{X}'\mathbf{X}\hat{\boldsymbol{\beta}} + \hat{\mathbf{u}}'\hat{\mathbf{u}} \tag{1.7}$$

Subtracting $n\bar{y}^2$ ($\bar{y}$ is the sample mean) from both sides, **and assuming the regression includes an intercept** (so that $\bar{\hat{y}} = \bar{y}$ and the residuals sum to zero, which is what makes this centered decomposition and the R^2 interpretation below hold),

$$\mathbf{y}'\mathbf{y} - \mathbf{n}\bar{\mathbf{y}}^{\mathbf{2}} = (\hat{\boldsymbol{\beta}}'\mathbf{X}'\mathbf{X}\hat{\boldsymbol{\beta}} - \mathbf{n}\bar{\mathbf{y}}^{\mathbf{2}}) + \hat{\mathbf{u}}'\hat{\mathbf{u}} \tag{1.8}$$

We decompose the total sum of squares into two parts: sum of squares due to error (noise), and sum of squares explained by the linear regression.

The R^2 is defined by

$$R^2 = 1 - \frac{SSR}{SST} \tag{1.9}$$

SST is the total (centered) sum of squares of y, $\mathbf{y'y - n\bar{y}^2 = \sum_t (y_t - \bar{y})^2}$.

R^2 is simply the proportion of the variation of Y that can be attributed to the variation of X. R^2, however, will never decrease with the addition of any variable to the set of regressors. R^2 stays exactly the same only in the knife-edge case that the added variable is orthogonal to the current residuals; in practice it always rises a little, even for an irrelevant variable. The adjusted R^2, however, takes account of the addition of any new regressors:

$$\bar{R}^2 = 1 - \frac{SSR/(n-k)}{SST/(n-1)} \tag{1.10}$$

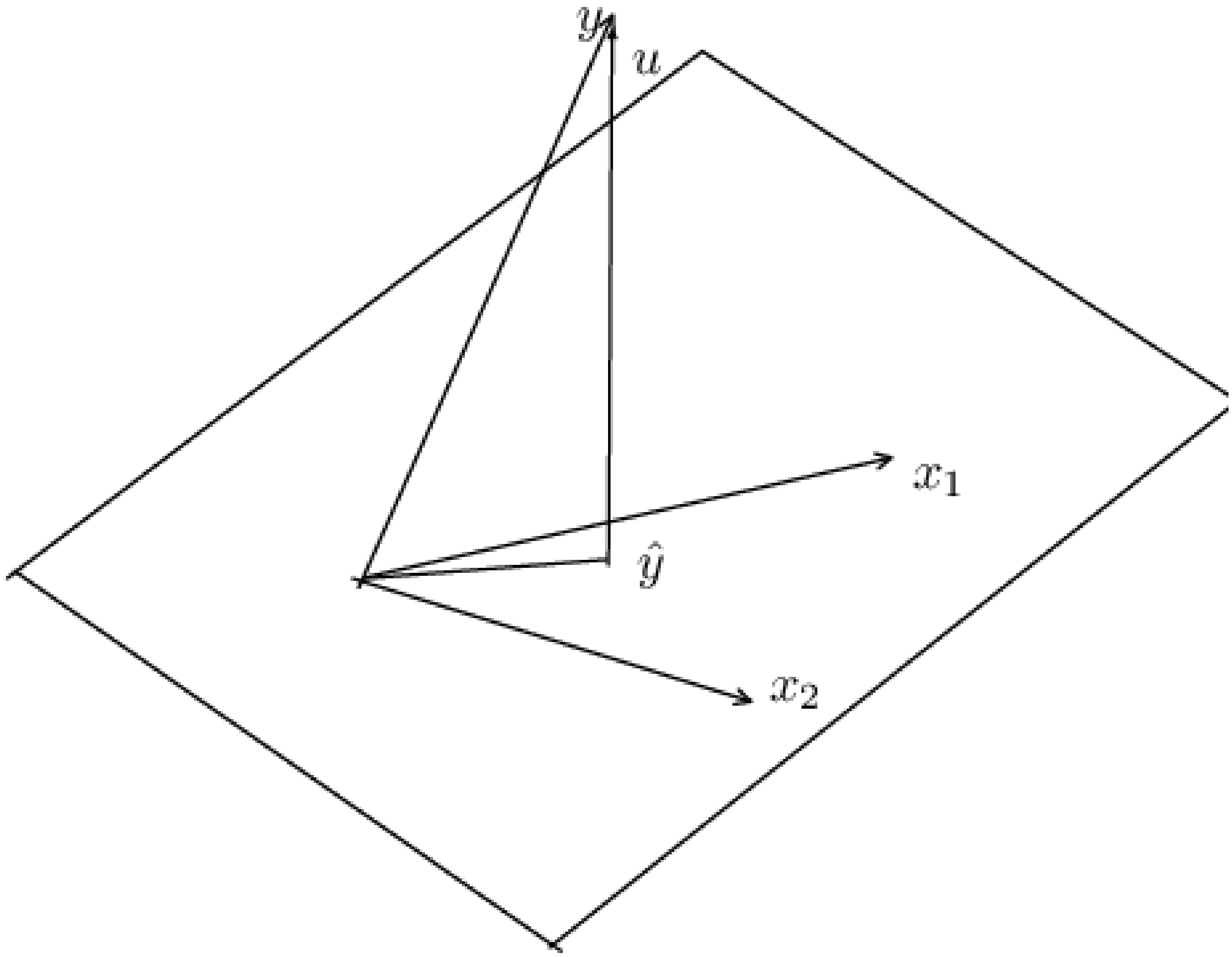
1.2 Example

Let's look at an example of OLS of car's mpg on disp, hp and wt.

```r
lm1 <- lm(mpg~disp+hp+wt, data=mtcars)
summary(lm1)
```

```

Call:
lm(formula = mpg ~ disp + hp + wt, data = mtcars)

Residuals:
   Min     1Q Median     3Q    Max 
-3.891 -1.640 -0.172  1.061  5.861 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept) 37.105505   2.110815  17.579  < 2e-16 ***
disp        -0.000937   0.010350  -0.091  0.92851    
hp          -0.031157   0.011436  -2.724  0.01097 *  
wt          -3.800891   1.066191  -3.565  0.00133 ** 
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.639 on 28 degrees of freedom
Multiple R-squared:  0.8268,	Adjusted R-squared:  0.8083 
F-statistic: 44.57 on 3 and 28 DF,  p-value: 8.65e-11
```

1.3 The Geometry of Least Squares



For simplicity, let's assume there are two explanatory variables x_1 and x_2. x_1 and x_2 form a plane. What OLS does is to project y onto this plane.

In mathematical terms, all linear combinations of these two vectors define a two-dimensional subspace of Euclidean space $\mathbf{E}^n$. This is called the column space of $\mathbf{X}$. The least-square principle is to choose β to make $\hat{y}$, which belongs to the subspace of $\mathbf{X}$, as close as possible to y.

When we estimate a linear regression model, we map the y into a vector of fitted values $\mathbf{X}\hat{\boldsymbol{\beta}}$ and a vector of residuals $\hat{\mathbf{u}} = \mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}$. Geometrically, these are examples of orthogonal projections. A projection is a mapping that takes each point of $\mathbf{E}^n$ into a point in a subspace of $\mathbf{E}^n$. An orthogonal projection maps any point into the point of the subspace that is closest to it.

An orthogonal projection can be performed by premultiplying the vector to be projected by a projection matrix. In the case of OLS, the two projection matrices that yield the vector of fitted values and the vector of residuals, are

$$\mathbf{P_X} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' \tag{1.11}$$

$$\mathbf{M_X} = \mathbf{I} - \mathbf{P_X} = \mathbf{I} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' \tag{1.12}$$

From this, we see that the effects of two projection matrices.

$$\mathbf{P_X y} = \mathbf{X}\hat{\beta} = \hat{\mathbf{y}} \tag{1.13}$$

$$\mathbf{M_X y} = (\mathbf{I} - \mathbf{P_X})\mathbf{y} = \mathbf{y} - \mathbf{P_X y} = \mathbf{y} - \mathbf{X}\hat{\beta} = \hat{\mathbf{u}} \tag{1.14}$$

That is, $\mathbf{P_X y}$ projects $\mathbf{y}$ onto $\mathbf{X}$ and makes it $\hat{\mathbf{y}}$; $\mathbf{M_X y}$ makes it $\hat{\mathbf{u}}$.

In the picture, that corresponds to the fact that the projection of $\mathbf{y}$ onto the plane of $\mathbf{X}$ creates two parts: $\hat{\mathbf{y}}$ and $\hat{\mathbf{u}}$.

1.4 OLS as Conditional Expectation

The derivation above shows how to compute $\hat{\boldsymbol{\beta}}$ from a sample. We now ask a different question: what does OLS estimate at the population level?

1.4.1 The Conditional Expectation Function

For a random pair (Y, X), the conditional expectation function (CEF) is $E[Y \mid X]$, a function of X alone. Any random variable Y can be decomposed as

$$Y = E[Y \mid X] + \varepsilon, \qquad E[\varepsilon \mid X] = 0. \tag{1.15}$$

This is not an assumption; it is a definition. The residual $\varepsilon = Y - E[Y \mid X]$ is mean-independent of X by the construction of conditional expectation.

The CEF has an optimality property: among all functions $g(X)$, $E[Y \mid X]$ minimizes the mean squared prediction error,

$$E[Y \mid X] = \arg\min_{g} \; E\big[(Y - g(X))^2\big]. \tag{1.16}$$

That is, if we could use any function of X to predict Y, the CEF would be the best predictor in the mean squared error sense. The proof follows from the decomposition above: for any g,

$$E\big[(Y - g(X))^2\big] = E\big[(Y - E[Y \mid X])^2\big] + E\big[(E[Y \mid X] - g(X))^2\big] + 2\,E\big[\varepsilon\,(E[Y \mid X] - g(X))\big]. \tag{1.17}$$

The cross term vanishes by iterated expectations: $E[Y \mid X] - g(X)$ is a function of X, so $E\big[\varepsilon\,(E[Y \mid X] - g(X))\big] = E\big[E[\varepsilon \mid X]\,(E[Y \mid X] - g(X))\big] = 0$. The first term does not depend on g, and the second is nonnegative and equals zero when $g = E[Y \mid X]$.

1.4.2 The Best Linear Predictor

In practice we rarely know the CEF, and we often restrict attention to linear predictors. The best linear predictor (BLP) of Y given X is the coefficient vector β that solves

$$\beta = \arg\min_b \; E\big[(Y - X'b)^2\big]. \tag{1.18}$$

Taking the first-order condition,

$$\frac{\partial}{\partial b} E\big[(Y - X'b)^2\big] = -2\, E\big[X(Y - X'b)\big] = 0, \tag{1.19}$$

which gives the population normal equation

$$E[XX']\,\beta = E[XY]. \tag{1.20}$$

If $E[XX']$ is nonsingular, the BLP coefficient is

$$\beta = \left(E[XX']\right)^{-1} E[XY]. \tag{1.21}$$

This is the population analog of the sample formula $\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-\mathbf{1}}\mathbf{X}'\mathbf{y}$. The population version replaces sample cross-products with their expectations.

The first-order condition can equivalently be written as the population orthogonality condition:

$$E\big[X(Y - X'\beta)\big] = 0. \tag{1.22}$$

This says that the linear prediction error $Y - X'\beta$ is uncorrelated with every component of X. It is the population version of the sample condition $\mathbf{X}'\hat{\mathbf{u}} = \mathbf{0}$ that we derived from the matrix normal equations.

1.4.3 The BLP Approximates the CEF

The BLP and the CEF are connected by a key result: the BLP is the best linear approximation to the CEF. To see this, write

$$\beta = \left(E[XX']\right)^{-1} E[XY] = \left(E[XX']\right)^{-1} E\big[X\, E[Y \mid X]\big], \tag{1.23}$$

where the second equality uses the law of iterated expectations. The BLP coefficient depends on Y only through the CEF. OLS is fitting a linear function to the CEF, not directly to Y.

When the CEF is linear in X — that is, $E[Y \mid X] = X'\beta$ for some β — the BLP recovers the CEF exactly. The linear model is correctly specified and OLS estimates the true conditional expectation.

When the CEF is nonlinear, the BLP still provides the minimum mean squared error linear approximation to it. The OLS coefficient retains a well-defined interpretation: it is the slope of the best linear predictor, even if the world is nonlinear. This is why OLS remains useful in settings where the true CEF is unknown and possibly complicated.

1.4.4 From Population to Sample

The population formulas become sample formulas when we replace expectations with sample averages. For a sample of n observations,

$$E[XX'] \approx \frac{1}{n}\sum_{i=1}^{n} X_i X_i' = \frac{1}{n}\mathbf{X}'\mathbf{X} \tag{1.24}$$

$$E[XY] \approx \frac{1}{n}\sum_{i=1}^{n} X_i Y_i = \frac{1}{n}\mathbf{X}'\mathbf{y} \tag{1.25}$$

Substituting into the population BLP formula gives

$$\hat{\boldsymbol{\beta}} = \left(\frac{1}{n}\mathbf{X}'\mathbf{X}\right)^{-1}\frac{1}{n}\mathbf{X}'\mathbf{y} = (\mathbf{X}'\mathbf{X})^{-\mathbf{1}}\mathbf{X}'\mathbf{y}, \tag{1.26}$$

which is the OLS estimator derived from the matrix normal equations. The $1/n$ factors cancel. This shows what OLS does: it uses sample analogs to estimate the population BLP, which is the best linear approximation to the conditional expectation function $E[Y \mid X]$.

1.5 Properties of OLS estimator

1.5.1 Unbiasedness

In a linear model

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{u}, \tag{1.27}$$

the OLS estimator is

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-\mathbf{1}}\mathbf{X}'\mathbf{y} \tag{1.28}$$

Since $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{u}$,

$$\hat{\boldsymbol{\beta}} = \boldsymbol{\beta} + (\mathbf{X}'\mathbf{X})^{-\mathbf{1}}\mathbf{X}'\mathbf{u}. \tag{1.29}$$

This makes

$$\mathrm{E}(\hat{\boldsymbol{\beta}}|\mathbf{X}) = \boldsymbol{\beta} + (\mathbf{X}'\mathbf{X})^{-\mathbf{1}}\mathbf{X}'\mathbf{E}(\mathbf{u}|\mathbf{X}). \tag{1.30}$$

The condition that makes the OLS estimator unbiased is:

$$\mathrm{E}(\mathbf{u}|\mathbf{X}) = \mathbf{0}, \tag{1.31}$$

that is, all explanatory variables which form the columns of $\mathbf{X}$ are exogenous. This condition is weaker than the independence condition that u and X are independent. This says that given $\mathbf{X}$, the expected value of $\mathbf{u}$ is zero; it implies that the model is correctly specified. That is, $\mathbf{y}$ is a linear function of $\mathbf{X}$.

In the context of cross-sectional data, this assumption is plausible. However, when we have time series data, the assumption becomes strong, because it assumes that the entire series of $\mathbf{X}$ has no relationship with the error term. In a time series context, this is hard to satisfy. The OLS estimator is biased if this condition is not satisfied.

For example, suppose we have a model

$$y_t = \beta_1 + \beta_2 y_{t-1} + u_t, \quad u_t \sim \mathrm{IID}(0, \sigma^2). \tag{1.32}$$

In this simple model, even if we assume that y_{t-1} and u_t are uncorrelated, OLS estimator is still biased. That is because $\mathrm{E}(\mathbf{u}|\mathbf{X}) = \mathbf{0}$ is not satisfied: y_{t-1} depends on u_{t-1}, u_{t-2} and so on.

There is a weaker condition – weaker than the strict exogeneity above, and in fact the weakest of the conditions in this chapter:

$$\mathrm{E}(\mathbf{X}\mathbf{u}) = \mathbf{0}, \tag{1.33}$$

This is a moment condition, and it is what GMM estimators are built on. It is a separate assumption from getting the functional form right: a correctly specified linear equation can still have $E(\mathbf{X}\mathbf{u}) \neq 0$, which is exactly the endogeneity problem of a later chapter.

1.5.2 Consistency

For OLS estimator to be consistent, a much weaker condition is needed:

$$\mathrm{E}(u_t|X_t) = 0, \tag{1.34}$$

This condition is much weaker since it only assumes that the mean of current error term does not depend on the current predictors. Even a model with lagged dependent variable can easily satisfy this condition. This condition is called **contemporaneous exogeneity**. It is the weakest of three exogeneity conditions worth keeping apart, which is precisely why it is the one to invoke for consistency:

- **Strict exogeneity**, $E[u_t \mid X_1, \ldots, X_n] = 0$: the error in every period is mean-independent of the regressors in *every* period. This is what unbiasedness requires, and what the lagged-dependent-variable example above violates.
- **Predeterminedness**, $E[u_t \mid \mathcal{F}_t] = 0$, where $\mathcal{F}_t = \sigma(X_1, \ldots, X_t)$ is the regressor history through period t – equivalently, X_t is orthogonal to the current and all *future* errors, while being allowed to correlate with *past* errors. This is the relevant condition in models with a lagged dependent variable.
- **Contemporaneous exogeneity**, $E[u_t \mid X_t] = 0$.
- **Contemporaneous orthogonality**, $E[X_t u_t] = 0$ – mere zero correlation. This is weakest of all, and it is what consistency actually needs: together with a rank condition and a law of large numbers – which for dependent data needs the weak-dependence conditions of the MLE chapter – it gives $\operatorname{plim} n^{-1}\mathbf{X}'\mathbf{u} = \mathbf{0}$ and hence $\operatorname{plim} \hat{\beta} = \beta$. Contemporaneous *exogeneity* is a strictly stronger, sufficient condition – it constrains the whole conditional mean rather than a single covariance – so the $E[\mathbf{X}\mathbf{u}] = \mathbf{0}$ introduced under unbiasedness above sits at the *bottom* of this ladder, not between the others.

Each condition implies the next, because conditioning on a larger information set is a more restrictive requirement, not a less restrictive one. The ladder needs the information sets to be nested, which is why $\mathcal{F}_t$ is defined above as regressor history: $\sigma(X_t) \subseteq \sigma(X_1, \dots, X_t) \subseteq \sigma(X_1, \dots, X_n)$. Widen $\mathcal{F}_t$ to include past errors or other series and strict exogeneity no longer implies predeterminedness, because then neither information set contains the other. By the law of iterated expectations,

$$E[u_t \mid X_t] = E\big[E[u_t \mid \mathcal{F}_t] \bigm| X_t\big] = 0, \tag{1.35}$$

so predeterminedness delivers contemporaneous exogeneity but not conversely. In the time series example, then, the OLS estimator is biased, but can still be consistent, if we are willing to assume no contemporaneous correlation.

Note: general linear hypothesis testing, the Chow test for structural change, and heteroskedasticity- and autocorrelation-consistent (HAC) standard errors (White, Newey-West) are all standard OLS extensions, but are covered in the Maximum Likelihood chapter alongside this guide's other estimation-inference material rather than repeated here.

2 The Method of Maximum Likelihood

If we make the assumption that the error terms are normally distributed, the maximum likelihood estimators (MLE) coincide with the least square estimators. But MLE can also be applied to a wide variety of models other than regression models, and it generally generates estimators with excellent asymptotic properties. The major disadvantage of MLE is that it requires stronger distributional assumptions than does the method of moments.

2.1 Basic Concepts

Maximum likelihood estimation is to find the set of parameters which makes the current sample most likely given the statistical model.

Suppose we have a sample of n independent and identically distributed (IID) observations. Each of them comes from some distribution function $f(\cdot, \boldsymbol{\theta})$, where parameters $\boldsymbol{\theta}$ is fixed. The joint distribution of all the data points would be

$$f(x_1, x_2, \dots, x_n|\theta) = f(x_1|\theta) \times f(x_2|\theta) \times \dots \times f(x_n|\theta). \tag{2.1}$$

It is referred to as the likelihood function of the model for the given data set.

Then it's natural to estimate θ by picking the θ that maximize this joint distribution function. A parameter vector $\hat{\boldsymbol{\theta}}$ at which the likelihood takes on its maximum value is called a maximum likelihood estimate, or MLE, of the parameters.

Usually the log form of the likelihood function is preferred for computational purposes.

2.2 MLE for a linear model

Consider the classical normal linear model:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{u}, \quad \mathbf{u} \sim \mathbf{N}(\mathbf{0}, \boldsymbol{\sigma}^{\mathbf{2}}\mathbf{I}) \tag{2.2}$$

In this case, we have two parameters to estimate, σ and $\boldsymbol{\beta}$. The distribution of $\mathbf{y}|\mathbf{X}$ is

$$f(\mathbf{y} \mid \mathbf{X}, \beta, \sigma) = (2\pi\sigma^2)^{-n/2} \exp\left(-\frac{(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})'(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})}{2\sigma^2}\right) \tag{2.3}$$

The log-likelihood function is

$$l(\mathbf{y}, \boldsymbol{\beta}, \boldsymbol{\sigma}) = -\frac{n}{2}\log(2\pi) - \frac{n}{2}\log(\sigma^2) - \frac{1}{2\sigma^2}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})'(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) \tag{2.4}$$

Maximizing this function gives exactly the OLS estimator for $\boldsymbol{\beta}$. That is algebra, and it holds whatever the true error distribution happens to be: the Gaussian log-likelihood is a decreasing function of the sum of squared residuals, so whatever minimizes one maximizes the other. For σ^2 the MLE divides by n rather than $n-k$, so it differs in finite samples and agrees asymptotically. What normality buys is not the formula but its standing. Under normality the OLS estimator *is* the maximum likelihood estimator and the model-based variance is valid. Without normality the same formula is a quasi-MLE: still consistent for $\boldsymbol{\beta}$ under the usual conditions, but its variance needs the sandwich correction discussed below.

2.3 Asymptotic Properties of MLE

1. Consistency

$$\text{plim}(\hat{\boldsymbol{\theta}}) = \boldsymbol{\theta}. \tag{2.5}$$

This is to say that MLE estimator can get arbitrarily close to the true parameter if the sample size gets to infinity.

2. Asymptotic Normality

$$\sqrt{\mathbf{n}}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}) \xrightarrow{d} N(\mathbf{0}, \mathbf{I^{-1}}(\boldsymbol{\theta})), \tag{2.6}$$

where $\mathbf{I}(\boldsymbol{\theta})$ below is the *per-observation* information. Equivalently, in finite-sample terms,

$$\hat{\boldsymbol{\theta}} \approx N(\boldsymbol{\theta}, \mathbf{I_n^{-1}}(\boldsymbol{\theta})), \qquad \mathbf{I_n}(\boldsymbol{\theta}) = \mathbf{n}\,\mathbf{I}(\boldsymbol{\theta}), \tag{2.7}$$

with $\mathbf{I_n}(\boldsymbol{\theta})$ the *full-sample* information. The $\sqrt{n}$ scaling (equivalently, dividing the full-sample information by n) is what makes the statement well-defined; writing $\hat{\theta} \sim N(\theta, I^{-1}(\theta))$ without specifying which information is meant is ambiguous.

This states that the asymptotic distribution of $\hat{\boldsymbol{\theta}}$ is normal with mean $\boldsymbol{\theta}$ and variance given by the inverse of the information, where the per-observation information matrix $\mathbf{I}(\boldsymbol{\theta})$ is defined using the single-observation log-likelihood contribution l_t (so that the full-sample log-likelihood is $l = \sum_{t=1}^{n} l_t$):

$$\mathbf{I}(\boldsymbol{\theta}) = E[(\frac{\partial l_t}{\partial \theta})(\frac{\partial l_t}{\partial \theta})'] = -E[(\frac{\partial^2 l_t}{\partial \theta \partial \theta'})] \tag{2.8}$$

The information matrix is the expected value of the outer product of the gradient of a single observation's log-likelihood contribution, or the negative value of the expected value of its Hessian (second derivative). Using the *full-sample* l here instead of l_t would give the full-sample information $I_n(\theta) = n\,I(\theta)$, not $I(\theta)$ itself.

3. Asymptotic efficiency.

If $\hat{\boldsymbol{\theta}}$ is the MLE estimator of θ and $\mathbf{V}$ denotes its *asymptotic* variance matrix – the variance of the limiting distribution of $\sqrt{n}(\hat{\theta} - \theta)$, not of $\hat{\boldsymbol{\theta}}$ itself, the two differing by a factor of n exactly as with the information matrix above – then

$$\sqrt{n}(\hat{\theta} - \theta) \to^d N(\mathbf{0}, \mathbf{V}), \qquad \mathbf{V} = \mathbf{I^{-1}}(\boldsymbol{\theta}). \tag{2.9}$$

If $\tilde{\mathbf{V}}$ denotes the asymptotic variance matrix of any other *regular* consistent, asymptotically normal estimator, then $\tilde{\mathbf{V}} - \mathbf{V}$ is a positive semidefinite matrix. That is, the MLE is the best estimator, in terms of asymptotic variance, in that class. The regularity qualifier is not pedantry: without it the statement is false, since superefficient estimators such as Hodges' can beat the bound at a single point of the parameter space, at the cost of much worse behaviour in a neighbourhood of it.

4. Invariance.

If $\hat{\theta}$ is the MLE of θ and $g(\theta)$ is a continuous function of θ, then $g(\hat{\theta})$ is the MLE of $g(\theta)$.

2.4 When the Model Is Wrong: Quasi-MLE and M-Estimation

2.4.1 What MLE Actually Minimizes

The asymptotic properties above assume that the true density belongs to the parametric family $f(y \mid \theta)$. In practice this is rarely literally true. What does MLE estimate when the model is wrong?

Write $g(y)$ for the true density and $f(y \mid \theta)$ for the parametric model. Maximizing the log-likelihood is equivalent to minimizing the Kullback-Leibler divergence from the true density to the model:

$$\mathrm{KL}(g \,\|\, f(\cdot \mid \theta)) = \int g(y) \log \frac{g(y)}{f(y \mid \theta)}\, dy = E_g[\log g(Y)] - E_g[\log f(Y \mid \theta)]\,. \tag{2.10}$$

The first term does not depend on θ, so minimizing KL over θ is the same as maximizing $E_g[\log f(Y \mid \theta)]$, which is the population version of the sample log-likelihood. The MLE converges to the value θ^* that makes $f(\cdot \mid \theta)$ closest to g in the KL sense, whether or not g belongs to the family. This θ^* is called the pseudo-true value.

2.4.2 The Information Matrix Inequality

Under correct specification, the two forms of the information matrix are equal:

$$J(\theta) = E\left[\frac{\partial l_t}{\partial \theta}\frac{\partial l_t}{\partial \theta}'\right] = -E\left[\frac{\partial^2 l_t}{\partial \theta\, \partial \theta'}\right] = H(\theta). \tag{2.11}$$

This is the information matrix equality. Under misspecification it fails: $J(\theta^*) \neq H(\theta^*)$ in general. The asymptotic variance of the MLE is no longer $I^{-1}(\theta)$ but the sandwich

$$V = H(\theta^*)^{-1}\, J(\theta^*)\, H(\theta^*)^{-1}. \tag{2.12}$$

When the model is correctly specified, $J = H$ and the sandwich collapses to $H^{-1} = I^{-1}$, recovering the standard result. When it is not, H^{-1} is simply the wrong variance – inconsistent, rather than merely too small. There is no general ordering between H^{-1} and the sandwich: it can be too small in some parameter directions and too large in others.

2.4.3 Quasi-Maximum Likelihood Estimation

The sandwich variance makes MLE usable even under misspecification, provided we are clear about what parameter is being estimated. A quasi-maximum likelihood estimator (QMLE) is an MLE computed from a likelihood that may be wrong, together with a sandwich variance that accounts for the misspecification.

The leading case in econometrics is Poisson QMLE for count or nonnegative data. The Poisson log-likelihood for observation t is

$$l_t = y_t \log \mu_t - \mu_t - \log(y_t!), \quad \mu_t = \exp(\mathbf{x}_t' \beta). \tag{2.13}$$

The score equation $\sum_t (y_t - \mu_t)\, \mathbf{x}_t = 0$ depends only on the conditional mean $E[Y_t \mid \mathbf{x}_t] = \mu_t$, not on whether Y_t is actually Poisson. If the conditional mean is correctly specified, the QMLE $\hat{\beta}$ is consistent for β regardless of the true distribution of Y_t — the data need not even be counts. The variance, however, must be estimated with the sandwich, since the Poisson variance assumption $\mathrm{Var}(Y_t \mid \mathbf{x}_t) = \mu_t$ will generally be wrong. A fuller treatment is in the Count Data chapter.

The same logic applies to logit and probit for binary data: as long as the conditional mean $P(Y = 1 \mid X) = F(X'\beta)$ is correctly specified, the MLE is consistent for β even if the error distribution is not exactly logistic or normal. In this sense, binary response MLE is already quasi-MLE.

2.4.4 M-Estimators

The QMLE perspective generalizes further. An M-estimator is any estimator defined as the solution to

$$\sum_{t=1}^{n} \psi(y_t, \mathbf{x}_t, \theta) = 0, \tag{2.14}$$

where ψ is a known vector-valued function. MLE is the special case where $\psi = \partial l_t / \partial \theta$, the score. OLS is the case $\psi(y_t, \mathbf{x}_t, \beta) = \mathbf{x}_t(y_t - \mathbf{x}_t'\beta)$, the population orthogonality condition from the OLS chapter. GMM generalizes further by allowing more equations than parameters; the GMM chapter develops this.

All M-estimators share the same asymptotic structure under regularity conditions:

$$\sqrt{n}\,(\hat{\theta} - \theta^*) \xrightarrow{d} N(0,\ A^{-1}\, B\, A^{-1\prime})\,, \tag{2.15}$$

where $A = E[\partial \psi / \partial \theta']$ and $B = E[\psi \psi']$. This is the sandwich again. When ψ is a score from a correctly specified likelihood, $A = -H$, $B = J = H$, and the sandwich reduces to H^{-1}.

The M-estimator framework unifies OLS, MLE, QMLE, and GMM as special cases of the same asymptotic theory. The sandwich variance is the general formula; the simpler I^{-1} or $\sigma^2(X'X)^{-1}$ are special cases that hold under correct specification.

2.5 Example

Let's look at the example of regression of car's mpg on disp, hp and wt. This time we use MLE. (R has a mle() function which I cannot make it work on this example. So I am using optim().)

```r
ols.lf <- function(theta, y, X) {
  beta <- theta[-1]
  sigma2 <- theta[1]
  # Return a large penalty rather than NA: L-BFGS-B's line search can
  # transiently evaluate points that violate the box constraint, and
  # `fn` returning NA there causes a fatal "L-BFGS-B needs finite values
  # of 'fn'" error. A finite penalty keeps the optimizer on track instead.
  if (sigma2 <= 0) return(1e10)
  n <- nrow(X)
  e <- y - X%*%beta
  logl <- ((-n/2)*log(2*pi)) - ((n/2)*log(sigma2)) - ((t(e)%*%e)/(2*sigma2))
  return(-logl)
}
X <- cbind(1,mtcars[,c('disp','hp','wt')])
y <- mtcars$mpg

# Deliberately naive starting values (sigma2 = 1, slopes at +/-1): the point
# of the exercise is that the optimizer finds the OLS solution without being
# handed it. For harder likelihoods than this one, good starting values matter
# a great deal -- they reduce the risk of converging to a spurious local
# maximum, or of failing to converge at all.
optim(c(1,1,-1,-1,-1), method="L-BFGS-B", fn=ols.lf, lower=c(1e-6,-Inf,-Inf,-Inf,-Inf), upper=
```

```
$par
[1]  6.0935580702 37.1058998652 -0.0009352804 -0.0311594264 -3.8010038304

$value
[1] 74.32149

$counts
function gradient
     126      126

$convergence
[1] 0

$message
[1] "CONVERGENCE: REL_REDUCTION_OF_F <= FACTR*EPSMCH"
```

```r
# it should match the OLS estimator
```

```r
lm1 <- lm(mpg~disp+hp+wt, data=mtcars)
summary(lm1)
```

```

Call:
lm(formula = mpg ~ disp + hp + wt, data = mtcars)

Residuals:
   Min     1Q Median     3Q    Max 
-3.891 -1.640 -0.172  1.061  5.861 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept) 37.105505   2.110815  17.579  < 2e-16 ***
disp        -0.000937   0.010350  -0.091  0.92851    
hp          -0.031157   0.011436  -2.724  0.01097 *  
wt          -3.800891   1.066191  -3.565  0.00133 ** 
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.639 on 28 degrees of freedom
Multiple R-squared:  0.8268,	Adjusted R-squared:  0.8083 
F-statistic: 44.57 on 3 and 28 DF,  p-value: 8.65e-11
```

2.6 Hypotheses Testing

2.6.1 Linear Hypotheses Testing

Linear hypothesis can be framed as

$$\mathbf{H_0}: \quad \mathbf{R}\boldsymbol{\beta} - \mathbf{r} = \mathbf{0}, \tag{2.16}$$

where $\mathbf{R}$ is a $q \times k$ matrix of known constants with $q \leq k$ and $\text{rank}(\mathbf{R}) = q$, and $\mathbf{r}$ is a q-vector of known constants. The restrictions may number as many as the parameters – testing that every coefficient is zero is the case $q = k$ – but they must be linearly independent.

We assume that u_i are normally distributed.

$$\mathbf{u} \sim \mathbf{N}(\mathbf{0}, \boldsymbol{\sigma}^2\mathbf{I}) \tag{2.17}$$

Even if this assumption does not hold, we still have asymptotic normality for $\hat{\boldsymbol{\beta}}$, so the tests below remain valid asymptotically.

Since linear combination of normal variables are also normally distributed,

$$\begin{aligned}\hat{\boldsymbol{\beta}} &\sim \mathbf{N}(\boldsymbol{\beta}, \boldsymbol{\sigma^2}(\mathbf{X'X})^{-1}) \\ \mathbf{R}\hat{\boldsymbol{\beta}} &\sim \mathbf{N}(\mathbf{R}\boldsymbol{\beta}, \boldsymbol{\sigma^2}\mathbf{R}(\mathbf{X'X})^{-1}\mathbf{R'}) \\ \mathbf{R}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}) &\sim \mathbf{N}(\mathbf{0}, \boldsymbol{\sigma^2}\mathbf{R}(\mathbf{X'X})^{-1}\mathbf{R'})\end{aligned} \tag{2.18}$$

Under null hypothesis,

$$(\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r}) \sim \mathbf{N}(\mathbf{0}, \boldsymbol{\sigma^2}\mathbf{R}(\mathbf{X'X})^{-1}\mathbf{R'}) \tag{2.19}$$

Therefore,

$$(\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r})'[\boldsymbol{\sigma^2}\mathbf{R}(\mathbf{X'X})^{-1}\mathbf{R'}]^{-1}(\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r}) \sim \chi^2(q) \tag{2.20}$$

Here we don't know σ^2. However, it can be shown that

$$\frac{\hat{\mathbf{u}}'\hat{\mathbf{u}}}{\boldsymbol{\sigma^2}} \sim \chi^2(n-k) \tag{2.21}$$

Then we have

$$\frac{(\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r})'[\mathbf{R}(\mathbf{X'X})^{-1}\mathbf{R'}]^{-1}(\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r})/\mathbf{q}}{\hat{\mathbf{u}}'\hat{\mathbf{u}}/(\mathbf{n}-\mathbf{k})} \sim F(q, n-k) \tag{2.22}$$

This is an example of the F test for any linear hypotheses testing. t test will be a special case of this.

2.6.2 The three classical tests: LR, Wald and LM

The F test above leans on normal errors. The general machinery for testing $\mathbf{H_0} : \mathbf{R}\boldsymbol{\theta} - \mathbf{r} = \mathbf{0}$ after maximum likelihood needs only the asymptotic results already established, and it comes in three forms. Write $\hat{\theta}_u$ for the unrestricted MLE, $\hat{\theta}_r$ for the MLE subject to the restriction, $s(\theta) = \partial\ell/\partial\theta$ for the score, and $\mathbf{I_n}(\boldsymbol{\theta}) = \mathbf{n}\,\mathbf{I}(\boldsymbol{\theta})$ for the full-sample information.

Likelihood ratio. Compare the two maximized log-likelihoods:

$$LR = 2[\ell(\hat{\theta}_u) - \ell(\hat{\theta}_r)] \xrightarrow{d} \chi^2(q). \tag{2.23}$$

Wald. Ask whether the *unrestricted* estimate is far from satisfying the restriction, measuring distance in units of its own estimated variance:

$$W = (\mathbf{R}\hat{\theta}_u - \mathbf{r})'\big[\mathbf{R}\,\mathbf{I}_n^{-1}(\hat{\theta}_u)\,\mathbf{R}'\big]^{-1}(\mathbf{R}\hat{\theta}_u - \mathbf{r}) \xrightarrow{d} \chi^2(q). \tag{2.24}$$

Lagrange multiplier (score). Ask whether the score, evaluated at the *restricted* estimate, is far from zero – if the restriction is harmless, the restricted point should already be nearly a maximum:

$$LM = s(\hat{\theta}_r)'\,\mathbf{I}_n^{-1}(\hat{\theta}_r)\,s(\hat{\theta}_r) \xrightarrow{d} \chi^2(q). \tag{2.25}$$

The three are asymptotically equivalent under the null and have the same local power, so the choice among them is one of convenience: **Wald needs only the unrestricted fit, LM needs only the restricted fit, and LR needs both.** That is why LM is the natural choice for specification testing, where the restricted model is the easy one to estimate (White's heteroskedasticity test and the Breusch-Godfrey serial-correlation test are both LM tests run as auxiliary regressions), while Wald is what software reports by default, since the unrestricted fit is what was just computed.

In the classical normal linear model the three are monotone transformations of the same F statistic above, so nothing is lost by using F there. Two cautions otherwise. In finite samples the three can disagree, sometimes enough to straddle a conventional critical value, and there is a known ordering $W \geq LR \geq LM$ in the linear model under normality. And the Wald statistic is **not invariant to nonlinear reparameterization**: testing $\gamma = 1$ and testing $\log \gamma = 0$ are the same hypothesis but give different Wald statistics, whereas LR is invariant because it compares likelihood *values*, which reparameterization does not change. When a hypothesis can be written several algebraically equivalent ways, prefer LR.

2.7 Test of Structural Change (Chow test)

Chow test is a test between two groups of observations. For example, we have data on consumption function of prewar period and postwar period. It is natural to think of a test on whether consumption function parameters differ between prewar and postwar periods. The null hypothesis is that there is no difference.

Let $\mathbf{y_i}$, $\mathbf{X_i}$ $(i = 1, 2)$ indicate the appropriate partitioning of the data. The unrestricted model may be written

$$\begin{bmatrix} \mathbf{y_1} \\ \mathbf{y_2} \end{bmatrix} = \begin{bmatrix} \mathbf{X_1} & \mathbf{0} \\ \mathbf{0} & \mathbf{X_2} \end{bmatrix} \begin{bmatrix} \boldsymbol{\beta_1} \\ \boldsymbol{\beta_2} \end{bmatrix} + u \quad u \sim N(0, \sigma^2 I) \tag{2.26}$$

The null hypothesis of no structural break is

$$\mathbf{H_0} : \quad \boldsymbol{\beta_1} = \boldsymbol{\beta_2}. \tag{2.27}$$

Running OLS on two equations separately, we have

$$\begin{bmatrix} \hat{\boldsymbol{\beta}}_\mathbf{1} \\ \hat{\boldsymbol{\beta}}_\mathbf{2} \end{bmatrix} = \begin{bmatrix} \mathbf{X_1'X_1} & \mathbf{0} \\ \mathbf{0} & \mathbf{X_2'X_2} \end{bmatrix}^{-1} \begin{bmatrix} \mathbf{X_1'y_1} \\ \mathbf{X_2'y_2} \end{bmatrix} = \begin{bmatrix} (\mathbf{X_1'X_1})^{\mathbf{-1}}\mathbf{X_1'y_1} \\ (\mathbf{X_2'X_2})^{\mathbf{-1}}\mathbf{X_2'y_2} \end{bmatrix} \tag{2.28}$$

Restricted model under null hypothesis

$$\begin{bmatrix} \mathbf{y_1} \\ \mathbf{y_2} \end{bmatrix} = \begin{bmatrix} \mathbf{X_1} \\ \mathbf{X_2} \end{bmatrix} \boldsymbol{\beta} + \mathbf{u} \tag{2.29}$$

The test of the null hypothesis is given by

$$\mathbf{F} = \frac{(\hat{\mathbf{u}}_*'\hat{\mathbf{u}}_* - \hat{\mathbf{u}}'\hat{\mathbf{u}})/\mathbf{k}}{\hat{\mathbf{u}}'\hat{\mathbf{u}}/(\mathbf{n}-\mathbf{2k})} \sim F(k, n-2k) \tag{2.30}$$

where $\hat{u}_*$ is the residual of the restricted model, $\hat{u}$ is the stacked residual of two unrestricted models.

For example, for prewar period (suppose there is n_1 observations):

$$y_1 = \beta_0 + \beta_1 x_1 + \beta_2 x_2 \tag{2.31}$$

For postwar period (suppose there is n_2 observations):

$$y_2 = \gamma_0 + \gamma_1 x_1 + \gamma_2 x_2 \tag{2.32}$$

The null hypothesis of no structural break is

$$\mathbf{H_0}: \quad \boldsymbol{\beta_0} = \boldsymbol{\gamma_0}, \boldsymbol{\beta_1} = \boldsymbol{\gamma_1}, \boldsymbol{\beta_2} = \boldsymbol{\gamma_2}. \tag{2.33}$$

Running OLS on two equations separately, we have two sums of squares of error (SSR_1 and SSR_2). The unrestricted error sum of squares is

$$SSR_U = SSR_1 + SSR_2 \tag{2.34}$$

Then run the regression on the stacked sample of $n_1 + n_2$ observations and get SSR_R.

Then

$$\frac{(SSR_R - SSR_U)/k}{SSR_U/(n_1+n_2-2k)} \sim F_{k,n_1+n_2-2k}. \tag{2.35}$$

k is number of restrictions; in this example, it is 3.

The other way to do the same test is easier: simply include interaction terms between group dummy and explanatory variables in the regression.

For our earlier example, we would estimate the model:

$$y = \alpha_0 + \alpha_1 x_1 + \alpha_2 x_2 + \eta_0 d + \eta_1 x_1 d + \eta_2 x_2 d \tag{2.36}$$

Here d is a dummy variable for postwar period. The Chow test is simply a test on the hypothesis that all the coefficients involving d are zero. A regression of the above model with $n+m$ observations and a F test are easy to do.

$$\mathbf{H_0}: \quad \boldsymbol{\eta_0} = \boldsymbol{\eta_1} = \boldsymbol{\eta_2} = \mathbf{0}. \tag{2.37}$$

2.7.1 An example of Chow test in R

Here we use an example from the “strucchange” library.

```
## Example 7.4 from Greene (1993), "Econometric Analysis"
## Chow test on Longley data
data("longley")
library(strucchange)
## use structural change test from the library.
sctest(Employed ~ Year + GNP.deflator + GNP + Armed.Forces, data = longley,
  type = "Chow", point = 7)
```

```
    Chow test

data:  Employed ~ Year + GNP.deflator + GNP + Armed.Forces
F = 3.9268, p-value = 0.06307
```

```
## which is equivalent to segmenting the regression via
fac <- factor(c(rep(1, 7), rep(2, 9)))
## here fac is the factor for segmenting it at point 7.
fm0 <- lm(Employed ~ Year + GNP.deflator + GNP + Armed.Forces, data = longley)
fm1 <- lm(Employed ~ fac/(Year + GNP.deflator + GNP + Armed.Forces), data = longley)
## Here anova is returning the F test (Chow test). Equivalent to sctest results.
anova(fm0, fm1)
```

```
Analysis of Variance Table

Model 1: Employed ~ Year + GNP.deflator + GNP + Armed.Forces
Model 2: Employed ~ fac/(Year + GNP.deflator + GNP + Armed.Forces)
  Res.Df    RSS Df Sum of Sq      F  Pr(>F)
1     11 4.8987
2      6 1.1466  5    3.7521 3.9268 0.06307 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
## estimates from Table 7.5 in Greene (1993)
summary(fm0)
```

```
Call:
lm(formula = Employed ~ Year + GNP.deflator + GNP + Armed.Forces,
    data = longley)

Residuals:
```

```
    Min      1Q  Median      3Q     Max 
-0.9058 -0.3427 -0.1076  0.2168  1.4377 

Coefficients:
               Estimate Std. Error t value Pr(>|t|)   
(Intercept)   1.169e+03  8.359e+02   1.399  0.18949   
Year         -5.765e-01  4.335e-01  -1.330  0.21049   
GNP.deflator -1.977e-02  1.389e-01  -0.142  0.88940   
GNP           6.439e-02  1.995e-02   3.227  0.00805 **
Armed.Forces -1.015e-04  3.086e-03  -0.033  0.97436   
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6673 on 11 degrees of freedom
Multiple R-squared:  0.9735,	Adjusted R-squared:  0.9639 
F-statistic: 101.1 on 4 and 11 DF,  p-value: 1.346e-08
```

```
summary(fm1)
```

```
Call:
lm(formula = Employed ~ fac/(Year + GNP.deflator + GNP + Armed.Forces), 
    data = longley)

Residuals:
     Min       1Q   Median       3Q      Max 
-0.47717 -0.18950  0.02089  0.14836  0.56493 

Coefficients:
                    Estimate Std. Error t value Pr(>|t|)   
(Intercept)        1.678e+03  9.390e+02   1.787  0.12413   
fac2               2.098e+03  1.786e+03   1.174  0.28473   
fac1:Year         -8.352e-01  4.847e-01  -1.723  0.13563   
fac2:Year         -1.914e+00  7.913e-01  -2.419  0.05194 . 
fac1:GNP.deflator -1.633e-01  1.762e-01  -0.927  0.38974   
fac2:GNP.deflator -4.247e-02  2.238e-01  -0.190  0.85576   
fac1:GNP           9.481e-02  3.815e-02   2.485  0.04747 * 
fac2:GNP           1.123e-01  2.269e-02   4.951  0.00258 **
fac1:Armed.Forces -2.467e-03  6.965e-03  -0.354  0.73532   
fac2:Armed.Forces -2.579e-02  1.259e-02  -2.049  0.08635 . 
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.4372 on 6 degrees of freedom
Multiple R-squared:  0.9938,	Adjusted R-squared:  0.9845 
F-statistic: 106.9 on 9 and 6 DF,  p-value: 6.28e-06
```

2.8 Heteroskedasticity and Autocorrelation Consistent Standard Errors

2.8.1 Variance estimator under homoscedasticity

In OLS regression, the variance-covariance matrix of $\hat{\beta}$ is

$$\mathrm{var}(\hat{\boldsymbol{\beta}}) = \mathbf{E}[(\hat{\boldsymbol{\beta}}-\boldsymbol{\beta})(\hat{\boldsymbol{\beta}}-\boldsymbol{\beta})'] = \mathbf{E}[(\mathbf{X}'\mathbf{X})^{-\mathbf{1}}\mathbf{X}'\mathbf{u}\mathbf{u}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-\mathbf{1}}] = \boldsymbol{\sigma}^{\mathbf{2}}(\mathbf{X}'\mathbf{X})^{-\mathbf{1}} \tag{2.38}$$

if

$$\mathrm{var}(\mathbf{u}) = \boldsymbol{\sigma}^{\mathbf{2}}\mathbf{I}$$

{#eq-mle-33}.

That is, the error term has mean zero and constant variance (homoscedasticity). The pairwise correlation between error terms is always zero (no serial correlation).

The estimation of σ^2:

$$s^2 = \frac{\hat{\mathbf{u}}'\hat{\mathbf{u}}}{n-k} \tag{2.39}$$

is an unbiased estimator of σ^2 (proof omitted).

The standard estimate of the variance-covariance matrix of the OLS parameter estimates under the assumption of IID errors is

$$\hat{\mathrm{Var}}(\hat{\beta}) = s^2(\mathbf{X}'\mathbf{X})^{-\mathbf{1}} \tag{2.40}$$

2.8.2 White's estimator

We made strong assumption that the error terms of the regression model are IID when we estimate the variance-covariance matrix of OLS estimator. Under this assumption, the usual estimator of variance-covariance matrix of $\hat{\boldsymbol{\beta}}$ is consistent. Now let's relax this assumption to only independent but not identically distributed. Again, the linear regression models is

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{u}, \quad \mathbf{E}(\mathbf{u}) = \mathbf{0}, \quad \mathbf{E}(\mathbf{u}\mathbf{u}') = \boldsymbol{\Omega}, \tag{2.41}$$

where $\boldsymbol{\Omega}$ is the error variance-covariance matrix with diagonal elements being σ_t^2 for t^{th} element, off-diagonal elements being zero. In other words, the error terms are heteroscedastic.

The variance-covariance matrix of the OLS estimator $\hat{\boldsymbol{\beta}}$ is equal to

$$\begin{aligned} \mathrm{Var}(\hat{\boldsymbol{\beta}}) &= \mathbf{E}[(\hat{\boldsymbol{\beta}}-\boldsymbol{\beta})(\hat{\boldsymbol{\beta}}-\boldsymbol{\beta})'] \\ &= [(\mathbf{X}'\mathbf{X})^{-\mathbf{1}}\mathbf{X}'(\mathbf{E}(\mathbf{u}\mathbf{u}'))\mathbf{X}(\mathbf{X}'\mathbf{X})^{-\mathbf{1}}] \\ &= (\mathbf{X}'\mathbf{X})^{-\mathbf{1}}\mathbf{X}'\boldsymbol{\Omega}\mathbf{X}(\mathbf{X}'\mathbf{X})^{-\mathbf{1}} \end{aligned} \tag{2.42}$$

If we know σ_t^2, then we would be able to estimate this "sandwich covariance matrix". But we don't.

$$\mathrm{Var}(\hat{\boldsymbol{\beta}}) = \frac{\mathbf{1}}{\mathbf{n}}[\frac{\mathbf{1}}{\mathbf{n}}(\mathbf{X}'\mathbf{X})]^{-\mathbf{1}}[\frac{\mathbf{1}}{\mathbf{n}}\mathbf{X}'\boldsymbol{\Omega}\mathbf{X}][\frac{\mathbf{1}}{\mathbf{n}}(\mathbf{X}'\mathbf{X})]^{-\mathbf{1}} \tag{2.43}$$

Let y_t denote the tth observation on the dependent variable, and $x_t' = [1x_{2t} \cdots x_{kt}]$ denote the tth row of the $\mathbf{X}$ matrix. Then

$$\mathbf{X'\Omega X} = \sum_{\mathbf{t=1}}^{\mathbf{n}} \boldsymbol{\sigma}_{\mathbf{t}}^{\mathbf{2}} \mathbf{x_t x_t'} \tag{2.44}$$

The White estimator replaces the unknown σ_t^2 by $\hat{u}_t^2$, the estimated OLS residuals. This provides a consistent estimator of the variance matrix for the OLS coefficient vector and is particularly useful since it does not require any specific assumptions about the form of the heteroscedasticity.

Therefore,

$$\begin{aligned} \hat{\mathrm{Var}}(\hat{\boldsymbol{\beta}}) &= \frac{\mathbf{1}}{\mathbf{n}}[\frac{\mathbf{1}}{\mathbf{n}}(\mathbf{X'X})]^{-\mathbf{1}}[\frac{\mathbf{1}}{\mathbf{n}}\mathbf{X'\hat{\Omega}X}][\frac{\mathbf{1}}{\mathbf{n}}(\mathbf{X'X})]^{-\mathbf{1}} \\ \hat{\boldsymbol{\Omega}} &= \mathbf{diag}(\hat{\mathbf{u}}_{\mathbf{1}}^{\mathbf{2}}, \hat{\mathbf{u}}_{\mathbf{2}}^{\mathbf{2}}, \cdots, \hat{\mathbf{u}}_{\mathbf{n}}^{\mathbf{2}}) \end{aligned} \tag{2.45}$$

2.8.3 Newey-West estimator

White's estimator deals with the situation that we have heteroskedasticity (a diagonal Ω) of unknown form. When we have serial correlation of unknown form (a non-diagonal Ω), we can estimate the variance-covariance matrix by a heteroskedasticity and autocorrelation consistent, or HAC, estimator. Newey-West estimator is the most popular HAC estimator.

Given a time series data set, suppose we are interested in estimating the mean vector (suppose we have more than one variable) and its variance. We know that given IID data, we can apply central limit theorem: sample mean is a consistent estimator of the population mean and it's variance can be calculated since asymptotically the sample mean conforms to a normal distribution and the variance can be estimated, relatively easily. However, in the case of time series data, autocorrelation usually exists. We may be concerned the CLT may not work in this case.

Fortunately, autocorrelation does not by itself defeat inference, though the conditions need care. Suppose $\mathbf{y_t}$ is covariance-stationary (the covariance is not a function of time) *and* its autocovariances are absolutely summable, $\sum_{v=-\infty}^{\infty} \|\Gamma_v\| < \infty$. Then, as in Hamilton (1994), the sample mean satisfies:

$$\bar{\mathbf{y}}_{\mathbf{t}} \to \boldsymbol{\mu}, \tag{2.46}$$

$$\mathbf{S} = \lim_{T\to\infty} \mathbf{T} \cdot \mathbf{E}[(\bar{\mathbf{y}}_{\mathbf{T}} - \boldsymbol{\mu})(\bar{\mathbf{y}}_{\mathbf{T}} - \boldsymbol{\mu})'] = \sum_{\mathbf{v}=-\infty}^{\infty} \boldsymbol{\Gamma}_{\mathbf{v}}. \tag{2.47}$$

where $\boldsymbol{\Gamma}_{\mathbf{v}}$ is the variance-covariance matrix for $\mathbf{y_t}$ and $\mathbf{y_{t-v}}$.

The first one says the law of large numbers still holds. The second gives the long-run variance used for the standard error, and absolute summability is what makes that sum exist.

Covariance stationarity on its own is not enough for either statement. It can fail even the law of large numbers. Let $y_t = z + \varepsilon_t$, where z is drawn once and then held fixed and ε_t is white noise. Every autocovariance equals $\mathrm{Var}(z)$, so the process is covariance-stationary, the autocovariances are not summable, and $\bar{y}_T$ converges to z rather than to μ. Asymptotic normality asks for more again: a mixing, near-epoch dependence, or martingale-difference condition, together with moment restrictions. Covariance stationarity buys the variance formula, not the central limit theorem.

If the data were generated by a vector MA(q) process, then

$$\mathbf{S} = \sum_{v=-q}^{q} \Gamma_v. \tag{2.48}$$

A natural estimate is

$$\hat{\mathbf{S}} = \hat{\mathbf{\Gamma}}_{\mathbf{0}} + \sum_{\mathbf{v=1}}^{\mathbf{q}} (\hat{\mathbf{\Gamma}}_{\mathbf{v}} + \hat{\mathbf{\Gamma}}'_{\mathbf{v}}), \tag{2.49}$$

where

$$\hat{\mathbf{\Gamma}}_{\mathbf{v}} = (\mathbf{1/T}) \sum_{\mathbf{t=v+1}}^{\mathbf{T}} (\mathbf{y_t} - \bar{\mathbf{y}})(\mathbf{y_{t-v}} - \bar{\mathbf{y}})'. \tag{2.50}$$

This gives a consistent estimate of $\mathbf{S}$; however, it sometimes is not positive semidefinite.

Newey-West (1987) suggested putting in a weight:

$$\hat{\mathbf{S}} = \hat{\mathbf{\Gamma}}_{\mathbf{0}} + \sum_{v=1}^{q} (1 - \frac{v}{q+1})(\hat{\mathbf{\Gamma}}_{\mathbf{v}} + \hat{\mathbf{\Gamma}}'_{\mathbf{v}}), \tag{2.51}$$

Here q is a truncation bandwidth, not a true MA order. For an MA(q) process the two coincide, but the Newey-West estimator is used precisely when the dependence has no known finite order: the Bartlett weights $1 - v/(q+1)$ downweight higher lags and keep $\widehat{S}$ positive semidefinite, and consistency comes from letting q grow with T, slowly enough that the poorly estimated high-order autocovariances do not dominate.

Consider a linear regression model:

$$y_t = \mathbf{x_t'}\boldsymbol{\beta} + u_t \tag{2.52}$$

Suppose we have the OLS estimator $\mathbf{b_T}$, then

$$\sqrt{T}(\mathbf{b_T} - \boldsymbol{\beta}) = [(1/T)\sum_{t=1}^{T} \mathbf{x_t x_t'}]^{-1}[(1/\sqrt{T})\sum_{t=1}^{T} \mathbf{x_t u_t}] \tag{2.53}$$

The first term converges in probability to some constant. The second term is $\sqrt{T}$ times the sample mean of the vector $\mathbf{x_t u_t}$, which is exactly the form to which a central limit theorem applies.

Under general conditions,

$$\sqrt{T}(\mathbf{b_T} - \boldsymbol{\beta}) \to^{L} N(0, Q^{-1}SQ^{-1}) \tag{2.54}$$

where $Q = \operatorname{plim}(1/T)\sum_{t=1}^{T} \mathbf{x_t x_t'} = \mathrm{E}[\mathbf{x_t x_t'}]$, and S can be estimated by

$$\hat{\mathbf{S}}_{\mathbf{T}} = \hat{\mathbf{\Gamma}}_{\mathbf{0T}} + \sum_{v=1}^{q} (1 - \frac{v}{q+1})(\hat{\mathbf{\Gamma}}_{\mathbf{v,T}} + \hat{\mathbf{\Gamma}}'_{\mathbf{v,T}}), \tag{2.55}$$

where

$$\hat{\Gamma}_{v,T} = (1/T) \sum_{t=v+1}^{T} (x_t \hat{u}_{t,T} \hat{u}_{t-v,T} x'_{t-v}), \tag{2.56}$$

where $\hat{u}_{t,T}$ is the OLS residual for data t in a sample of size T.

Overall, the variance of $\mathbf{b_T}$ is approximated by

$$\hat{\Sigma}_{NW} = [\sum_{t=1}^{T} x_t x_t']^{-1} [\sum_{t=1}^{T} \hat{u}_t^2 x_t x_t' + \sum_{v=1}^{q} (1 - \frac{v}{q+1}) \sum_{t=v+1}^{T} (x_t \hat{u}_{t,T} \hat{u}_{t-v,T} x_{t-v}' + x_{t-v} \hat{u}_{t-v,T} \hat{u}_{t,T} x_t')] [\sum_{t=1}^{T} x_t x_t']^{-1} \tag{2.57}$$

This estimation obviously depends on the selection of q, the lag length beyond which the autocorrelation of $x_t u_t$ and $x_{t-v} u_{t-v}$ is treated as negligible. One common rule of thumb sets $q = 0.75 \cdot T^{\frac{1}{3}}$; it is a convention rather than a theorem, and other growth rates are in use. Newey-West (1994) has suggested a way to automatically select the bandwidth q. Here we omitted the discussion. Both Stata and R now also implement Newey-West (1994) estimator, with no need to specify q.

3 GLS

3.1 Introduction

Since heteroscedasticity and serial correlation affect both linear and nonlinear regression models in the same way, there is no harm in limiting our attention to the simpler linear case.

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{u}, \quad \mathbf{E}(\mathbf{u}) = \mathbf{0}, \quad \mathbf{E}(\mathbf{u}\mathbf{u}') = \boldsymbol{\Omega}, \tag{3.1}$$

where $\boldsymbol{\Omega}$, the variance-covariance matrix of the error term, is a positive definite $n \times n$ matrix. If $\boldsymbol{\Omega}$ is equal to $\sigma^2\mathbf{I}$, then it is just the linear regression model without heteroscedasticity and serial correlation. If $\boldsymbol{\Omega}$ is diagonal with nonconstant diagonal elements, then the error terms are still uncorrelated, but there is heteroscedasticity. If $\boldsymbol{\Omega}$ is not diagonal, then u_i and u_j are correlated. In next section, we obtain an efficient estimator for the vector β by transforming the regression so that it satisfies the conditions of Gauss-Markov Theorem. This efficient estimator is called the Generalized Least Squares, or GLS, estimator.

3.2 The GLS Estimator

Since $\boldsymbol{\Omega}$, the variance-covariance matrix of the error term, is a positive definite $n \times n$ matrix, there always exist full-rank $n \times n$ matrices (usually triangular) $\boldsymbol{\Psi}$ such that

$$\boldsymbol{\Omega}^{-1} = \boldsymbol{\Psi}\boldsymbol{\Psi}'. \tag{3.2}$$

Premultiplying (the equation above) by $\boldsymbol{\Psi}'$ gives

$$\boldsymbol{\Psi}'\mathbf{y} = \boldsymbol{\Psi}'\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\Psi}'\mathbf{u}. \tag{3.3}$$

The OLS estimator from regression (the equation above) is

$$\hat{\beta}_{\mathbf{GLS}} = (\mathbf{X}'\boldsymbol{\Psi}\boldsymbol{\Psi}'\mathbf{X})^{-1}\mathbf{X}'\boldsymbol{\Psi}\boldsymbol{\Psi}'\mathbf{y} = (\mathbf{X}'\boldsymbol{\Omega}^{-1}\mathbf{X})^{-1}\mathbf{X}'\boldsymbol{\Omega}^{-1}\mathbf{y} \tag{3.4}$$

The transformed error term has an identity variance-covariance matrix:

$$\mathbf{E}(\boldsymbol{\Psi}'\mathbf{u}\mathbf{u}'\boldsymbol{\Psi}) = \boldsymbol{\Psi}'\mathbf{E}(\mathbf{u}\mathbf{u}')\boldsymbol{\Psi} = \boldsymbol{\Psi}'\boldsymbol{\Omega}\boldsymbol{\Psi} = \mathbf{I} \tag{3.5}$$

Since the transformed model satisfies OLS assumptions, we have

$$\begin{aligned} \mathrm{Var}(\hat{\boldsymbol{\beta}}_{\mathbf{GLS}}) &= (\mathbf{X}'\boldsymbol{\Psi}\boldsymbol{\Psi}'\mathbf{X})^{-1} \\ &= (\mathbf{X}'\boldsymbol{\Omega}^{-1}\mathbf{X})^{-1} \end{aligned} \tag{3.6}$$

3.3 Weighted Least Squares

It is easy to obtain GLS estimates when the error terms are heteroscedastic but uncorrelated, which means $\mathbf{\Omega}$ is diagonal. Let ω_t^2 denote the tth diagonal element of $\mathbf{\Omega}$. Then $\mathbf{\Psi}$ can be chosen as the diagonal matrix with tth diagonal element ω_t^{-1}. For a typical observation, regression can be written as

$$\omega_t^{-1} y_t = \omega_t^{-1} \mathbf{x}_t' \beta + \omega_t^{-1} u_t. \tag{3.7}$$

This is called weighted least squares, or WLS. It is worth keeping two notions of "weight" distinct. The transformed variables y_t and $\mathbf{x_t'}$ are *scaled* by ω_t^{-1}; equivalently, the WLS objective weights each squared residual by the inverse variance ω_t^{-2} (the square of the scaling factor). Either way, observations whose error variance is large are given low weight, and observations for which it is small are given high weight.

3.3.1 Feasible Generalized Least Squares

In many cases it is reasonable to suppose that $\mathbf{\Omega}$ depends in a known way on a vector of unknown parameters $\boldsymbol{\gamma}$. If so, it may be possible to estimate consistently, so as to obtain $\mathbf{\Omega}(\hat{\boldsymbol{\gamma}})$. This type of procedure is called feasible generalized least squares, or feasible GLS. But we'll have to specify the error term as some function of some known variables.

A typical FGLS procedure has three steps. First, run OLS and obtain the residuals $\hat{u}_t$. Second, use those residuals to estimate the parameters γ of the assumed error structure – for example:

- **Heteroscedasticity of known form**, e.g. $\omega_t^2 = \exp(\mathbf{z}_t'\gamma)$: regress $\log(\hat{u}_t^2)$ on $\mathbf{z_t}$ to estimate γ, then set $\hat{\omega}_t^2 = \exp(\mathbf{z}_t'\hat{\gamma})$ and apply WLS as above. (The intercept of that auxiliary regression is inconsistent, because $E[\log(u_t^2/\omega_t^2)] \neq 0$ – it is about -1.27 for normal errors. This does not matter for WLS: a constant added to every $\log \hat{\omega}_t^2$ scales all weights by a common factor, which leaves $\hat{\beta}$ unchanged. It does matter if the fitted $\hat{\omega}_t^2$ are wanted as variance estimates in their own right.)

- **AR(1) serial correlation**, $u_t = \rho u_{t-1} + \epsilon_t$: regress $\hat{u}_t$ on $\hat{u}_{t-1}$ to get $\hat{\rho}$, then quasi-difference each variable as $\tilde{y}_t = y_t - \hat{\rho} y_{t-1}$ for $t = 2, \dots, n$ (and similarly for X), so that the transformed error is approximately i.i.d.

  The first observation is where **Cochrane-Orcutt** and **Prais-Winsten** part company, and the $\mathbf{\Psi}$ framework above says which is right. For AR(1) errors $\boldsymbol{\Omega} = \frac{\boldsymbol{\sigma_\epsilon^2}}{\mathbf{1-\boldsymbol{\rho}^2}} \left[\boldsymbol{\rho}^{|\mathbf{i-j}|}\right]$, and its Cholesky factor makes the first row of $\mathbf{\Psi}'$ equal to $\sqrt{1-\rho^2}$ times the first unit vector. Full GLS therefore keeps observation 1, rescaled as $\tilde{y}_1 = \sqrt{1-\hat{\rho}^2}\, y_1$; that is Prais-Winsten. Cochrane-Orcutt simply drops observation 1. Both are consistent and asymptotically equivalent, but Cochrane-Orcutt throws away a data point, and how much that costs takes some care. The direct weight on observation 1 is $\sqrt{1-\rho^2}$, which *shrinks* as $|\rho| \to 1$, so that row's mechanical contribution falls rather than rises. What grows is its relative importance: quasi-differencing strips variation out of the remaining rows as well, and with trending or highly persistent regressors the single untransformed level observation ends up carrying a disproportionate share of the information about the level. So the loss is worth caring about in short samples, and with trending regressors when ρ is near 1 – not because that row is weighted heavily, but because the others are weighted lightly. Prefer Prais-Winsten.

Third, run OLS on the transformed (weighted or quasi-differenced) data to get the FGLS estimator $\hat{\beta}_{FGLS}$. FGLS is consistent and asymptotically efficient under two requirements, not one: the assumed *form* of $\mathbf{\Omega}$ must be correct, and $\hat{\gamma}$ must be consistent for γ within that form – alongside the usual exogeneity, rank and moment conditions. A consistent $\hat{\gamma}$ inside a misspecified covariance model buys nothing. Either way FGLS is generally biased in finite samples, since $\hat{\Omega}$ depends on the first-stage OLS residuals.

3.4 Efficiency of GLS

3.4.1 The Gauss-Markov Theorem for GLS

The GLS estimator is the best linear unbiased estimator (BLUE) when $\mathbf{\Omega}$ is known. To see this, consider any other linear unbiased estimator $\tilde{\boldsymbol{\beta}} = \mathbf{C}\mathbf{y}$ for some matrix $\mathbf{C}$ satisfying $\mathbf{C}\mathbf{X} = \mathbf{I}$. Write $\mathbf{C} = (\mathbf{X}'\mathbf{\Omega}^{-1}\mathbf{X})^{-1}\mathbf{X}'\mathbf{\Omega}^{-1} + \mathbf{D}$, where $\mathbf{D}$ is the difference from the GLS weighting. Unbiasedness requires $\mathbf{C}\mathbf{X} = \mathbf{I}$, which means $\mathbf{D}\mathbf{X} = \mathbf{0}$. The variance of $\tilde{\boldsymbol{\beta}}$ is

$$
\begin{aligned}
\mathrm{Var}(\tilde{\boldsymbol{\beta}}) &= \mathbf{C}\,\mathbf{\Omega}\,\mathbf{C}' \\
&= (\mathbf{X}'\mathbf{\Omega}^{-1}\mathbf{X})^{-1} + \mathbf{D}\,\mathbf{\Omega}\,\mathbf{D}'.
\end{aligned} \tag{3.8}
$$

The cross terms vanish because $\mathbf{D}\mathbf{X} = \mathbf{0}$ implies $\mathbf{D}\,\mathbf{\Omega}\,\mathbf{\Omega}^{-1}\mathbf{X} = \mathbf{D}\mathbf{X} = \mathbf{0}$, so $\mathbf{D}\,\mathbf{\Omega}\,\mathbf{\Omega}^{-1}\mathbf{X}(\mathbf{X}'\mathbf{\Omega}^{-1}\mathbf{X})^{-1} = \mathbf{0}$. Since $\mathbf{D}\,\mathbf{\Omega}\,\mathbf{D}'$ is positive semidefinite,

$$
\mathrm{Var}(\tilde{\boldsymbol{\beta}}) - \mathrm{Var}(\hat{\boldsymbol{\beta}}_{\mathbf{GLS}}) = \mathbf{D}\,\mathbf{\Omega}\,\mathbf{D}' \geq \mathbf{0}. \tag{3.9}
$$

No linear unbiased estimator can have a smaller variance than GLS.

When $\mathbf{\Omega} = \boldsymbol{\sigma}^2\mathbf{I}$, GLS reduces to OLS, and this is the ordinary Gauss-Markov theorem: OLS is BLUE under homoscedasticity and no serial correlation.

3.4.2 GLS versus OLS with Robust Standard Errors

The Gauss-Markov result says GLS is more efficient than OLS when $\mathbf{\Omega}$ is known. In practice $\mathbf{\Omega}$ is never known, and we face a choice: estimate $\mathbf{\Omega}$ and use FGLS, or use OLS with a sandwich variance estimator (White or Newey-West, as in the MLE chapter) that does not require specifying the form of $\mathbf{\Omega}$.

Both approaches give consistent estimates of β, since OLS is consistent under the same conditions whether or not errors are heteroscedastic or serially correlated. The difference is efficiency. GLS exploits the structure of $\mathbf{\Omega}$ to weight observations optimally; OLS ignores that structure and compensates afterward with a robust variance estimate.

The tradeoff turns on how well $\mathbf{\Omega}$ can be modeled:

When the error structure is well understood — for example, heteroscedasticity that follows a known function of observables, or AR(1) serial correlation whose form is dictated by the data generating process — FGLS can deliver substantial efficiency gains. The efficiency improvement is largest when the heterogeneity in error variances is large, since GLS downweights noisy observations that OLS treats equally.

When the error structure is unknown or hard to model correctly, FGLS carries a specification risk. If $\hat{\boldsymbol{\Omega}}$ is badly estimated, the FGLS estimator can be less efficient than OLS, and the FGLS standard errors can be misleading because they assume the estimated error structure is correct. OLS with robust standard errors avoids this risk: the point estimates are the same regardless of $\boldsymbol{\Omega}$, and the sandwich variance is consistent under unknown heteroscedasticity, and under serial correlation weak enough to satisfy the HAC conditions – summable autocovariances and a bandwidth that grows slowly with the sample. "Robust" is not the same as robust to dependence of any form.

The modern default in applied work is to report OLS with robust standard errors, reserving GLS for settings where there is strong prior information about the error structure or where the efficiency gain is large enough to justify the specification risk. Panel data models are an important exception: the random effects estimator is a form of FGLS, and there the structure of $\boldsymbol{\Omega}$ (an equicorrelated block for each unit) is implied by the model rather than estimated freely.

4 Endogeneity

When we discuss the linear model we assume that the regressors are exogenous. That word covers conditions of different strength – full independence, mean independence, and mere zero correlation – which the OLS chapter sets out as a ladder. What instrumental variables work with is the weakest of them, the moment condition $E[Zu] = 0$. However, there could be reasons to believe that some regressors are correlated with the error term. In that case we call those regressors endogenous.

Under the classical assumptions OLS estimators are unbiased and consistent. One key assumption is that the regressors have to be uncorrelated with the error term. If this condition does not hold, OLS estimators are biased and inconsistent.

When one independent variable does not satisfy this condition, we say this variable is endogenous. The estimates of other coefficients are then usually biased as well, but not always: the contamination travels through the correlation between regressors, so a coefficient on a regressor orthogonal to the endogenous one is left alone. With an orthogonal design the damage stays in the endogenous coefficient.

The most popular cure for endogeneity is to use instrumental variables.

4.1 Instrumental Variables (IV)

The basic idea of IV is to use an exogenous variable (or exogenous variables) which is correlated with the endogenous independent variable as an "instrument" for the endogenous variable.

Suppose the linear regression model

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{u}, \quad \mathbf{E}(\mathbf{u}\mathbf{u}') = \boldsymbol{\sigma}^{\mathbf{2}}\mathbf{I}, \tag{4.1}$$

at least one of the explanatory variables in the $n \times k$ matrix $\mathbf{X}$ is assumed not to be predetermined with respect to the error terms, or say, endogenous.

Suppose we have a set of variables $\mathbf{Z}$, an $n \times l$ matrix of instruments , which satisfies the moment condition

$$\mathrm{E}[\mathbf{Z}'(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})] = \mathbf{0}. \tag{4.2}$$

That is, $\mathbf{Z}$ is uncorrelated with the error term.

Here is what we do for two-stage least squares (2sls):

Stage 1: Regress each of the variables in the $\mathbf{X}$ matrix on $\mathbf{Z}$ to obtain a matrix of fitted values $\hat{\mathbf{X}}$,

$$\hat{\mathbf{X}} = \mathbf{Z}(\mathbf{Z}'\mathbf{Z})^{-\mathbf{1}}\mathbf{Z}'\mathbf{X} = \mathbf{P_Z}\mathbf{X} \tag{4.3}$$

This is essentially to get the part of $\mathbf{X}$ that is correlated with $\mathbf{Z}$. Or to say, to project $\mathbf{X}$ on to $\mathbf{Z}$.

Stage 2: Regress $\mathbf{y}$ on $\hat{\mathbf{X}}$ to obtain the estimated $\boldsymbol{\beta}$

$$\hat{\boldsymbol{\beta}}_{\mathbf{2sls}} = (\hat{\mathbf{X}}'\hat{\mathbf{X}})^{-\mathbf{1}}(\hat{\mathbf{X}}'\mathbf{y}) = (\mathbf{X}'\mathbf{P_Z}\mathbf{X})^{-\mathbf{1}}(\mathbf{X}'\mathbf{P_Z}\mathbf{y}) = \hat{\boldsymbol{\beta}}_{\mathbf{IV}} \tag{4.4}$$

Standard errors:

$$Var[\hat{\boldsymbol{\beta}}_{\mathbf{2sls}}] = \hat{\boldsymbol{\sigma}}^{\mathbf{2}}(\hat{\mathbf{X}}'\hat{\mathbf{X}})^{-\mathbf{1}} = \hat{\boldsymbol{\sigma}}^{\mathbf{2}}(\mathbf{X}'\mathbf{P_Z}\mathbf{X})^{-\mathbf{1}} \tag{4.5}$$

where $\hat{\sigma}^2 = \hat{\mathbf{u}}'\hat{\mathbf{u}}/N$. This is the homoskedastic large-sample variance; finite-sample formulas typically apply a degrees-of-freedom correction ($N-k$ in place of N), and when errors are heteroskedastic or clustered the robust/clustered IV covariance matrix replaces $\hat{\sigma}^2(X'P_Z X)^{-1}$ with a sandwich form.

Note that $\hat{\mathbf{u}} = \mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}_{\mathbf{2sls}}$.

If we do it manually by two steps, the second step will report a wrong standard error. That is because the second stage regression will report standard errors based on $\hat{\mathbf{u}} = \mathbf{y} - \hat{\mathbf{X}}\hat{\boldsymbol{\beta}}_{\mathbf{2sls}}$. Therefore, it's always recommended to ask the statistical program to do a 2sls for you, since presumably that will give you correct standard errors.

Geometry Illustration:

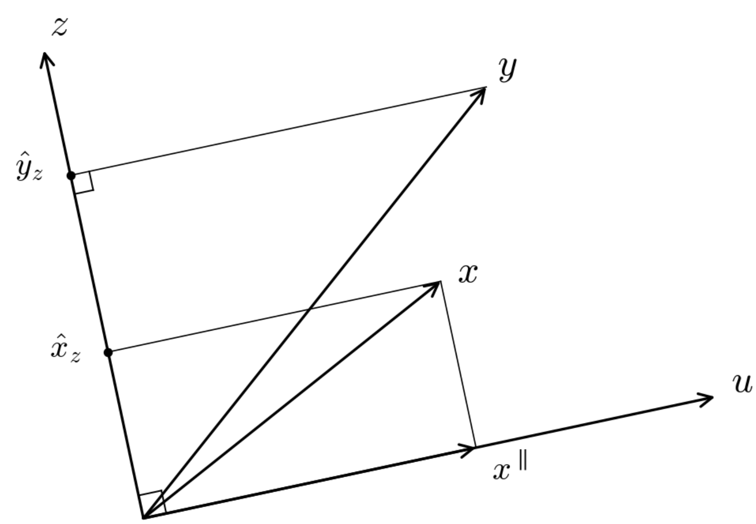


Figure 4.1: IV geometry: the instrument z is perpendicular to the error u, so projecting x and y onto z isolates the exogenous variation. The IV estimator $\hat{\beta}_{IV} = \hat{y}_z/\hat{x}_z$ recovers β without using the endogenous component $x^{\parallel}$.

Suppose the simplest case: $y = \beta_0 + \beta_1 x + u$. Take all variables in deviations from their means, so the intercept drops out and $z'\iota = 0$; this keeps the algebra below about the slope alone. Split x into the part that lies along the error and the part that does not,

$$x = x^{\perp} + x^{\parallel}, \tag{4.6}$$

where $x^{\parallel}$ is the orthogonal projection of x onto u and $x^{\perp}$ is what is left over. Two facts hold *by construction* and are what make the picture work: $x^{\perp}$ is perpendicular to u, and $x^{\perp}$ is perpendicular to $x^{\parallel}$. Endogeneity is simply the statement that $x^{\parallel} \neq 0$. (This is a decomposition of x, not an extra assumption about it – every x admits it.)

Now let z be a vector perpendicular to u with a non-zero projection onto x (relevance: z actually predicts the regressor). Project both y and x onto z, and write $\hat{y}_z$ and $\hat{x}_z$ for those two projections. The instrumental variables estimator is the ratio of their signed projection coefficients – signed, because lengths are never negative and β_1 may be,

$$\hat{\beta}_{IV} = \frac{\hat{y}_z}{\hat{x}_z} = \frac{z'y}{z'x}, \tag{4.7}$$

since $z'y = \beta_1 z'x + z'u$. The figure draws z exactly perpendicular to u, which sets $z'u = 0$ and removes the second term. Read that as the idealisation it is. Exogeneity is a statement about the population, $E[z_i u_i] = 0$, and it does not make $z'u$ vanish in a realised sample. What we have in a sample of size n is

$$\hat{\beta}_{IV} = \beta_1 + \frac{z'u}{z'x}, \tag{4.8}$$

so $\hat{\beta}_{IV}$ is consistent because $n^{-1}z'u \to 0$ while $n^{-1}z'x$ converges to something non-zero – not because the numerator is identically zero. The geometry shows why IV works; it does not make IV exact.

Staying inside the idealised picture for a moment: because the two right triangles involved are similar, $\hat{\beta}_{IV}$ is the same number you would get from the *infeasible* regression of y on the exogenous part alone: that regression returns

$$\frac{x^{\perp\prime}y}{x^{\perp\prime}x^{\perp}} = \beta_1 \frac{x^{\perp\prime}x}{x^{\perp\prime}x^{\perp}} + \frac{x^{\perp\prime}u}{x^{\perp\prime}x^{\perp}} = \beta_1, \tag{4.9}$$

where the second term vanishes because $x^{\perp} \perp u$ and the first collapses to β_1 because $x^{\perp} \perp x^{\parallel}$, so $x^{\perp\prime}x = x^{\perp\prime}x^{\perp}$. That infeasible regression is the one we would run if we could observe the decomposition. We cannot – which is exactly why we need the instrument.

```r
## DGP: data$y <- data$x + data$z + data$u
library(MASS)
set.seed(66)
nobs=10000
nDim = 3
sdxx = 1
sdww=1
sdzz=1

## here we have three variables x,z,w.
## z is the omitted variable,x and z are correlated,  w is the instrument, which is correlated
```

```r
crxz=.6
crzw=0
crxw=.5

covarMat = matrix( c(sdxx^2, crxz, crxw, crxz, sdzz^2, crzw,  crxw, crzw, sdww^2 ) , nrow=nDim
covarMat
```

```
     [,1] [,2] [,3]
[1,]  1.0  0.6  0.5
[2,]  0.6  1.0  0.0
[3,]  0.5  0.0  1.0
```

```r
data  = data.frame(mvrnorm(n=nobs, mu=rep(0,nDim), Sigma=covarMat ))
names(data) <- c('x','z','w')
data$u <- rnorm(nobs,0,1)
# dgp
data$y <- data$x +  data$z + data$u

lm <- lm(y~x, data=data)
lm.full <- lm(y~ x + z, data=data)
tsls.model <- ivreg(y ~ x | w, data=data)
# lm is biased
summary(lm)
```

```
Call:
lm(formula = y ~ x, data = data)

Residuals:
    Min      1Q  Median      3Q     Max
-5.3506 -0.8612  0.0123  0.8504  4.9096

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.004284   0.012843  -0.334    0.739
x            1.596114   0.013079 122.035   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.284 on 9998 degrees of freedom
Multiple R-squared:  0.5983,    Adjusted R-squared:  0.5983
F-statistic: 1.489e+04 on 1 and 9998 DF,  p-value: < 2.2e-16
```

```r
# lm.full is good
summary(lm.full)
```

```
Call:
lm(formula = y ~ x + z, data = data)

Residuals:
    Min      1Q  Median      3Q     Max
-3.7267 -0.6633 -0.0179  0.6742  3.6018

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 0.001857   0.009973   0.186    0.852
x           0.987181   0.012629  78.166   <2e-16 ***
z           1.017559   0.012542  81.129   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.997 on 9997 degrees of freedom
Multiple R-squared:  0.7578,    Adjusted R-squared:  0.7577
F-statistic: 1.564e+04 on 2 and 9997 DF,  p-value: < 2.2e-16
```

```r
# tsls is good.
summary(tsls.model)
```

```
Call:
ivreg(formula = y ~ x | w, data = data)

Residuals:
      Min        1Q    Median        3Q       Max
-6.144530 -0.962464  0.003414  0.944728  5.340560

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.01153    0.01433   0.804    0.421
x            0.95129    0.03007  31.637   <2e-16 ***

Diagnostic tests:
                  df1  df2 statistic p-value
Weak instruments    1 9998    3074.7  <2e-16 ***
Wu-Hausman          1 9997     807.8  <2e-16 ***
Sargan              0   NA        NA      NA
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.431 on 9998 degrees of freedom
Multiple R-Squared: 0.5007, Adjusted R-squared: 0.5006
Wald test:  1001 on 1 and 9998 DF,  p-value: < 2.2e-16
```

4.2 Control Function Approach

A second way to do an IV regression is also two-step approach: Regress $\mathbf{X}$ (endogenous) on $\mathbf{Z}$, get the residual: $\hat{\mathbf{v}} = \mathbf{X} - \mathbf{Z}(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'\mathbf{X}$, then regress y on $\mathbf{X}$ and $\hat{v}$ to get $\hat{\beta}_{IV}$.

The difference between this approach and the 2sls approach is that in 2sls we regress y on $\hat{\mathbf{X}}$; in the control function approach, we regress y on $\mathbf{X}$ and $\hat{v}$. They should both give you the same coefficient estimates. There are advantages of using the control function approach.

An example to show how to do 2sls manually, or using control function approach.

```
## DGP: data$y <- data$x + data$z + data$u

lm1 <- lm(y~x, data=data)
lm2 <- lm(x~w, data=data)
tsls.manual <- lm(data$y ~ lm2$fitted.values)
summary(tsls.manual)
```

```
Call:
lm(formula = data$y ~ lm2$fitted.values)

Residuals:
    Min      1Q  Median      3Q     Max 
-8.6485 -1.3401 -0.0203  1.3543  7.6974 

Coefficients:
                  Estimate Std. Error t value Pr(>|t|)    
(Intercept)        0.01153    0.01977   0.583     0.56    
lm2$fitted.values  0.95129    0.04147  22.937   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.974 on 9998 degrees of freedom
Multiple R-squared:  0.04999,	Adjusted R-squared:  0.04989 
F-statistic: 526.1 on 1 and 9998 DF,  p-value: < 2.2e-16
```

```
# control function approach
tsls.control <- lm(data$y ~ data$x + lm2$residuals)
summary(tsls.control)
```

```
Call:
lm(formula = data$y ~ data$x + lm2$residuals)

Residuals:
    Min      1Q  Median      3Q     Max 
```

```
-5.1997 -0.8329  0.0131  0.8370  4.7457

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    0.01153    0.01237   0.932    0.351
data$x         0.95129    0.02594  36.670   <2e-16 ***
lm2$residuals  0.84313    0.02966  28.423   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.235 on 9997 degrees of freedom
Multiple R-squared:  0.6284,    Adjusted R-squared:  0.6283
F-statistic:  8451 on 2 and 9997 DF,  p-value: < 2.2e-16
```

This approach can be used to do a simple endogeneity test. First, you can do a simple test of endogeneity of **X**. For example:

```
reg x_endog x* z*
predict v_x, resid
reg y x_endog x* v_x
test v_x
```

Obviously, this test is based on **Z** being exogenous.

Residual inclusion extends to *some* non-linear models – logit, probit, Poisson – under additional control-function assumptions, but it is not automatically valid for "any glm". The construction is model-specific: the appropriate first-stage residual (a plain linear residual or a generalized residual), the identification conditions, and the standard errors all differ for continuous, binary, and nonlinear endogenous regressors, and inserting a linear first-stage residual into an arbitrary GLM is generally not correct. The generated-regressor step also inflates uncertainty, so standard errors need to account for the estimated first stage (e.g. by bootstrap). For example, the "eteffects" procedure in Stata (version 14) uses a control-function approach to get endogenous treatment effects for several outcome types.

4.3 Durbin-Wu-Hausman Test

4.3.1 Idea

In econometric modeling, there are often questions on endogeneity. Do we know how to test whether an independent variable is endogenous statistically? The answer is: sort of, but not really. We cannot do endogeneity test without a valid instrument. Therefore, we have to have strong argument for a valid instrument first before we can do endogeneity test.

With endogenous variables on the right-hand side of the equation, we need to use instrumental variable (IV) regression for consistent estimation. However, with IV regression, we lose efficiency: the asymptotic variance of the IV estimator is larger, and can be much larger than the OLS estimator. Therefore, we gain consistency, but lose efficiency, by using IV estimator when there is an endogeneity problem.

Now we have a familiar scenario (if you are familiar with Hausman test for fixed effect and random effect estimator for panel data): Suppose we have the null hypothesis as the regressor being exogenous. We have an efficient estimator under null hypothesis yet inconsistent under alternative hypothesis (OLS estimator). We also have a consistent estimator under both null and alternative (IV estimator).

Similar to panel data setting, we have the Hausman test statistic as:

$$H = (\hat{\beta}_c - \hat{\beta}_e)' D^{-} (\hat{\beta}_c - \hat{\beta}_e) \tag{4.10}$$

where $D = \mathrm{Var}[\hat{\beta}_c] - \mathrm{Var}[\hat{\beta}_e]$, $^{-}$ is the generalized inverse, $\hat{\beta}_c$ is the consistent estimator (in this case the IV estimator) and $\hat{\beta}_e$ is the efficient estimator (in this case OLS estimator).

H conforms to χ^2_k asymptotically, where k is the number of endogenous variables.

This test is to compare the IV estimator and the OLS estimator: if it's close, then OLS estimator is fine (fail to reject null that OLS is consistent, or say the variable is exogenous). If it's large, then IV estimator is needed, although we lose some efficiency. This test is based on the assumption that the instruments are exogenous. If that is in question, then it's pointless to do the test, since the IV estimator cannot guarantee consistency either.

4.3.2 Implementation in Stata

In Stata, there are different ways to do it:

Do a regular Hausman test:

```
ivreg y x1 (x2=x3 x4)
estimates store iv
reg y x1 x2
hausman iv ., constant sigmamore
```

Or, simply use "ivendog" in Stata.

4.4 Identification

Identification in a regression equation means that all parameters can be uniquely estimated. The relevant counting rule (the *order condition*) is most cleanly stated in terms of *excluded* instruments. Let k_e be the number of endogenous regressors and l_e the number of excluded instruments (instruments not already included as exogenous regressors). A necessary condition is $l_e \geq k_e$: there must be at least as many excluded instruments as endogenous regressors. (Equivalently, counting the included exogenous regressors as their own instruments, the total number of instruments must be at least the total number of regressors.) If $l_e = k_e$ we have exact-identification. If $l_e > k_e$, we have over-identification.

Over-identification generates more efficient estimates, given the assumption of instruments being exogenous. The other advantage of over-identification is that over-identification tests can be done to test the adequacy of instruments.

Under the null hypothesis that all the instruments are uncorrelated with the error term, an LM statistic $N \times R^2$ conforms to $\chi^2(r)$ distribution, $r = l_e - k_e$, the number of excess instruments, or say, the number of overidentifying restrictions. If we reject the null, then we should be concerned about the exogeneity of the whole set of the instruments. This test is called Sargan's test in IV context, and (Hansen's) J test in GMM context. The two names are not quite interchangeable: Sargan's $N \times R^2$ form assumes homoskedastic errors, while Hansen's J, built from the efficient GMM weighting, is the heteroskedasticity-robust counterpart and the one to use whenever the standard errors are robust.

It is worth being exact about what the test delivers, because it is easy to over-read. It is a joint test of the overidentifying restrictions. Failing to reject does not certify the instruments: the test has no power against violations that all of them share, because instruments wrong in the same way agree on the same wrong answer. Rejection, for its part, does not say *which* moment condition failed. A passing J is the absence of one particular kind of evidence against the instruments, not evidence for them.

What the J test or Sargan's test does is to test the whole set of instruments being exogenous or not. There is another test for testing exogeneity for a subset of instruments. It's call a C test or a difference-in-Sargan test. The idea is to calculate the difference between two Sargan's statistics (or Hansen's J in GMM setting); one is with the whole set of instruments, the other one without the suspected instruments. The null is that the suspect instruments are exogenous; or orthogonal to the error term. Obviously to conduct the C test, we'll have to have at least one extra instrument more than the number of endogenous variables.

To understand it better, we look at how to implement Sargan's test manually: For the 2SLS estimator, the test statistic is Sargan's statistic, typically calculated as $N \times R^2$ from a regression of the IV residuals on the full set of instruments.

```
. ivregress 2sls rent pcturban (hsngval = faminc i.region)
. estat overid

  Tests of overidentifying restrictions:

  Sargan (score) chi2(3) =  11.2877  (p = 0.0103)
  Basmann chi2(3)        =  12.8294  (p = 0.0050)

. predict res, residual

. reg res pcturban faminc i.region

. disp e(N)*e(r2)
11.287665
```

4.4.1 Implementation in Stata

In Stata, there are different ways to do over-identification test, ivreg2 reports a comprehensive set of tests; overid command does the over-identification test after the ivreg command.

ivreg2 with gmm option returns J test; it reports Sargan's test without this option.

ivreg2 also reports the C test statistic, with ortho(). If the C test rejects and the J test without the suspect instruments does not, the evidence points at the suspect instruments – conditional on the retained ones being valid. That condition does real work: a difference test cannot tell an invalid suspect from an invalid maintained instrument, so it locates the problem only when the maintained set is sound.

4.5 Weak Instruments

4.5.1 Problem with the cure

An instrument needs to satisfy two criteria: orthogonality and relevance. We need instruments to be orthogonal to the error term. We can verify the orthogonality condition by Sargan's test if there are extra instruments.

It turns out instrument relevance is important too: if instruments are weak, the usual large-sample approximations are poor. It helps to keep three cases apart. With fixed, nonzero first-stage relevance, IV remains *consistent* but can be badly *biased* in finite samples (and biased toward OLS), with the normal approximation working poorly. Under weak-instrument (local-to-zero) asymptotics, the limiting distribution is *nonstandard* and not centered in the usual way, so conventional standard errors and t-statistics mislead. Only in the limiting case of complete irrelevance ($\Pi = 0$) is the parameter unidentified.

To see the problem, suppose

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{u}, \quad \mathbf{E}(\mathbf{u}\mathbf{u}') = \boldsymbol{\sigma}_{\mathbf{u}}^{\mathbf{2}}\mathbf{I}, \tag{4.11}$$

$$\mathbf{X} = \mathbf{Z}\boldsymbol{\Pi} + \mathbf{v}, \quad \mathbf{E}(\mathbf{v}\mathbf{v}') = \boldsymbol{\sigma}_{\mathbf{v}}^{\mathbf{2}}\mathbf{I}, \tag{4.12}$$

and

$$\mathrm{E}(\mathbf{Z}\mathbf{u}) = \mathbf{0}. \tag{4.13}$$

We can see here $\mathbf{Z}$ is exogenous. However, the model does not say anything about relevance. To illustrate the problem caused by weak instrument, suppose we have only one endogenous variable and one instrument.

$$\hat{\beta}_{2sls} = \frac{\mathbf{Z}'\mathbf{y}}{\mathbf{Z}'\mathbf{X}} = \frac{\mathbf{Z}'(\mathbf{X}\boldsymbol{\beta} + \mathbf{u})}{\mathbf{Z}'\mathbf{X}} = \beta + \frac{\mathbf{Z}'\mathbf{u}}{\mathbf{Z}'\mathbf{X}}. \tag{4.14}$$

If $\mathbf{Z}$ is irrelevant, or, $\Pi = 0$, then

$$\hat{\beta}_{2sls} - \beta = \frac{\mathbf{Z'u}}{\mathbf{Z'v}} = \frac{\frac{1}{\sqrt{n}}\sum_{i=1}^{n} Z_i u_i}{\frac{1}{\sqrt{n}}\sum_{i=1}^{n} Z_i v_i} \xrightarrow{d} \frac{z_u}{z_v}, \tag{4.15}$$

where (normalizing $\mathrm{Var}(Z_i) = 1$, which is harmless since any common scale factor cancels in the ratio)

$$\begin{bmatrix} z_u \\ z_v \end{bmatrix} \sim N(0, \begin{bmatrix} \sigma_u^2 & \sigma_{uv} \\ \sigma_{uv} & \sigma_v^2 \end{bmatrix}). \tag{4.16}$$

Therefore, if $\mathbf{Z}$ is irrelevant, β_{2sls} is inconsistent. Also, the distribution of the bias is Cauchy-like (the ratio of correlated normals).

This is a case where the cure might be worse than the disease itself: the bias can be big comparing to the bias an OLS estimate suffers.

4.5.2 Weak Instrument Tests

There are a variety of weak-instruments tests proposed. Most of them are based on so-called weak-instruments asymptotics and a new parameter called "concentration parameter" $\mu^2 = \Pi' Z' Z \Pi / \sigma_v^2$. Sample size only enters the distribution through μ^2.

With weak-instruments asymptotics, IV estimators are no longer consistent, and they are not normal asymptotically. Most test statistics (J test, etc.) do not have normal or χ^2 distributions anymore.

Now I list the following tests in the order of recommended level by James Stock:

1. Moreira (2003) conditional likelihood ratio test (CLR).

Advantages of this test: a. Nearly optimal among invariant similar tests (Andrews, Moreira and Stock 2006). b. Implemented in Stata as condivreg.

Disadvantages:

a. Complicated. Only developed so far for one endogenous variable case.

2. Stock-Yogo bias method and size method.

Stock and Yogo (2005) provide critical values for both methods: one is to control the size of bias, the other one is to control the size of a Wald test of $\beta = \beta_0$. Bias method is more frequently used. In the case of multiple endogenous variables, the Cragg-Donald statistics is used to compare with the critical values. It is implemented in Stata as part of the *ivreg2* command, but it's only available for the situation there are at least two excluded variables (meaning the number of instruments minus the number of endogenous variables).

3. Anderson-Rubin confidence intervals.

In the model of

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{u}, \quad \mathbf{E}(\mathbf{u}\mathbf{u}') = \boldsymbol{\sigma}_{\mathbf{u}}^{\mathbf{2}}\mathbf{I}, \tag{4.17}$$

$$\mathbf{X} = \mathbf{Z}\boldsymbol{\Pi} + \mathbf{v}, \quad \mathbf{E}(\mathbf{v}\mathbf{v}') = \boldsymbol{\sigma}_{\mathbf{v}}^{\mathbf{2}}\mathbf{I}, \tag{4.18}$$

The null hypothesis $H_0 : \beta = \beta_0$. Anderson-Rubin statistic is the F statistic in the regression of $y - X\beta_0$ on Z, the F test that the coefficient vector on Z is zero (under the model that coefficient equals $\Pi(\beta - \beta_0)$, so it is zero exactly when $\beta = \beta_0$):

$$AR(\beta_0) = \frac{(y - X\beta_0)' P_Z (y - X\beta_0)/l}{(y - X\beta_0)' M_Z (y - X\beta_0)/(N - l)} \tag{4.19}$$

where l is the number of instruments. The idea of AR confidence interval is to collect all values of β_0 at which we fail to reject $H_0 : \beta = \beta_0$.

4. First-stage F test.

The familiar rule of thumb is $F > 10$, and its scope is worth knowing. It comes from the single-endogenous-regressor case with homoskedastic errors, where it corresponds roughly to a bound on the relative bias of 2SLS. The formal Stock-Yogo critical values depend on how many instruments there are and on which criterion – relative bias or size distortion – is being controlled, so one number does not carry over to every design. Under heteroskedasticity or clustering the relevant statistic is the effective first-stage F of Montiel Olea and Pflueger rather than the conventional one. And when identification is weak the dependable route is not a higher threshold but weak-instrument-robust inference – the Anderson-Rubin and CLR tests in this list, whose size does not depend on instrument strength.

5. Kleibergen's LM test.

This test is dominated by the CLR test, thus no longer the optimal test to use.

6. First-stage R^2, or partial R^2, etc., are not recommended.

4.6 When is endogeneity a problem and what can we say from an IV regression

Let's consider a model of housing price on no. of rooms and square footage.

$$P = \alpha R + \beta F + \epsilon \tag{4.20}$$

Should square footage really be there? We are not interested in, given square footage, how much money we need to pay for an additional room. We are generally interested in how much money we need to pay for an additional footage. However, if we do not include footage, then we "sort of" have endogeneity problem due to omitted variable.

Another example would be education's impact on wage. Say IQ is an omitted variable. Do we want to say something about education's impact on wage controlling on IQ, or not?

So it depends on what questions to ask. If I am a builder and ask what an extra bedroom on the already built house would bring to me, I need to control for footage. If I am generally asking what an extra bedroom would cost, then I am implying what that bedroom and associated footage would cost me. In that case, without controlling for footage is not a problem, since the question is: what an additional bedroom (AND the associated average square footage) would cost me.

In the wage and education example, if I am asking: what's the effect of education on wage? I implicitly ask what would be my pay raise if I go to college (given IQ, of course), for example. Then we are concerned that this model of

$$wage = \alpha edu + \epsilon \tag{4.21}$$

will generate biased estimate of α, because we are not controlling for IQ. But if I am an employer and interested in hiring a person, what would I have to pay extra to hire a college graduate vs. a high school graduate? Then I don't have to include that IQ variable, because implicitly I am asking: what extra I need to pay for that college graduate who comes with a higher IQ, and everything else that is associated with higher education.

In addition, what an IV regression really gives us is part of the causal inference of X (suspected to be endogenous) on y, namely the part induced by Z. For example, we are interested in the relationship between college education and wage. If we use distance to college as an instrument, then our inference is the effect of college education from those who decide to go because of proximity of the college.

This is the Local Average Treatment Effect (LATE) of Imbens and Angrist (1994): with a binary instrument and heterogeneous treatment effects, IV identifies the causal effect for *compliers* — those whose treatment status is shifted by the instrument — rather than for the full population. The formal derivation, including the monotonicity assumption and the complier/always-taker/never-taker decomposition, is in the Causal Econometrics Guide.

5 Generalized Method of Moments (GMM)

5.1 The Method of Moments (MOM)

A population moment γ can be defined as the expectation of some continuous function g of a random variable x:

$$\gamma = \mathrm{E}[g(x)] \tag{5.1}$$

On the other hand, a sample moment is the sample version of the population moment in a particular sample:

$$\hat{\gamma} = \frac{1}{n}\sum[g(x)] \tag{5.2}$$

5.2 OLS as a moment problem

Consider the simple linear regression

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{u}, \quad \mathbf{u} \sim \mathbf{IID}(\mathbf{0}, \boldsymbol{\sigma^2}). \tag{5.3}$$

If the model is correctly specified, then

$$\mathbf{E}(\mathbf{X}'\mathbf{u}) = \mathbf{0}. \tag{5.4}$$

The MOM principle suggests that we replace the left-hand side with its sample analog $\frac{1}{n}\mathbf{X}'(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})$.

Since we know that the true $\boldsymbol{\beta}$ sets the population moment equal to zero in expectation, it seems reasonable to assume that a good choice of $\hat{\boldsymbol{\beta}}$ would be one that sets the sample moment to zero. The MOM procedure suggests an estimate of $\boldsymbol{\beta}$ that solves

$$\frac{1}{n}\mathbf{X}'(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}) = \mathbf{0}. \tag{5.5}$$

The MOM estimator is

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{\mathbf{-1}}\mathbf{X}'\mathbf{y}, \tag{5.6}$$

which is the same as the OLS estimator.

5.3 IV as a moment problem

Consider the simple linear regression

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{u}, \quad \mathbf{u} \sim \mathbf{IID}(\mathbf{0}, \boldsymbol{\sigma}^{\mathbf{2}}). \tag{5.7}$$

If one or more of the regressors is endogenous, then

$$\mathbf{E}(\mathbf{X}'\mathbf{u}) \neq \mathbf{0}. \tag{5.8}$$

We have to find an instrumental variable $\mathbf{Z}$ which is

$$\mathbf{E}(\mathbf{Z}'\mathbf{u}) = \mathbf{0}. \tag{5.9}$$

Or,

$$\mathbf{E}(\mathbf{Z}'(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})) = \mathbf{0}. \tag{5.10}$$

The sample analogy of this is

$$\frac{1}{n}\mathbf{Z}'(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}) = \mathbf{0}. \tag{5.11}$$

When the model is *exactly identified* – the number of instruments equals the number of regressors, so $\mathbf{Z}'\mathbf{X}$ is square and nonsingular – the sample moment condition has a unique solution, the simple IV estimator

$$\hat{\boldsymbol{\beta}} = (\mathbf{Z}'\mathbf{X})^{-\mathbf{1}}\mathbf{Z}'\mathbf{y}. \tag{5.12}$$

When the model is *overidentified* – more instruments than regressors – $\mathbf{Z}'\mathbf{X}$ is no longer square and we cannot set all sample moments to zero simultaneously. We instead minimize a quadratic form in the moments, which gives the two-stage least squares (2SLS) estimator

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{P_Z}\mathbf{X})^{-\mathbf{1}}\mathbf{X}'\mathbf{P_Z}\mathbf{y}, \qquad \mathbf{P_Z} = \mathbf{Z}(\mathbf{Z}'\mathbf{Z})^{-\mathbf{1}}\mathbf{Z}', \tag{5.13}$$

the special case of the GMM estimator below with weighting matrix $(\mathbf{Z}'\mathbf{Z})^{-\mathbf{1}}$.

5.4 The Generalized Method of Moments

The expectation $\mathrm{E}(Y^r)$ for any $r = 1, 2, ...$ is called the r^{th} (raw) moment of Y. The expectation $\mathrm{E}[(Y - \mathrm{E}(Y))^r]$ is called the r^{th} centered moment of Y.

The mean is the first raw moment.

The variance is the second centered moment.

The third centered moment measures the skewness of the distribution.

The fourth centered moment measures the kurtosis of the distribution. Interpreted as a measure of "fatness of tails".

The standardized kurtosis is

$$k = \frac{E[(Y - E(Y))^4]}{E[(Y - E(Y))^2]^2}. \tag{5.14}$$

For a normal distribution, $k = 3$.

For a t distribution with $v > 4$ degrees of freedom, $k = 3 + 6/(v-4) > 3$, i.e., the t distribution has fatter tails than a normal distribution (the excess kurtosis $6/(v-4)$ shrinks toward 0 as v grows).

The distribution function of a random variable captures all information about the random variable. If the moment generating function exists in a neighborhood of zero, the full set of moments also determines the distribution (without such a condition this can fail: the lognormal distribution is not determined by its moments).

This matters directly for GMM: GMM only ever matches a finite set of moment conditions, and if the underlying distribution is not uniquely pinned down by its (even infinite) sequence of moments, then no amount of moment-matching – however many moment conditions we add – can recover the full data-generating distribution. This is precisely the sense in which GMM asks for less than MLE: MLE assumes we know the full distributional family and estimates its parameters (recovering the whole distribution when correctly specified), while GMM only requires correctly specified moment conditions and only ever identifies the parameters entering those moments, not the full distribution. When the distributional assumption behind MLE is correct, MLE is efficient and gives you more (the whole distribution); when it is not, GMM's weaker requirements make it more robust.

This distinction underlies the relative strengths and weaknesses of ML and GMM.

5.5 GMM

The statistical model takes the general form

$$E[m(Y_i;\theta_0)] = 0 \tag{5.15}$$

where - $Y_1, \cdots, Y_n$ are random variables from which the sample $y_1, \cdots, y_n$ is drawn, - $m(Y, \theta)$ is a function specifying the model, - θ_0 is the "true value" of the parameter.

$E[m(Y_i;\theta_0)] = 0$ are called the population moment conditions.

Two ideas behind GMM:

1. Replace the population mean $E[.]$ with the sample mean calculated from the observed sample $y_1, \cdots, y_n$.
2. Since $E[m(Y_i;\theta_0)] = 0$, choose $\hat{\theta}_{GMM}$ to make $\frac{1}{n}\sum_{i=1}^{n} m(y_i;\hat{\theta}_{GMM})$ as close to zero as possible.

Define the notation

$$\bar{m}(\theta) = \frac{1}{n}\sum_{i=1}^{n} m(y_i;\theta). \tag{5.16}$$

$\hat{\theta}_{GMM}$ is chosen to make $\bar{m}(\theta)'\bar{m}(\theta)$ as close to zero as possible.

More generally, $\hat{\theta}_{GMM}$ is chosen to minimize $\bar{m}(\theta)'W\bar{m}(\theta)$ for some weighting matrix W.

The choice of W matters only under overidentification. With as many moment conditions as parameters, all sample moments can be driven to zero exactly and W drops out – which is why the just-identified IV estimator above did not need one. When there are more moments than parameters, the efficient choice is $W = S^{-1}$, where S is the asymptotic variance of the sample moments, $S = \lim_{n\to\infty} \mathrm{Var}(\sqrt{n}\,\bar{m}(\theta_0))$. With that weighting,

$$\sqrt{n}(\hat{\theta}_{GMM} - \theta_0) \xrightarrow{d} N(0, (G'S^{-1}G)^{-1}), \qquad G = \mathrm{E}\left[\frac{\partial m(Y_i;\theta_0)}{\partial \theta'}\right], \tag{5.17}$$

which is the GMM counterpart of the information-matrix bound of the previous chapter. Since S depends on θ_0, in practice one starts from any consistent weighting (often the identity, or $(\mathbf{Z}'\mathbf{Z})^{-\mathbf{1}}$ – scalar multiples of $\mathbf{W}$ leave the minimizer unchanged, so with $\bar{m}$ defined as a sample average this can equally be written $[n^{-1}\mathbf{Z}'\mathbf{Z}]^{-1}$), estimates $\hat{S}$ from the resulting residuals, and re-minimizes: two-step GMM. Stopping at $W = (\mathbf{Z}'\mathbf{Z})^{-\mathbf{1}}$ is exactly 2SLS, which is therefore efficient GMM only under homoskedasticity.

5.5.1 An example

Let's see an example with GMM, using the same simulated data as before. We have the same situation as before, X is endogenous. We are doing GMM version of 2sls.

Here I use R's "gmm" library, which makes things easy. Its first two arguments are `g` and `x`: with the formula interface, `g` is the model formula whose residual forms the moment function, and `x` is a formula giving the instruments. The moment condition in this example is $E(\mathbf{z}'\mathbf{u}) = \mathbf{0}$, where $\mathbf{z}$ is the instrument – the variable `w` in the simulated data – and u is the residual from the structural equation. Note that $\mathbf{W}$ above denotes the *weighting matrix* in $\bar{m}'W\bar{m}$, which is a different object entirely, so the instrument is written $\mathbf{z}$ here.

```
## DGP: data$y <- data$x + data$z + data$u

set.seed(66)
nobs=10000
nDim = 3
sdxx = 1
sdww=1
sdzz=1

## here we have three variables x,z,w.
## z is the omitted variable,x and z are correlated,  w is the instrument, which is correlated
crxz=.6
crzw=0
crxw=.5

covarMat = matrix( c(sdxx^2, crxz, crxw, crxz, sdzz^2, crzw,  crxw, crzw, sdww^2 ) , nrow=nDim
covarMat
```

```
     [,1] [,2] [,3]
[1,]  1.0  0.6  0.5
[2,]  0.6  1.0  0.0
[3,]  0.5  0.0  1.0
```

```
data  = data.frame(mvrnorm(n=nobs, mu=rep(0,nDim), Sigma=covarMat ))
names(data) <- c('x','z','w')
data$u <- rnorm(nobs,0,1)
# dgp
data$y <- data$x +  data$z + data$u

gmm.fit = gmm(y ~ x, x = ~w, data = data)
summary(gmm.fit)
```

```
Call:
gmm(g = y ~ x, x = ~w, data = data)


Method:  twoStep

Kernel:  Quadratic Spectral

Coefficients:
             Estimate     Std. Error   t value      Pr(>|t|)
(Intercept)   1.1531e-02   1.4151e-02   8.1481e-01   4.1518e-01
x             9.5129e-01   2.9971e-02   3.1740e+01  4.3442e-221

J-Test: degrees of freedom is 0
                J-test               P-value
Test E(g)=0:    1.16401567357162e-26 *******
```

```
# It returns the same estimates as the 2sls results.
tsls.model <- ivreg(y ~ x | w, data=data)
summary(tsls.model)
```

```
Call:
ivreg(formula = y ~ x | w, data = data)

Residuals:
      Min        1Q    Median        3Q       Max
-6.144530 -0.962464  0.003414  0.944728  5.340560

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.01153    0.01433   0.804    0.421
x            0.95129    0.03007  31.637   <2e-16 ***

Diagnostic tests:
                  df1  df2 statistic p-value
```

```
Weak instruments    1 9998    3074.7  <2e-16 ***
Wu-Hausman          1 9997     807.8  <2e-16 ***
Sargan              0   NA        NA      NA
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.431 on 9998 degrees of freedom
Multiple R-Squared: 0.5007, Adjusted R-squared: 0.5006
Wald test:  1001 on 1 and 9998 DF,  p-value: < 2.2e-16
```

Two things about that agreement are worth being explicit about, because it is easy to over-generalise from it. First, GMM and 2SLS coincide *numerically* here because the model is just-identified: one instrument (`w`) for one endogenous regressor (`x`), so $\mathbf{Z}'\mathbf{X}$ is square and the sample moments can be set to zero exactly, whatever the weighting matrix. The output says so – the *J*-test reports "degrees of freedom is 0", and its value ($\sim 10^{-26}$) is numerical noise rather than evidence. Second, `gmm()` is using an optimal two-step weighting with a heteroskedasticity-and-autocorrelation-consistent kernel (note `Method: twoStep` and `Kernel: Quadratic Spectral` in the output). Under *over*-identification that choice would generally **not** reproduce 2SLS, for exactly the reason given above: $W = (\mathbf{Z}'\mathbf{Z})^{-1}$ is the efficient weighting only under homoskedasticity.

In OLS case, it would be $E(Xu) = 0$.

```r
gmm.ols = gmm(y ~ x, x = ~x, data = data)
summary(gmm.ols)
```

```
Call:
gmm(g = y ~ x, x = ~x, data = data)


Method:  twoStep

Kernel:  Quadratic Spectral

Coefficients:
             Estimate     Std. Error   t value      Pr(>|t|)
(Intercept)   -0.0042841    0.0126399   -0.3389314    0.7346614
x              1.5961141    0.0128364  124.3424084    0.0000000

J-Test: degrees of freedom is 0
                J-test               P-value
Test E(g)=0:    2.66671052451529e-28 *******
```

```r
# It returns the same estimates as the OLS results.
ols <- lm(y ~ x, data=data)
summary(ols)
```

```
Call:
lm(formula = y ~ x, data = data)

Residuals:
    Min      1Q  Median      3Q     Max 
-5.3506 -0.8612  0.0123  0.8504  4.9096 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept) -0.004284   0.012843  -0.334    0.739    
x            1.596114   0.013079 122.035   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.284 on 9998 degrees of freedom
Multiple R-squared:  0.5983,    Adjusted R-squared:  0.5983 
F-statistic: 1.489e+04 on 1 and 9998 DF,  p-value: < 2.2e-16
```

5.5.2 Overidentification, the weighting matrix, and the J test

The example above cannot show what GMM adds, because it is just-identified. One instrument for one endogenous regressor leaves nothing for the weighting matrix to do and nothing for a specification test to check. Two things have to change. We need more instruments than endogenous regressors, and we need errors that are not homoskedastic – under homoskedasticity $(\mathbf{Z}'\mathbf{Z})^{-1}$ already is the efficient weighting, so 2SLS and efficient GMM agree and the comparison is empty.

The design keeps $y = x + z + u$ with z omitted from the regression, so the composite error is $z + u$ and x is endogenous through its correlation with z. There are now two instruments. Both w_1 and w_2 correlate 0.5 with x and 0.2 with each other, and neither correlates with z, so both are valid. The error is drawn with standard deviation $\exp(0.8w_1)$, which leaves $E[u \mid w] = 0$ intact while making the variance depend on an instrument. That is exactly the case where the efficient weighting differs from $(\mathbf{Z}'\mathbf{Z})^{-1}$. The true coefficient on x is 1.

```r
set.seed(66)
nobs <- 10000

# cor(x,z)=.6; cor(x,w1)=cor(x,w2)=.5; cor(z,w1)=cor(z,w2)=0; cor(w1,w2)=.2
covarMat <- matrix(c(1,  .6, .5, .5,
                     .6, 1,  0,  0,
                     .5, 0,  1,  .2,
                     .5, 0,  .2, 1), nrow = 4)
d <- data.frame(mvrnorm(n = nobs, mu = rep(0, 4), Sigma = covarMat))
names(d) <- c("x", "z", "w1", "w2")

# the error variance depends on w1, so E[u|w] = 0 still holds but 2SLS is
# no longer the efficient GMM estimator
```

```r
d$u <- rnorm(nobs, 0, exp(0.8 * d$w1))
d$y <- d$x + d$z + d$u

fit_ident <- gmm(y ~ x, x = ~ w1 + w2, data = d, wmatrix = "ident")
fit_eff   <- gmm(y ~ x, x = ~ w1 + w2, data = d)
tsls      <- ivreg(y ~ x | w1 + w2, data = d)

data.frame(
  estimator = c("GMM, identity weighting", "GMM, efficient two-step",
                "2SLS, classical SE", "2SLS, robust SE"),
  b_x = round(c(coef(fit_ident)["x"], coef(fit_eff)["x"],
                coef(tsls)["x"], coef(tsls)["x"]), 4),
  se = round(c(sqrt(diag(vcov(fit_ident)))["x"], sqrt(diag(vcov(fit_eff)))["x"],
               sqrt(diag(vcov(tsls)))["x"],
               sqrt(diag(vcovHC(tsls, type = "HC0")))["x"]), 4),
  row.names = NULL)
```

```
                estimator    b_x     se
1 GMM, identity weighting 1.0152 0.0463
2 GMM, efficient two-step 0.9887 0.0416
3      2SLS, classical SE 1.0157 0.0328
4         2SLS, robust SE 1.0157 0.0459
```

```r
specTest(fit_eff)
```

```
 ##  J-Test: degrees of freedom is 1  ##

                J-test   P-value
Test E(g)=0:    1.64870  0.19914
```

Three readings. The weighting matrix now matters: identity weighting gives 1.015 and the efficient two-step estimator gives 0.989, where in the just-identified case every weighting gave the same number. The efficiency gain is real but modest, and it shows up only against the *robust* 2SLS standard error, 0.042 against 0.046. And the classical 2SLS standard error, 0.033, is far too small – the earlier remark about homoskedasticity, arriving as a number instead of a caveat.

The J statistic is now a test rather than a formality. It is a joint specification test of the moment conditions, not a certificate of instrument validity. It has one degree of freedom – three moment conditions, from the constant and the two instruments, less the two parameters – and its value of 1.65 with a p-value of 0.20 gives no reason to doubt the moment conditions.

To see it earn its keep, make w_2 invalid by correlating it with the omitted z. Nothing else changes.

```r
covarMat2 <- covarMat
covarMat2[2, 4] <- covarMat2[4, 2] <- 0.4   # w2 now correlates with omitted z

set.seed(66)
d2 <- data.frame(mvrnorm(n = nobs, mu = rep(0, 4), Sigma = covarMat2))
names(d2) <- c("x", "z", "w1", "w2")
d2$u <- rnorm(nobs, 0, exp(0.8 * d2$w1))
d2$y <- d2$x + d2$z + d2$u

fit_bad <- gmm(y ~ x, x = ~ w1 + w2, data = d2)
coef(fit_bad)["x"]
```

```
       x
1.595747
```

```r
specTest(fit_bad)
```

```

 ##  J-Test: degrees of freedom is 1  ##

                J-test      P-value
Test E(g)=0:    1.4345e+02  4.6853e-33
```

The coefficient moves to 1.60 and the J statistic goes to 143, with a p-value of 5×10^{-33}. This is what overidentification buys. With one instrument there is no way to notice that it is invalid, because a single moment condition can always be satisfied exactly. With two, the instruments have to agree about β, and when one of them is contaminated they do not. Two limits are worth keeping in view. The test does not say *which* instrument is at fault, and it has no power against violations the instruments share: if w_1 and w_2 were both correlated with z in the same way they would agree on the same wrong answer, and J would pass.

5.6 A few concepts of conditioning

5.6.1 Independence

If X and Y are independent then

$$f(x, y) = f(x)f(y) \tag{5.18}$$

and hence

$$f(y|x) = f(y). \tag{5.19}$$

If X and Y are independent then

$$E[g(X)h(Y)] = E[g(X)] \cdot E[h(Y)] \tag{5.20}$$

and hence

$$Cov[g(X), h(Y)] = 0. \tag{5.21}$$

i.e. all functions of X and Y are uncorrelated.

5.6.2 Law of Iterated Expectations

$$E[Y] = E[E(Y|X)]. \tag{5.22}$$

5.6.3 Dependence Concepts

X, Y independent:

$$Cov[g(X), h(Y)] = 0 \tag{5.23}$$

X, Y uncorrelated:

$$Cov[X, Y] = 0 \tag{5.24}$$

$E[Y|X] = 0$:

$$Cov[g(X), Y] = 0 \tag{5.25}$$

These are listed in decreasing strength, and the implications run one way only: independence $\Rightarrow$ mean independence $\Rightarrow$ uncorrelatedness. The middle condition is what a *model of the conditional mean* needs. Neither converse holds: X and Y can be uncorrelated with $E[Y \mid X] \neq 0$ (take $Y = X^2 - 1$ with $X \sim N(0,1)$, so $Cov[X,Y] = E[X^3] = 0$ while $E[Y \mid X] = X^2 - 1$), and mean independence permits dependence through higher moments (any pure heteroskedasticity, $Y = X\epsilon$ with ϵ independent of X and mean zero). Keeping the three apart matters because the moment conditions below need only the *weakest* of them. $E[u_i] = 0$ and $E[X_i u_i] = 0$ are statements about means and covariances, nothing more. Conditional mean zero is sufficient for them and is not necessary, which is exactly why GMM asks for less than a conditional-mean model does.

5.6.4 Regression

A regression model is a model of $E[Y_i|X_i]$. For example,

$$Y_i = \beta_0 + \beta_1 X_i + u_i \tag{5.26}$$

where $E[u_i|X_i] = 0$.

5.6.5 GMM regression

The regression model

$$Y_i = \beta_0 + \beta_1 X_i + u_i, \quad E[u_i|X_i] = 0 \tag{5.27}$$

implies the moment condition

$$E[u_i] = 0 \quad \text{and} \quad E[X_i u_i] = 0 \tag{5.28}$$

That is,

$$E[Y_i - \beta_0 - \beta_1 X_i] = 0 \tag{5.29}$$

$$E[X_i(Y_i - \beta_0 - \beta_1 X_i)] = 0 \tag{5.30}$$

The sample moment conditions are

$$\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0 \tag{5.31}$$

$$\frac{1}{n}\sum_{i=1}^{n}x_i(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0 \tag{5.32}$$

These are just normal equations for OLS.

A characteristic of GMM: the specification of the model generates the estimator. i.e. only $E[Y_i|X_i] = \beta_0 + \beta_1 X_i$ is assumed.

Note there are no assumptions that u_i is homoscedastic, not autocorrelated or normally distributed. These properties affect the statistical properties of the GMM estimator, not its definition.

6 Censored, Truncated or Selected Data

6.1 Compare three different cases

A sample is truncated if some observations are systematically excluded from the sample. For example, suppose we are interested in relationship between people's income(y) and education(x). If we have observations of both y and x only for people whose income is above $20,000 per year, then we have a truncated sample. A sample is censored if no observations have been systematically excluded but some of information contained in them has been suppressed. In other words, in a truncated sample, you don't have observations which have been truncated; while in a censored sample, you do have observations but you don't have full information. The idea of "censoring" is that some data above or below a threshold are mis-reported at the threshold. If we use the same example of people's income, if people with income lower than $20,000 per year do not appear in the sample at all—we have no observations of either y or x for them—then we have a truncated sample. If we have observations of x for people whose income above AND below that limit, but no information on exactly how much their income is for those whose income below the limit (only thing we know is that it's lower than $20,000 per year), then we have a censored sample.

The problem of sample selection arises when no observations have been systematically excluded but some information has been suppressed, just like censoring. However, in sample selection problems, we have information on how the "selection" happens. In censoring, usually a constant "threshold" is set. Observations on dependent variable for those above (or below) that criteria are missing. Using the income example again, the threshold is set to be $20,000, we have no information on exactly how much their income is for those whose income below the limit (only thing we know is that it's lower than $20,000 per year). When we have a third variable that determines the censoring situation, then we call it sample selection. For example, we have observed only union member's information on income, but we observe information on education for everybody, including union members or non-members. In this example, union membership is an additional variable that "selects" the sample which are involved in the estimation of effect of education on income.

6.2 An illustration for truncated or censored data

Consider the model

$$y_t^0 = \beta_1 + \beta_2 x_t + u_t,\ u_t \sim NID(0, \sigma^2), \tag{6.1}$$

where y_t^0 is a latent variable. We actually observe y_t, which differs from y_t^0 , because it's either truncated or censored.

Suppose that censorship or truncation occurs whenever y_t^0 is less than 0. Clearly, the larger the error term u_t, the larger is y_t^0 and thus the greater must be the probability that $y_t^0 \geq 0$. This probability also depends on x_t. So for the sample we observe, u_t does not have conditional mean 0 and is not uncorrelated with x_t. OLS using truncated or censored samples (OLS of y_t on x_t) yields biased and inconsistent estimators. Normally we would like to draw inference on the population which is represented by the full sample. Shown in the figure below, ideally we would have the OLS regression line (if we had the full sample), which is very close to the "true" regression line (which is the mechanism generates the data). We would have the "OLS censored sample" line if we run OLS on the censored sample; we would have "OLS truncated sample" if we run OLS on the truncated sample. It shows that both of them are severely biased from the "true" regression line.

Here is a graph to illustrate from Davidson and MacKinnon.

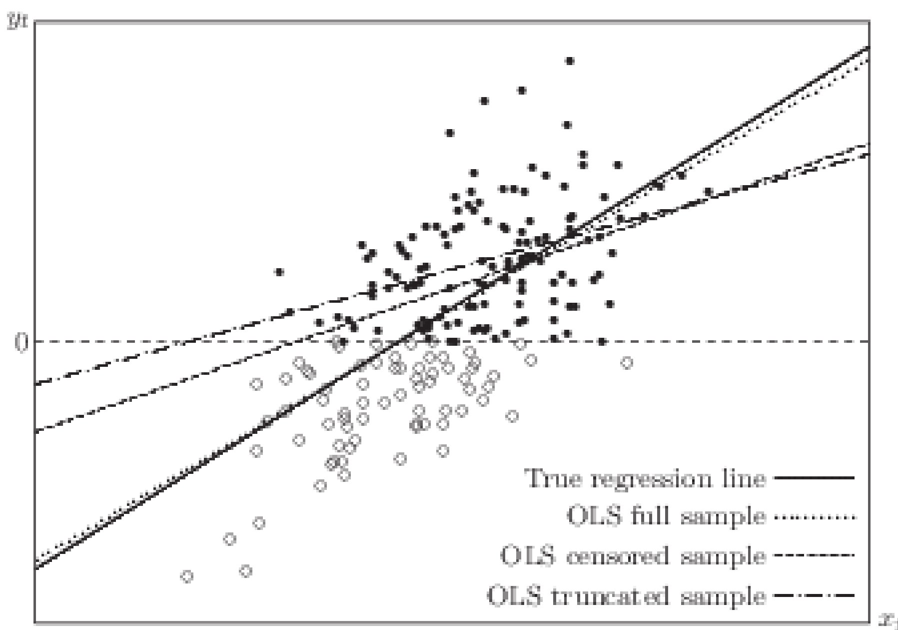


Figure 6.1: Effects of truncation and censoring from Davidson and MacKinnon

6.3 Truncated Models

For truncated data, a consistent estimator comes from maximum likelihood estimator (MLE). If we assume the error terms in the latent variable model has a known distribution, then MLE estimator can be applied. The most popular choice is Gaussian.

$$\begin{aligned} \Pr(y_t^0 \geq 0) &= \Pr(\mathbf{X}_t\beta + u_t \geq 0) \\ &= 1 - \Pr(u_t/\sigma < -\mathbf{X}_t\beta/\sigma) \\ &= 1 - \Phi(-\mathbf{X}_t\beta/\sigma) = \Phi(\mathbf{X}_t\beta/\sigma) \end{aligned} \tag{6.2}$$

The density of y_t is proportional to the density of y_t^0 when $y_t^0 \geq 0$ and y_t is observed. It is 0 elsewhere. The factor of proportionality, which is needed to ensure that the density integrates to unity, is the inverse of the probability that $y_t^0 \geq 0$. The density of y_t can be written as

$$\frac{\sigma^{-1}\phi((y_t - \mathbf{X}_t\beta)/\sigma)}{\Phi(\mathbf{X}_t\beta/\sigma)} \tag{6.3}$$

This implies the log-likelihood function,

$$\ell(\mathbf{y}, \boldsymbol{\beta}, \sigma) = -\frac{n}{2}\log(2\pi) - n\log(\sigma) - \frac{1}{2\sigma^2}\sum_{t=1}^{n}(y_t - \mathbf{X}_t\beta)^2 - \sum_{t=1}^{n}\log\Phi(\mathbf{X}_t\beta/\sigma). \tag{6.4}$$

This can be estimated by MLE. In Stata, truncreg is the command to do truncated regression model.

6.4 Censored Models

The most popular censored model is Tobit model.

$$\begin{aligned} y_t^0 &= \mathbf{X}_t\beta + u_t, \quad u_t \sim NID(0, \sigma^2) \\ y_t &= y_t^0 \text{ if } y_t^0 > 0;\ y_t = 0 \text{ otherwise.} \end{aligned} \tag{6.5}$$

We see that

$$\begin{aligned} \Pr(y_t = 0) &= \Pr(y_t^0 \leq 0) = \Pr(\mathbf{X}_t\beta + u_t \leq 0) \\ &= \Pr(u_t/\sigma < -\mathbf{X}_t\beta/\sigma) \\ &= \Phi(-\mathbf{X}_t\beta/\sigma) \end{aligned} \tag{6.6}$$

The contribution to the log-likelihood function made by observations with $y_t = 0$ is

$$\ell_t(y_t, \beta, \sigma) = \log\Phi(-\mathbf{X}_t\beta/\sigma). \tag{6.7}$$

If y_t is positive, the contribution to the log-likelihood is the logarithm of the density,

$$\log(\frac{1}{\sigma}\phi((y_t - \mathbf{X}_t\beta)/\sigma)). \tag{6.8}$$

The log-likelihood function of the tobit model is

$$\sum_{y_t=0}\log\Phi(-\mathbf{X}_t\beta/\sigma) + \sum_{y_t>0}\log(\frac{1}{\sigma}\phi((y_t - \mathbf{X}_t\beta)/\sigma)) \tag{6.9}$$

This can be estimated by MLE. In Stata, tobit or intreg can be used for censoring models.

6.5 Sample Selection

The sample selection models differ from censored model in that it involves a different variable (from y itself) to determine the censorship (selection).

Suppose that y_t^0 and z_t^0 are two latent variables, generated by the bivariate process

$$\begin{bmatrix} y_t^0 \\ z_t^0 \end{bmatrix} = \begin{bmatrix} \mathbf{X}_t\beta \\ \mathbf{W}_t\gamma \end{bmatrix} + \begin{bmatrix} u_t \\ v_t \end{bmatrix}, \quad \begin{bmatrix} u_t \\ v_t \end{bmatrix} \sim NID(\mathbf{0}, \begin{bmatrix} \sigma^2 & \rho\sigma \\ \rho\sigma & 1 \end{bmatrix}) \tag{6.10}$$

Here ρ is the correlation between u_t and v_t; since $\mathrm{Var}(u_t) = \sigma^2$ and $\mathrm{Var}(v_t) = 1$, their covariance is $\rho\sigma$.

We observe y_t and z_t:

$$y_t = y_t^0 \text{ if } z_t^0 > 0;\; y_t \text{ unobservable otherwise}; z_t = 1 \text{ if } z_t^0 > 0;\; z_t = 0 \text{ otherwise.} \tag{6.11}$$

There are two types of observations, ones we observe $y_t = y_t^0$ and $z_t = 1$, along with both $\mathbf{X_t}$ and $\mathbf{W_t}$, and ones we observe only $z_t = 0$ and $\mathbf{W_t}$.

Each observation contributes to the likelihood function by

$$I(z_t = 0)\mathrm{Pr}(z_t = 0) + I(z_t = 1)\mathrm{Pr}(z_t = 1)f(y_t^0|z_t = 1), \tag{6.12}$$

Under the normality assumption,

$$u_t = \rho\sigma v_t + e_t \tag{6.13}$$

where e_t is independent of $v_t \sim N(0,1)$, with $\mathrm{Var}(e_t) = \sigma^2(1-\rho^2)$. A useful fact about the standard normal distribution is that

$$\mathrm{E}(v_t|v_t > -x) = \lambda(x) = \frac{\phi(x)}{\Phi(x)} \tag{6.14}$$

and the function $\lambda(x)$ is called the inverse Mills ratio.

The log-likelihood function can be shown to be

$$\sum_{z_t=0} \log\Phi(-\mathbf{W}_t\gamma) + \sum_{z_t=1} \log(\frac{1}{\sigma}\phi((y_t - \mathbf{X}_t\beta)/\sigma)) + \sum_{z_t=1} \log\Phi(\frac{\mathbf{W}_t\gamma + \rho(y_t - \mathbf{X}_t\beta)/\sigma}{(1-\rho^2)^{1/2}}) \tag{6.15}$$

So this model can be estimated by MLE.

It is also popular to use Heckman's two-step method.

Heckman's method is based on the fact that the original latent model can be rewritten as

$$y_t = \mathbf{X}_t\beta + \rho\sigma v_t + e_t \tag{6.16}$$

Here the error term u_t is divided into two parts, one perfectly correlated with v_t, and one independent of v_t.

In the first step, an ordinary probit model is used to obtain consistent estimates $\hat{\gamma}$ of the parameters of the selection equation.

In the second step, the unobserved v_t is replaced by the selectivity regressor $\frac{\phi(\mathbf{W}_t\hat{\gamma})}{\Phi(\mathbf{W}_t\hat{\gamma})}$.

Therefore, the regression becomes

$$y_t = \mathbf{X}_t\beta + \rho\sigma\frac{\phi(\mathbf{W}_t\hat{\gamma})}{\Phi(\mathbf{W}_t\hat{\gamma})} + e_t \tag{6.17}$$

Or,

$$y_t = \mathbf{X}_t\beta + \rho\sigma\hat{\lambda}_t + e_t \tag{6.18}$$

(Strictly, conditional on selection the error also contains $\rho\sigma(v_t - \lambda_t)$, which is why the second-step OLS standard errors require correction; the coefficient $\rho\sigma$ on $\hat{\lambda}_t$ is unaffected.)

One thing to note in Heckman model is that in many situations, $\hat{\lambda}$ is believed to be highly collinear with $\mathbf{X_t}$, if $\mathbf{W_t}$ is the same as $\mathbf{X_t}$. However, the model can still be estimated due to the nonlinearity of the model. Nevertheless, it becomes a regular practice to require $\mathbf{W_t}$ contains at least one extra variable than $\mathbf{X_t}$, which is sometimes called exclusion restriction. In many situations, $\mathbf{W_t}$ contains all $\mathbf{X_t}$ variables, and at least one more variable. The reason to include all the $\mathbf{X_t}$ variables is more mundane than it is often made to sound: whatever plausibly affects selection belongs in the selection equation, and the determinants of the outcome usually do. What identification turns on is the other direction – at least one variable in $\mathbf{W_t}$ kept *out* of $\mathbf{X_t}$, affecting selection but not the outcome. Without such an exclusion, $\hat{\lambda}$ is a nonlinear function of the same $\mathbf{X_t}$'s and identification rests entirely on that nonlinearity.

In Stata, heckman is the command to do sample selection models. It has options to do a Heckman two-step estimation or MLE estimation. MLE is usually preferred, on the understanding that its efficiency advantage holds *if* the joint normality it assumes is right. The two-step estimator is less efficient but leans less heavily on the full likelihood, which makes it easier to diagnose when the specification is in doubt.

6.6 Switching Regression (Treatment-Effects Model)

There is another situation that is similar to sample selection model: we observe y for $z = 1$ and $z = 0$. In the example of effect of education on income, we observe both union member's income and non-member's income. In this case, we have switching regression model or treatment-effect model. The treatment effect model estimates the effect of an endogenous binary treatment z_t (treatment, program participation, etc.) on a continuous, fully-observed variable, y_t, conditional on the independent variables x_t and w_t.

Suppose z_t^0 is a latent variable.

$$\begin{bmatrix} y_t \\ z_t^0 \end{bmatrix} = \begin{bmatrix} \mathbf{X}_t\beta \\ \mathbf{W}_t\gamma \end{bmatrix} + \delta \begin{bmatrix} z_t \\ 0 \end{bmatrix} + \begin{bmatrix} u_t \\ v_t \end{bmatrix}, \quad \begin{bmatrix} u_t \\ v_t \end{bmatrix} \sim NID(\mathbf{0}, \begin{bmatrix} \sigma^2 & \rho\sigma \\ \rho\sigma & 1 \end{bmatrix}) \tag{6.19}$$

We observe y_t and z_t:

$$z_t = 1 \text{ if } z_t^0 > 0; \;\; z_t = 0 \text{ otherwise.} \tag{6.20}$$

Notice that the only difference between this model and the selection model is that we observe y_t when $z_t = 0$ and $z_t = 1$. It's not a selection, but a regime switching.

One caveat about the name. The specification above lets treatment shift only the *intercept*: there is a single β and a single δ, so the treatment effect is δ for everyone. That is the treatment-effects model. A **switching regression** in the fuller sense lets the whole coefficient vector differ by regime,

$$y_t = z_t(\mathbf{X}_t\beta_1 + u_{1t}) + (1 - z_t)(\mathbf{X}_t\beta_0 + u_{0t}), \qquad z_t^0 = \mathbf{W}_t\gamma + v_t, \tag{6.21}$$

with u_{1t} and u_{0t} each allowed to correlate with v_t (two selection parameters, ρ_1 and ρ_0, rather than one). This is the Roy model, and the treatment effect $\mathbf{X}_t(\beta_1 - \beta_0)$ then varies with $\mathbf{X_t}$ instead of being a constant. The distinction matters in practice: if the returns to education genuinely differ between union and non-union jobs, the common-slope model above will report a single δ that is some average of them and will not reveal the heterogeneity. Fitting it requires two outcome equations rather than one intercept shift.

A word on software, since the two models above are easy to conflate. Stata's etregress fits the *common-slope* specification – the endogenous treatment-effects model with a single δ – not the two-regime switching regression with separate β_1 and β_0. For the fuller Roy-type model, Stata users generally reach for the user-written movestay (endogenous switching regression), and SAS's qlim covers a range of limited-dependent and selection specifications. The default in each case is MLE.

7 Discrete and Limited Dependent Variables

7.1 Binary Response Models

7.1.1 Probit and Logit

Let P_t denote the probability that $y_t = 1$ conditional on the information set Ω, which consists of exogenous and predetermined variables. A binary response model serves to model this conditional expectation. Since the values are 0 or 1, it is clear that P_t is also the expectation of y_t conditional on Ω_t:

$$P_t \equiv \Pr(y_t = 1|\Omega_t) = \mathrm{E}(y_t|\Omega_t). \tag{7.1}$$

Here X_t is a set of regressors. A linear probability model would set $P_t = X_t\beta$ directly, which requires the fitted values to stay in $[0, 1]$ – an awkward constraint, since the linear index $X_t\beta$ is unbounded. Logit and probit avoid this problem by modeling $P_t = F(X_t\beta)$, mapping the unbounded index through a CDF-valued transformation into $[0, 1]$.

We ensure that $0 \leq P_t \leq 1$ by specifying that

$$P_t \equiv \Pr(y_t = 1|\Omega_t) = F(\mathbf{X_t}\boldsymbol{\beta}). \tag{7.2}$$

$F(x)$ is a transformation function, which has the same characteristics as the CDF of a probability distribution.

Two popular choices of $F(x)$ are Gaussian (probit) and Logistic (logit).

The less familiar logistic function is

$$\Lambda(x) = \frac{e^x}{1+e^x} \tag{7.3}$$

The logit model is most easily derived by assuming that

$$\log\left(\frac{P_t}{1-P_t}\right) = \mathbf{X_t}\boldsymbol{\beta} \tag{7.4}$$

which says the logarithm of the odds (the ratio of the two probabilities) is equal to $\mathbf{X_t}\boldsymbol{\beta}$. Therefore,

$$P_t = \frac{\exp(\mathbf{X}_t\beta)}{1+\exp(\mathbf{X}_t\beta)} = \Lambda(\mathbf{X}_t\beta) \tag{7.5}$$

7.1.2 MLE for binary data

The likelihood for an observation t is the probability that $y_t = 1$ if $y_t = 1$, or the probability that $y_t = 0$ if $y_t = 0$. The logarithm of the appropriate probability is then the contribution to the loglikelihood made by observation t. Therefore, if $\mathbf{y}$ is an n-vector with typical element y_t, the loglikelihood function for $\mathbf{y}$ can be written as

$$\ell(\mathbf{y}, \boldsymbol{\beta}) = \sum_{t=1}^{n} (y_t \log F(\mathbf{X_t}\boldsymbol{\beta}) + (1 - y_t) \log(1 - F(\mathbf{X_t}\boldsymbol{\beta}))) \tag{7.6}$$

For the logit and probit models, this function is globally concave with respect to β, so there are no local maxima to worry about. Concavity is not the whole story, though. For the MLE to exist and be unique and *finite* we also need $\mathbf{X}$ to have full column rank, and the data must not be separable. If some linear combination of the regressors predicts the outcome perfectly – complete or quasi-complete separation – the likelihood keeps rising along that direction and the maximum is attained at no finite β, concave or not. Software reports this as coefficients diverging or standard errors blowing up. With full rank and no separation, the first-order conditions, or likelihood equations, uniquely define the MLE estimator $\hat{\beta}$. These likelihood equations can be written as

$$\sum_{t=1}^{n} \frac{(y_t - F(\mathbf{X_t}\boldsymbol{\beta}))f(\mathbf{X_t}\boldsymbol{\beta})x_{ti}}{F(\mathbf{X_t}\boldsymbol{\beta})(1 - F(\mathbf{X_t}\boldsymbol{\beta}))} = 0, \; i = 1, \dots, k. \tag{7.7}$$

where $f = F'$ is the density (e.g. ϕ for probit). For logit, $f = F(1 - F)$, so the numerator cancels the denominator and the score reduces to $\sum_t (y_t - F)x_{ti} = 0$.

Newton's Method can be used to find $\hat{\beta}$.

7.2 Models for More than Two Discrete Responses

7.2.1 The Ordered Probit

Ordered Probit can be easily derived from a latent variable model.

$$y_t^0 = \mathbf{X}_t\beta + u_t, \; u_t \sim NID(0, 1) \tag{7.8}$$

Suppose we observe y_t with three values.

$$\begin{cases} y_t = 0 & \text{if } y_t^0 < \gamma_1; \\ y_t = 1 & \text{if } \gamma_1 \le y_t^0 < \gamma_2; \\ y_t = 2 & \text{if } y_t^0 \ge \gamma_2. \end{cases} \tag{7.9}$$

Therefore,

$$\begin{aligned} \Pr(y_t = 0) &= \Pr(y_t^0 < \gamma_1) = \Pr(\mathbf{X}_t\beta + u_t < \gamma_1) \\ &= \Pr(u_t < \gamma_1 - \mathbf{X}_t\beta) = \Phi(\gamma_1 - \mathbf{X}_t\beta) \end{aligned} \tag{7.10}$$

Similarly,

$$\Pr(y_t = 2) = 1 - \Phi(\gamma_2 - \mathbf{X}_t\beta) \tag{7.11}$$

$$\Pr(y_t = 1) = \Phi(\gamma_2 - \mathbf{X}_t\beta) - \Phi(\gamma_1 - \mathbf{X}_t\beta) \tag{7.12}$$

These probabilities depend solely on the value of the index function and on the two threshold parameters.

Two normalizations are needed before any of this can be estimated, and they are easy to pass over. First, the *location*: an intercept in $\mathbf{X}_t\beta$ and both thresholds cannot be identified together, since adding a constant to the index and the same constant to γ_1 and γ_2 leaves every probability unchanged. Software resolves this by dropping the intercept or by fixing $\gamma_1 = 0$. Second, the *scale*: that is what setting $\mathrm{Var}(u_t) = 1$ above does, since multiplying the index, the thresholds and u_t by any positive constant is likewise unobservable. The analogous restriction for the multinomial logit below is $\beta^0 = 0$.

The loglikelihood function is

$$\ell(\beta, \gamma_1, \gamma_2) = \sum_{y_t=0} \log(\Phi(\gamma_1 - \mathbf{X}_t\beta)) + \sum_{y_t=2} \log(\Phi(\mathbf{X}_t\beta - \gamma_2)) + \sum_{y_t=1} \log(\Phi(\gamma_2 - \mathbf{X}_t\beta) - \Phi(\gamma_1 - \mathbf{X}_t\beta)) \tag{7.13}$$

7.2.2 The Multinomial Logit

When responses are unordered, two popular choices are multinomial logit and conditional logit. They are easy to conflate because they share the same softmax form, but they differ in what varies across the $J + 1$ alternatives $l = 0, \dots, J$: in multinomial logit the *regressors* are individual-specific and only the *coefficients* vary by alternative; in conditional logit it's the reverse.

Multinomial logit. Here the regressor vector $\mathbf{X_t}$ (individual-specific characteristics, e.g. age, income) is the same for every alternative, and it is the coefficient vector that is alternative-specific:

$$\Pr(y_t = l) = \frac{\exp(\mathbf{X}_t\beta^l)}{\sum_{j=0}^{J} \exp(\mathbf{X}_t\beta^j)} \quad \text{for } l = 0, \dots, J, \tag{7.14}$$

Here $\mathbf{X_t}$ is a row vector of length k of individual-specific covariates, and β^l is a k-vector of alternative-specific parameters. For identification one category is taken as the base with $\beta^0 = 0$ (so its numerator is 1).

Conditional logit. Here it is the regressors that vary by alternative (e.g. the price or characteristics of choice j itself), while the coefficient vector is common across alternatives:

$$\Pr(y_t = l) = \frac{\exp(\mathbf{W}_{tl}\beta)}{\sum_{j=0}^{J} \exp(\mathbf{W}_{tj}\beta)} \quad \text{for } l = 0, \dots, J, \tag{7.15}$$

Here $\mathbf{W}_{tj}$ is a row vector of length k of alternative-j-specific variables, and β is a single k-vector of parameters shared across all alternatives $j = 0, \dots, J$.

For binary and multinomial models alike, the coefficients β do not directly measure the effect of a unit change in X on the probability. That effect depends on where in the distribution the change occurs. The average marginal effect (AME) — averaging the partial effect over the observed distribution of covariates — is the standard summary. The full treatment of AME vs. marginal effects at the mean, including fixed-effects models, is in the Blog Book.

Writing *both* choice-varying regressors $\mathbf{W}_{tl}$ *and* choice-specific coefficients β^l into the same formula, as some treatments do, conflates the two models. It is worth being precise about what that costs, because the usual complaint – that such a specification is not identified – is not true. Choice probabilities depend only on utility *differences*, so $\Pr(y_t = l)$ is governed by $\mathbf{W}_{tj}\beta^j - \mathbf{W}_{tl}\beta^l$; when the $\mathbf{W}_{tj}$ vary independently across alternatives, those differences pin down each β^j separately, with no normalization required. It is a perfectly estimable model – it is what Stata's `asclogit` fits when alternative-specific coefficients are requested.

What the ambiguous formula really costs is clarity about *which normalization is in force*. Individual-specific regressors require one alternative's coefficients to be fixed ($\beta^0 = 0$), since only differences across alternatives are identified; alternative-specific regressors entering with a common coefficient require no such restriction. One caveat on "no normalization required" above: it concerns the slope coefficients β^j. If the specification also carries alternative-specific intercepts, those are identified only up to a location shift and one of them still has to be fixed. A formula that does not say which of the two it means leaves the reader unable to tell what has been normalized, and therefore unable to interpret the coefficients.

7.2.3 Independence of Irrelevant Alternatives

Both logit forms carry a restriction that comes from the softmax itself. Take the ratio of two choice probabilities. With individual-specific regressors,

$$\frac{\Pr(y_t = j)}{\Pr(y_t = l)} = \exp(\mathbf{X}_t(\beta^j - \beta^l)), \tag{7.16}$$

and with alternative-specific regressors and a common coefficient the ratio is $\exp((\mathbf{W}_{tj} - \mathbf{W}_{tl})\beta)$. Either way everything to do with the other alternatives cancels. The relative odds of j against l do not depend on what else is on the menu, or on the attributes of any third alternative. That is **independence of irrelevant alternatives**, or IIA.

Sometimes it is harmless. Often it is not. The standard illustration is travel mode. Suppose commuters split evenly between a car and a red bus, so the odds are 1:1 and each share is 1/2. Now add a blue bus, identical to the red one in everything but colour. IIA holds the car-to-red-bus odds at 1:1, and the two buses are interchangeable, so the model predicts three shares of 1/3. What we would expect is that the blue bus draws almost entirely from the red bus, leaving the car near 1/2. The model has no way to know that the two buses are close substitutes, because the softmax gives it no way to express that.

There are two standard escapes. **Nested logit** groups alternatives into nests – the two buses in one, the car in another – and allows correlation within a nest, so IIA holds inside nests but not across them. **Mixed logit**, also called random-coefficients logit, lets β vary across individuals; shared taste parameters then induce correlation across alternatives and IIA goes away, at the cost of simulation-based estimation. Which one to reach for depends on whether the substitution pattern you are worried about can be described by a grouping.

8 Count Data Models

8.1 Poisson Model

If the interested variable is a count, it is natural to model its conditional distribution as Poisson. What the regression assumes is the *distribution*, not that the counts were generated by a continuous-time Poisson process – that process is one way to arrive at the distribution, not a requirement of the model. The number of occurrences follows the Poisson distribution:

$$\Pr[Y = y] = \frac{e^{-\mu}\mu^y}{y!} \tag{8.1}$$

One of Poisson's properties is that it has equal variance as its mean:

$$\mathrm{E}[Y] = \mathrm{Var}[Y] = \mu \tag{8.2}$$

In a count data model such as Poisson regression model, we are modeling log of the expected counts:

$$\log(\mathrm{E}(Y)) = \mathbf{X_i'}\boldsymbol{\beta} \tag{8.3}$$

The log-likelihood function is

$$\log L(\beta) = \sum_{i=1}^{N} - \exp(X_i'\beta) + y_i X_i'\beta - \ln y_i! \tag{8.4}$$

To maximize it, the first order condition is

$$\sum_{i=1}^{N} (y_i - \exp(X_i'\beta))X_i' = 0 \tag{8.5}$$

We can use Newton-Raphson or other optimization algorithms to find the solution for the first order condition.

8.2 QMLE Poisson

8.2.1 Model

Note that in the first-order condition above, if X_i' includes a constant, then $(y_i - \exp(X_i'\beta))$ sums to zero. If $\mathrm{E}[y_i|X_i] = \exp(X_i'\beta)$, then the summation on the left-hand side has expectation of zero. Hence the only specification needed to apply it is the conditional expectation of Y given X. Even if the data is not Poisson-distributed, the estimator is still consistent. Therefore, this estimator is called the quasi-ML (QML) Poisson estimator.

Although the QML Poisson estimator is consistent under relatively weak condition, the regular variance estimator for the coefficients are not valid anymore.

If a stronger assumption is made that the data follows a Poisson distribution, then the error term has a variance which equals to the mean of Y. The estimator $\hat{\beta}_P$ (ML estimator) follows a normal distribution asymptotically with a variance matrix

$$\mathrm{Var}_{ML}[\hat{\beta}_P] = (\sum_{i=1}^{N} \mu_i X_i X_i')^{-1} \tag{8.6}$$

On the other hand, the variance matrix for the QML Poisson estimator $\hat{\beta}_{QML}$ is

$$\mathrm{Var}_{QML}[\hat{\beta}_{QML}] = (\sum_{i=1}^{N} \mu_i X_i X_i')^{-1} (\sum_{i=1}^{N} \omega_i X_i X_i') (\sum_{i=1}^{N} \mu_i X_i X_i')^{-1} \tag{8.7}$$

where $\omega_i = \mathrm{Var}[y_i|X_i]$ is the conditional variance of y_i.

8.2.2 Implementation

Poisson estimation is implemented in almost every statistical package. However, some of them may not work if you have a continuous dependent variable.

Stata's implementation of the Poisson model: "poisson" and "xtpoisson" do take a continuous dependent variable. Standard errors reported without any "vce" specification assume the Poisson variance equals the mean and are likely too small under overdispersion. For a cross-sectional Poisson treated as QMLE, the robust (sandwich) standard errors – the "robust" option – *are* exactly the adjustment needed for a misspecified variance, so they are valid even when the equidispersion assumption fails; the conditional mean only needs to be correctly specified. In the panel fixed-effects case, serial correlation within a unit is a separate issue that requires *clustered* standard errors. Older versions of "xtpoisson" did not offer a cluster-robust option, which is why the user-written program xtpqml was written to supply the cluster-robust standard errors of Wooldridge (1999) after xtpoisson.

8.3 Fixed-effect Poisson model

We specify a panel data count model as:

$$E[y_{it}|\alpha_i, X_{it}] = \alpha_i \exp(X_{it}'\beta) = \exp(\gamma_i + X_{it}'\beta). \tag{8.8}$$

It turns out that for fixed-effect Poisson model, just like in linear regression case, there is NO incidental parameter problem. Poisson MLE is the same for conditional and unconditional likelihood.(See Cameron and Trivedi, 2005, Page 805).

What the result does not do is relax the exogeneity requirement. Consistency of $\hat{\beta}$ needs $E[y_{it} \mid \alpha_i, X_{i1}, \dots, X_{iT}] = \alpha_i \exp(X_{it}'\beta)$ – strict exogeneity conditional on the unit effect, which rules out feedback from past outcomes to later regressors. The absence of an incidental-parameter problem is a statement about estimating β alongside the α_i, not a licence to ignore what the conditional mean assumes.

Therefore, the coefficient estimates are the same for conditional or unconditional fixed-effect Poisson model. That is, in Stata, "xtpoisson, fe" will return the same results as "xi: poisson i.group", is the same as "xtpqml, fe". The only difference is that "xtpqml, fe" returns "correct" standard errors for QMLE Poisson. You can also calculate standard errors with the other commands, using clustered standard errors, or bootstrapped standard errors.

Conditional fixed effect negative binomial model is also possible to be consistent for some specifications. But Poisson fixed effect is more popular since it is QMLE.

8.4 Negative Binomial model

The Poisson model has a restrictive property that the conditional variance equals the conditional mean. Often we'll see "overdispersion", that is, the variance exceeds the mean, in real data. The negative binomial model is a generalization of poisson model which allows for overdispersion by introducing an unobserved heterogeneity term. In a negative binomial model,

$$\mathrm{E}[Y|X, \tau] = \mu\tau = e^{X'\beta} \cdot \tau \tag{8.9}$$

This extra τ term follows a $Gamma(\theta, \theta)$ distribution, with $E(\tau) = 1$ and $Var(\tau) = 1/\theta$.

Conditional on X and τ, the distribution of Y is still poisson:

$$\Pr[Y = y|X, \tau] = \frac{e^{-\mu\tau}(\mu\tau)^y}{y!} \tag{8.10}$$

Conditional on X only, the distribution of Y is negative binomial:

$$\Pr[Y = y|X] = \frac{\theta^\theta \mu^y \Gamma(\theta + y)}{\Gamma(y+1)\Gamma(\theta)(\mu + \theta)^{\theta + y}} \tag{8.11}$$

This distribution still has conditional mean of μ, but variance is $\mu(1 + (1/\theta)\mu)$.

When $\alpha = 1/\theta$ approaches zero, the negative binomial model converges to poisson model.

8.5 Models for truncated counts

In some situations, all we observe are non-zero counts, because of the way data was collected. For example, we only observe people's visits to a park, only if they visit. To model how many times they visit, based on this kind of data set, we need to use zero-truncated count models.

$$\Pr[Y = y|y > 0, X] = \frac{\Pr(y|X)}{\Pr(y > 0|X)} = \frac{\Pr(y|X)}{1 - \exp(-\mu)} \tag{8.12}$$

where the last equality holds for the Poisson case, since $\Pr(y = 0|X) = e^{-\mu}$; for the negative binomial, $\Pr(y = 0|X) = (\theta/(\theta + \mu))^{\theta}$, so the truncation factor is $1 - (\theta/(\theta + \mu))^{\theta}$ instead. Thus we are modeling the counts given that we have a positive outcome. Both zero-truncated poisson ("ztp" in Stata) and zero-truncated negative binomial ("ztnb" in Stata) are estimated by Maximum-likelihood method.

We need to pay extra attention to the "over-dispersion" problem when dealing with zero-truncated data. When data is not truncated, coefficients estimated by poisson model is consistent, even if data is "overdispersed". However, if we have zero-truncated sample, then "over-dispersion" results in inconsistent coefficient estimation (Grogger and Carson, 1991). Therefore, "over-dispersion" needs to be tested before using a truncated *Poisson* model: under truncation the mean-variance restriction stops being innocuous, and violating it breaks consistency of the coefficients rather than only the standard errors.

Overdispersion is not an argument against the truncated negative binomial, though – that model exists precisely to accommodate it. What it does require is that the *form* of the overdispersion match the model. NB2 comes from a particular gamma-mixed Poisson, so overdispersion arising some other way leaves the truncated negative binomial misspecified in its turn. A test tells you that equidispersion fails; it does not tell you that the NB2 mixing story is the right replacement.

8.6 The hurdle regression model

Although negative binomial model relaxes the assumption of equal mean and variance, some researchers prefer modeling the process of generating zeros differently from the process of generating other values. Suppose zero counts are generated by a binary process:

$$\Pr[y = 0|X] = \frac{\exp(X\gamma)}{1 + \exp(X\gamma)} = \pi \tag{8.13}$$

Positive counts are generated by a truncated count process.

In this model, zero is a "hurdle" to get past before reaching positive counts. The hurdle model is estimated by two separate equations. It is easy to estimate. However, the marginal effect is tricky to calculate, since an independent variable, if appearing in both equations, will have effect through both equations.

$$\mathrm{E}(y|X) = [\pi \times 0] + (1 - \pi) \times \mathrm{E}(y|y > 0, X) = (1 - \pi) \times \mathrm{E}(y|y > 0, X) \tag{8.14}$$

In stata, there are commands such as "hplogit", which is a hurdle model with first equation logit, and second equation truncated poisson. It is the same as two separate estimations; namely "logit" first, then "ztp" on positive counts.

The difference between a hurdle model and a Heckman selection model is that in a Heckman model the two equations are jointly estimated and the outcome equation must account for selection bias. An exclusion restriction – a variable that affects selection but not the outcome – is strongly recommended for the selection equation. Strictly speaking, under joint normality the model is identified by the nonlinearity of the inverse Mills ratio even without an excluded variable, but identification then rests entirely on functional form and is fragile, so an exclusion restriction is recommended in practice. On the other hand, a hurdle model assumes the two processes are separate; there is no selection.

8.7 Zero-inflated count models

A zero-inflated count model also models two different processes. One is a binary process: generating zero or non-zeros. The second one is a count model. Note there is a different between zero-inflated and hurdle model: In hurdle model, the second process is a truncated count model (positive counts). In a zero-inflated model, a zero can come from either the binary process, or from the count process.

A second difference is that the estimation of zero-inflated model is a joint estimation of the two processes.

$$y_i \sim \begin{cases} 0 \text{ with probability } \psi \\ g(y_i|X_i)\text{with probability } 1-\psi \end{cases} \tag{8.15}$$

Then

$$P(Y_i = y_i|X_i, Z_i) = \begin{cases} \psi(\gamma' Z_i) + (1-\psi(\gamma' Z_i))g(0|X_i) \text{ if } y_i = 0 \\ (1-\psi(\gamma' Z_i))g(y_i|X_i) \text{ if } y_i > 0 \end{cases} \tag{8.16}$$

This says that every non-zero observation follows a count process $g(y_i|X_i)$. Every zero observation has two possible sources: from a binary process (with probability $\psi(\gamma' Z_i)$) and from a count process (with probability $1-\psi(\gamma' Z_i)$ into the count process and then with probability of $g(0|X_i)$ to be a zero). The count process can be a Poisson process ("zip" in Stata) or a Negative Binomial process ("zinb" in Stata).

8.8 An example of extra-zero count data model comparison

A companion simulation study (see the detailed write-up in this guide's companion `blog_book`, chapter "Which count data model to use") compares plain Poisson, Negative Binomial, zero-inflated Poisson (ZIP, both with a correctly and an incorrectly specified zero-process), and hurdle models across sample sizes and zero-inflation levels. The main findings: with large samples, plain Poisson QMLE is consistent even with unmodeled excess zeros, provided the *observed* conditional mean is correctly specified. That proviso is worth spelling out. Under zero inflation the observed mean is $E[Y \mid X] = (1-\pi(X))\,\mu(X)$, so fitting $\exp(\mathbf{X}\beta)$ recovers the slopes of μ only when π does not depend on X: constant inflation shifts the intercept by $\log(1-\pi)$ and leaves the slopes alone. If π varies with X, the term $\log(1-\pi(X))$ enters the mean function and the slopes are contaminated – independence of the latent zero and count *draws* does not rescue them. Where the inflation is constant, the excess zeros only inflate the standard errors (via overdispersion), not the point estimates, so robust/sandwich standard errors should be used rather than switching models. Zero-inflated Poisson with a correctly specified zero process performs best when that specification is available; hurdle and misspecified-zero-process ZIP perform similarly to plain Poisson; log-linear regression on $\log(y+1)$ is consistently the worst performer, since it targets a different (log-scale) conditional expectation than the models above.

8.9 Interpreting Coefficients and Marginal Effects in Nonlinear Models

This section is broader than count data: it covers how to read coefficients and compute marginal effects in nonlinear models generally, including the binary discrete-choice models of the Discrete and Limited Dependent Variables chapter as well as the count models above.

8.9.1 Binary Response Models

In a linear model

$$y_i = X_i'\beta + \epsilon, \tag{8.17}$$

$$E(y_i) = X_i'\beta \tag{8.18}$$

When y_i is binary (meaning it takes two values, 1 or 0), We can also do an OLS regression on it, but a more popular way is to use a non-linear model. The most popular choices for modeling binary response are logit model and probit model. Both of them use the same idea: use a link function to map the binary variable into a continuous variable which is a linear function of the predictors.

Suppose p_i is the probability that $y_i = 1$. Then

$$p_i = E(y_i) = g^{-1}(X_i'\beta), \tag{8.19}$$

where $g^{-1}()$ is a function that maps from the linear predictor to y_i, which is often called index function or inverse link function.

This says that the expectation of y_i (equivalently, p_i) is not a linear function of x_i's, but by using the link function, we are still able to model a transformed p_i as a linear function of x_i's:

$$g(p_i) = X_i'\beta. \tag{8.20}$$

$g^{-1}()$ can be a normal CDF then we have a probit model:

$$g^{-1}(z) = \Phi(z) = \int_{-\infty}^{z} \frac{1}{\sqrt{2\pi}} \exp[-(t^2/2)]dt, \tag{8.21}$$

or it can be a logit:

$$g^{-1}(z) = \Lambda(z) = e^z/(1+e^z). \tag{8.22}$$

8.9.2 Marginal Effects

Often times we are interested in the marginal effects of the predictors. In a linear model, it's easy: the coefficient on x_1 means when x_1 increase by one unit, how much will y increase.

In a binary response model,

$$\frac{\partial p}{\partial x} = \frac{\partial g^{-1}(X_i'\beta)}{\partial x} \tag{8.23}$$

In the case of probit:

$$\frac{\partial p}{\partial x} = \frac{\partial \Phi(X_i'\beta)}{\partial x} = \phi(X_i'\beta)\beta, \tag{8.24}$$

where $\phi()$ is the density function of normal distribution.

In the case of logit:

$$\frac{\partial p}{\partial x} = \frac{\partial \Lambda(X_i'\beta)}{\partial x} = \gamma(X_i'\beta)\beta, \tag{8.25}$$

where $\gamma()$ is the density function for logistic distribution:

$$\gamma(X_i'\beta) = \Lambda(X_i'\beta)(1 - \Lambda(X_i'\beta)) = p(1-p). \tag{8.26}$$

Therefore for logit, a quick approximation of the marginal effect is to plug in the sample proportion $\bar{p}$ and coefficient estimate $\hat{\beta}$:

$$\widehat{\frac{\partial p}{\partial x}} \approx \bar{p}(1-\bar{p})\hat{\beta} \tag{8.27}$$

This evaluates the effect at the mean probability; it is a rough rule of thumb, not the average marginal effect $\frac{1}{n}\sum_i \hat{p}_i(1-\hat{p}_i)\hat{\beta}$, which is what software reports.

The logit model can also be formulated as

$$\log(\frac{p}{1-p}) = \mathbf{X_i'}\boldsymbol{\beta} \tag{8.28}$$

which says the logarithm of the odds (the ratio of the two probabilities) is equal to $\mathbf{X_i'}\boldsymbol{\beta}$. Therefore,

$$p = \frac{\exp(X_i'\beta)}{1+\exp(X_i'\beta)} = \Lambda(X_i'\beta) \tag{8.29}$$

Often times we have discrete valued predictors, such as gender, or other dummy variables. In that case, it makes more sense to ask what's the effect of gender being 1 vs. 0 (female vs. male). What this question is can be formulated as:

$$E(p|x_1 = 1) - E(p|x_1 = 0). \tag{8.30}$$

In empirical analysis, it's often calculated by calculating the predicted probability by setting x_1 to 1 and by setting x_1 to 0. Then calculate the difference.

8.9.3 Count Data Models

In a count data model such as Poisson regression model or Negative Binomial model, we are modeling log of the expected counts:

$$\log(\mathrm{E}(y)) = \mathbf{X_i'}\boldsymbol{\beta} \tag{8.31}$$

Therefore the marginal effect of x_1, for example, on $\mathrm{E}(y)$ is

$$\frac{\partial \log \mathrm{E}(y)}{\partial x} = \hat{\beta}, \tag{8.32}$$

which means β is the marginal effect of x on log of the expected counts. Suppose the expected counts at x is μ_x, then

$$\frac{(\partial \mu_x)/\mu_x}{\partial x} = \hat{\beta}, \tag{8.33}$$

which says that β represents the marginal effect of percentage change in μ_x with respect to x.

Another way to formulate this is to use the Incidence Rate Ratio Interpretation. Suppose the expected counts at x is μ_x, and expected counts at $x+1$ is μ_{x+1}.

$$\log(\mu_x) = x'\beta. \tag{8.34}$$

Therefore,

$$\mu_x = \exp(x'\beta), \tag{8.35}$$

and

$$\frac{\mu_{x+1}}{\mu_x} = \exp(\beta). \tag{8.36}$$

This says that the exponentiated β is the incidence rate ratio of the expected counts, if x increases by 1.

To calculate the marginal effect of x on y (or $\mathrm{E}(y)$), we need to calculate

$$\frac{\partial \mu_x}{\partial x} = \hat{\beta}\mu_x = \hat{\beta}\exp(X_i'\hat{\beta}), \tag{8.37}$$

if we evaluate μ_x at predicted value.

8.9.4 How to calculate marginal effects in Stata

Stata's margins command is a powerful command to get the predicted value or marginal effects after an estimation command. Check Stata's manual for more details.

9 Panel Data Models

9.1 Background

Panel data are repeated observations on the same "unit" (we call that cross-sectional units) over time. Panel data models are used since it has advantages over cross-sectional models or time-series models. The major advantage it has over cross-sectional models is: **Controlling for individual heterogeneity**.

For example, Baltagi and Levin (1992) studies cigarette demand across 46 states for the years 1963-88. Consumption is modeled as a function of lagged consumption, price and income. These variables vary across states and time. However, there are other variables that may be time-invariant (z_i) or individual-invariant (w_t). Examples of z_i are religion and, in a panel of adults, completed education. For example, Utah, as a Mormon state, has very low cigarette demand due to religious reason. Generally we consider it does not change over time or change very little over time. Examples of w_t include cigarette commercials on national TV or radio. Panel data models are able to control for individual heterogeneity (or cross-sectional heterogeneity) while cross-sectional models are not. In many (or most) social science studies, there is "unobserved effects" which is embedded in the error term. In other words, we have "omitted variable" problem. Without controlling for it, the estimation results are generally biased and inconsistent.

Panel data give more informative data, more variability and less collinearity among variables. Often time series data suffers from multicollinearity, while panel data has more variability from its cross-sectional units.

The basic unobserved effects model can be written as:

$$y_{it} = \mathbf{x_{it}}\boldsymbol{\beta} + c_i + u_{it} \tag{9.1}$$

where i indexes "units" (can be people, firms, or households, etc.), and t indexes time periods.

Three kinds of regressor appear in this chapter, and it pays to name them once. $\mathbf{x}_{it}$ varies across units *and* time — price, income, and in most applications the treatment itself. z_i varies across units only, so within a unit it is fixed over time; religion and completed education above are the examples. w_t varies across time only, so in any period it is common to every unit; the national TV campaign is one. Which group a variable falls into is what decides almost everything that follows, including which coefficients fixed effects can estimate and how OLS weights the data.

Instruments do not appear until the panel turns dynamic. When they do, they are written $\mathbf{Z}$, which is a different object from the time-invariant z_i here — see the Endogeneity, GMM and Dynamic Panel Data chapters.

Comparing to regular cross-sectional models, there is an extra term c_i. c_i can be treated as a random effect or fixed effect. Traditionally it is distinguished by whether it is estimated as a random variable or a parameter. Modern econometrics tends to distinguish by the correlation between c_i and $\mathbf{x_{it}}$. If there is no correlation between c_i and $\mathbf{x_{it}}$, then it's a random effect; otherwise, it's a fixed effect.

It is true that in some cases it is hard to justify that there is no correlation between c_i and $\mathbf{x_{it}}$, which is one reason that people in economics tend to use fixed-effect models. However, fixed-effect models have its own limitations. One of them is: It is hard to justify that ALL the cross-sectional units' characteristics other than those already in the model do not change over time.

9.2 Random Effect Methods

A random effect model puts c_i into the error term, then estimate by FGLS (feasible GLS). Random effect model is sometimes called "Error-components model" since the overall error term is divided into an individual level error term (c_i) and individual-time level error term (u_{it}).

Under a random effect model, the variance-covariance matrix of the error term becomes:

$$\boldsymbol{\Omega} = \begin{bmatrix} \sigma_c^2 + \sigma_u^2 & \sigma_c^2 & \cdots & \sigma_c^2 \\ \sigma_c^2 & \sigma_c^2 + \sigma_u^2 & \cdots & \sigma_c^2 \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_c^2 & \sigma_c^2 & \cdots & \sigma_c^2 + \sigma_u^2 \end{bmatrix} \tag{9.2}$$

The random effect estimator is:

$$\hat{\beta}_{\mathbf{RE}} = (\sum_{\mathbf{i=1}}^{\mathbf{N}} \mathbf{X_i'} \hat{\boldsymbol{\Omega}}^{\mathbf{-1}} \mathbf{X_i})^{\mathbf{-1}} (\sum_{\mathbf{i=1}}^{\mathbf{N}} \mathbf{X_i'} \hat{\boldsymbol{\Omega}}^{\mathbf{-1}} \mathbf{y_i}) \tag{9.3}$$

It is a special case of GLS. Before calculating $\hat{\beta}_{RE}$, σ_c^2 and σ_u^2 need to be estimated. This is generally done by using residuals from within and between regressions (the Swamy-Arora method, the default in most software), or from pooled OLS.

9.3 Fixed Effect Methods

Random effect methods assume that c_i be orthogonal to $\mathbf{x_{it}}$. In many applications, the whole point of using panel data is to allow c_i be correlated with $\mathbf{x_{it}}$. We need fixed-effect models in those cases.

For the same model as in the equation above, the fixed-effect methods try to eliminate c_i using fixed effects transformation, or "within transformation". The FE transformation is to "de-mean" each observation by subtracting the group mean. Basically subtracting the group-mean equation below from the unit-level equation above:

$$\bar{y}_i = \mathbf{\bar{x}_i} \boldsymbol{\beta} + c_i + \bar{u}_i \tag{9.4}$$

where $\bar{y}_i$, etc., means group means. For example, if we have 50 states with state level data of cigarette consumption across years, then groups means states, and we have 50 group means which are state level average cigarette consumption across years.

The reason for "demeaning" is to remove c_i from final estimation. The "demeaned" regression equation is the final regression used in estimation:

$$(y_{it} - \bar{y}_i) = (\mathbf{x}_{it} - \bar{\mathbf{x}}_i)\beta + (u_{it} - \bar{u}_i). \tag{9.5}$$

The unit-specific effect c_i cancels because it is constant over t ($c_i - \bar{c}_i = 0$). Writing the within-transformed variables as $\ddot{y}_{it} = y_{it} - \bar{y}_i$ and $\ddot{\mathbf{x}}_{it} = \mathbf{x}_{it} - \bar{\mathbf{x}}_i$, OLS on the demeaned data gives the **fixed-effects (within) estimator** (here $\mathbf{x}_{it}$ is a $1 \times k$ *row* vector, so $\ddot{\mathbf{x}}_{it}'\ddot{\mathbf{x}}_{it}$ is the $k \times k$ outer product)

$$\hat{\beta}_{FE} = \left(\sum_i \sum_t \ddot{\mathbf{x}}_{it}'\ddot{\mathbf{x}}_{it} \right)^{-1} \sum_i \sum_t \ddot{\mathbf{x}}_{it}'\ddot{y}_{it}. \tag{9.6}$$

This is consistent for β even when c_i is correlated with $\mathbf{x}_{it}$, which is the key advantage of FE over random effects. Consistency still asks for the rest of the usual package: strict exogeneity of the idiosyncratic error conditional on the unit effect, $E[u_{it} \mid c_i, \mathbf{x}_{i1}, \dots, \mathbf{x}_{iT}] = 0$, enough within-unit variation for the demeaned cross-product to have full rank, and an asymptotic scheme – usually $N \to \infty$ with T fixed. What FE buys is freedom from correlation between c_i and the regressors. It buys nothing at all against correlation between u_{it} and them. The price is that any time-invariant regressor (the z_i above) is also swept out by the demeaning and cannot be estimated. The usual variance estimator is $\hat{V}(\hat{\beta}_{FE}) = \hat{\sigma}_u^2 (\sum_i \sum_t \ddot{\mathbf{x}}_{it}'\ddot{\mathbf{x}}_{it})^{-1}$ with $\hat{\sigma}_u^2$ using degrees of freedom $NT - N - k$; in practice one clusters by unit to allow serial correlation within i.

9.4 Fixed effects or random effects: the Hausman test

Random effects is more efficient (it uses both within and between variation) but is consistent only if c_i is uncorrelated with $\mathbf{x}_{it}$; fixed effects is consistent either way. The Hausman test compares the two estimators:

$$H = (\hat{\beta}_{FE} - \hat{\beta}_{RE})' \, [\hat{V}(\hat{\beta}_{FE}) - \hat{V}(\hat{\beta}_{RE})]^{-1} \, (\hat{\beta}_{FE} - \hat{\beta}_{RE}) \ \sim \ \chi^2_k \tag{9.7}$$

under the null that random effects is consistent. A large H (small p-value) favors fixed effects. In R this is `phtest(fe, re)` from the `plm` package; in Stata, `hausman fe re`.

Caveat: this classic form requires RE to be efficient. The formula above relies on $\hat{V}(\hat{\beta}_{FE}) - \hat{V}(\hat{\beta}_{RE})$ being positive semi-definite, which in turn requires the RE estimator to actually be the *efficient* estimator under the null – true only under the RE model's homoscedasticity and no-serial-correlation assumptions. If you are using robust or cluster-robust covariance matrices (the usual practice when errors are heteroscedastic or serially correlated), RE is generally no longer efficient, the covariance difference can fail to be positive definite, and the classic Hausman test statistic can be negative or otherwise ill-behaved. In that case, use a regression-based Hausman test instead: augment the RE (or pooled) regression with the within-unit means of the time-varying regressors

(Mundlak's approach) or with the demeaned regressors themselves (Wooldridge's regression-based test), and test the added coefficients jointly with a standard (robust) Wald test – this is valid under heteroscedasticity/clustering and reduces to the same idea as the classic Hausman test under its assumptions.

9.5 Collapse to unit means, or keep the long format?

A practical question comes up whenever we hold a panel. Should we average each unit over time and run one regression on the N resulting rows, or keep all NT rows and cluster the standard errors by unit? Many people treat this as a trade-off and assume the long format must carry more information. It usually does not. The answer turns entirely on where the regressors live.

Take the case where every regressor is time-invariant, so the model is $y_{it} = \mathbf{z}_i\gamma + c_i + u_{it}$. With a balanced panel, pooled OLS on the NT rows and OLS on the N unit means give the same $\hat{\gamma}$. Not close — identical. Random effects returns the same number as well. Wooldridge (2010) states the reason plainly: because c_i is there, new observations within a unit carry no information about γ beyond how they move the unit average $\bar{y}_i$. In effect we only have N useful pieces of information. Averaging discards nothing, because $\mathbf{z}_i$ was already constant over t and the within variation was never available to γ in the first place.

What does differ is the degrees of freedom. The unit-mean regression uses $N-k$; pooled OLS reports $NT-k$, and that is the mistake. Moulton (1990) is the classic illustration, and Donald and Lang (2007) show that t_{N-k} is the reference distribution that keeps the size right here – an exact result resting on the distributional and design assumptions of the group-level regression, not a universal small-sample repair. With few clusters and heteroskedasticity the practical routes are a cluster-robust variance with a few-cluster correction, or a cluster bootstrap. Clustering by unit repairs the standard error when N is large, but it does not repair the degrees of freedom when N is small.

Now put a time-varying $\mathbf{x}_{it}$ into the model. The two options stop agreeing, and not only for the coefficient on $\mathbf{x}_{it}$ — every coefficient moves, the one on $\mathbf{z}_i$ included. Averaging to unit means throws away all within variation, so what survives is the between estimator. Pooled OLS keeps both sources and forces a single coefficient on them, so it returns a variance-weighted average of the within and the between relationship. The two agree only when the within and between coefficients are equal, which is what the Hausman test is looking for in practice. Strictly the null is consistency of random effects – orthogonality of c_i and the regressor history – which *implies* equal within and between coefficients rather than being the same statement. Equality could hold under an alternative, and equality in one sample is not the null at all. So this is not a new problem. It is the FE-versus-RE question in different clothes.

Situation	Collapsed vs long
All regressors time-invariant, balanced panel	Same coefficients; only the degrees of freedom differ
Any time-varying regressor	Coefficients generally differ, including those on $\mathbf{z}_i$
Unbalanced panel	Generally differ, since pooled OLS weights each unit by T_i

The word to hold on to in the last two rows is *generally*. The differences come from the design, not from a theorem: a regressor orthogonal to the rest, or T_i variation that happens to leave the weighting unchanged, can leave particular coefficients coinciding. What you lose is the right to count on it.

The reconciliation is Mundlak's device, the same one used above for the regression-based Hausman test. Add the unit means $\bar{\mathbf{x}}_i$ to the long regression alongside $\mathbf{x}_{it}$, and the within and between variation carry separate coefficients: $\mathbf{x}_{it}$ picks up the within estimate, $\bar{\mathbf{x}}_i$ picks up the *difference* between the between and within estimates, so the two coefficients sum to what the collapsed regression reports. That second coefficient is the Hausman contrast. With a single time-varying regressor its t-statistic *is* the Hausman test; with several, the test is the joint Wald or F test on all the mean coefficients together, not any one t. Exact numerical agreement with the collapsed regression also depends on balance and on the weighting, so read it as the same comparison rather than an identity to check to the last digit. We keep the within variation and recover the between answer in one regression.

The useful way to remember it: collapse freely when every regressor is constant within the group, and never when one is not.

9.6 What the long format buys

The section above is mostly a warning. The other side is worth seeing directly, so we simulate panels where we know the truth. We use `fixest` here; in Stata the same regressions are `regress`, `xtreg, be` and `xtreg, fe`.

9.6.1 The equivalence, and where it fails

Take 1000 units over 6 periods, one time-invariant regressor z_i, and a true $\gamma = 2$.

```
set.seed(42)
N <- 1000; T <- 6

z   <- rnorm(N)
bal <- data.frame(id = rep(1:N, each = T), t = rep(1:T, N),
                  z  = rep(z, each = T), c = rep(rnorm(N), each = T))
bal$y <- 2 * bal$z + bal$c + rnorm(N * T)
agg   <- aggregate(cbind(y, z) ~ id, data = bal, FUN = mean)

print(c(long      = coef(feols(y ~ z, bal))[["z"]],
        collapsed = coef(feols(y ~ z, agg))[["z"]]), digits = 12)
```

```
         long     collapsed
1.99351677619 1.99351677619
```

The two agree to every digit printed. Now break the balance by dropping the later periods for units with high z_i, so T_i varies, and varies with the regressor.

```r
unb  <- bal[!(bal$z > 0.5 & bal$t > 2), ]
agg2 <- aggregate(cbind(y, z) ~ id, data = unb, FUN = mean)

print(c(long      = coef(feols(y ~ z, unb))[["z"]],
        collapsed = coef(feols(y ~ z, agg2))[["z"]]), digits = 12)
```

```
         long     collapsed
2.00682070731 2.01481052322
```

They no longer agree. Pooled OLS weights each unit by T_i; the collapsed regression counts each unit once. Neither is wrong — they answer different questions, and we should know which one we asked. That T_i is a special case of a general rule, set out in the next section.

9.6.2 What collapsing costs

Now the case that matters. Let x_{it} vary over time and make c_i correlated with $\bar{x}_i$, which is the situation fixed effects exists for. The truth is $\beta = 1$.

```r
mu <- rnorm(N)
d  <- data.frame(id = rep(1:N, each = T),
                 x  = rep(mu, each = T) + rnorm(N * T),
                 c  = rep(mu + rnorm(N), each = T))
d$y    <- 1 * d$x + d$c + rnorm(N * T)
d$xbar <- ave(d$x, d$id)
dagg   <- aggregate(cbind(y, x) ~ id, data = d, FUN = mean)

between <- feols(y ~ x,        data = dagg)
pooled  <- feols(y ~ x,        data = d, cluster = ~id)
within  <- feols(y ~ x | id,   data = d)
mundlak <- feols(y ~ x + xbar, data = d, cluster = ~id)

pull <- function(m, v) c(coef(m)[[v]], se(m)[[v]])
tab <- rbind(`between (collapsed)` = pull(between, "x"),
             `pooled OLS`          = pull(pooled,  "x"),
             `within / FE`         = pull(within,  "x"),
             `Mundlak: x`          = pull(mundlak, "x"),
             `Mundlak: xbar`       = pull(mundlak, "xbar"))
colnames(tab) <- c("estimate", "std. error")
round(tab, 3)
```

```
                    estimate std. error
between (collapsed)    1.845      0.035
pooled OLS             1.484      0.022
within / FE            0.995      0.014
Mundlak: x             0.995      0.014
Mundlak: xbar          0.850      0.036
```

Read the table down. The collapsed regression returns about 1.85, badly biased, because it is the between estimator and $\bar{x}_i$ carries c_i with it. Pooled OLS returns about 1.48 — closer, since some within variation is mixed in, but still inconsistent. The within estimator returns about 0.99, and only the long format can compute it: once we collapse there is nothing left to demean. Mundlak delivers both at once, exactly as described above — 0.99 on x_{it}, 0.85 on $\bar{x}_i$, and $0.99 + 0.85$ recovers the between estimate.

Mundlak's device is more than a test: it is an estimator in its own right. Wooldridge's correlated random effects (CRE) framework generalizes the idea, allowing unit-level heterogeneity to depend on the time-averages of the regressors while retaining the RE structure. The full CRE treatment, including extensions to nonlinear models, is in the Blog Book.

So the argument for the long format is not precision. It is that demeaning, the Hausman comparison, unit-specific slopes, lagged dependent variables and anything else built on within variation stop existing once the panel is collapsed. Dynamic models are the extreme case, and they get their own chapter, Dynamic Panel Data.

The rule behind that T_i, and behind the variance weighting in the between-versus-within comparison, is not specific to panels. It is set out in Interpreting OLS: Effect Weights and its companion, Outcome Weights.

10 Survival Models

10.1 Survival function

Survival analysis is also called time to event analysis or event history analysis. We use survival models to analyze data that have the following characteristics:

1. the dependent variable is the waiting time till the occurrence of some event;
2. some observations are censored; that is, some units are not in the data set anymore for some reason before an event happens. For some units we observe the event and the exact waiting time; for others it has not occurred, and all we know is that the waiting time exceeds the observation time. However, we believe if they stayed in the study, sooner or later an event would happen.
3. we are trying to explain (predict) the occurrence of an event with some predictors (explanatory variables).

Censoring is what makes this its own subject, and it carries an assumption that belongs before any estimator. Kaplan-Meier and the Cox partial likelihood below both require censoring to be **independent**, or noninformative: conditional on the covariates in the model, units censored at a given time must be neither more nor less likely to experience the event than comparable units still under observation. Put plainly, being censored may not carry information about when the event would have happened.

This is an identification assumption, not a technical footnote. It fails whenever units leave for reasons tied to the outcome – patients withdrawing because they are deteriorating, firms vanishing from a sample shortly before they fail, workers leaving a spell study just as re-employment arrives. Nothing in the data announces the failure and no sample size repairs it. The usual response is to bring the relevant variables into the model, since the assumption is conditional on covariates, and then to argue explicitly why the censoring that remains is unrelated to the event process.

Let T denote the time variable, a nonnegative, continuous random variable with PDF $f(t)$ and CDF $F(t)$, where t is a realization of T. The survival function is defined by

$$S(t) = Pr(T > t) = 1 - F(t) \tag{10.1}$$

The graph of $S(t)$ against t is known as the survival curve. If there is no censoring, then the survival function is easy to estimate by the ratio of the number of units surviving beyond t to the total number of units in the sample. The most common method to estimate the survival function with a survival data set containing censored observations is the Kaplan-Meier method. The basic idea is to use the product of a series of conditional probabilities.

First sort the event times from the smallest to the largest:

$$t_{(1)} \leq t_{(2)} \leq ... \leq t_{(r)} \tag{10.2}$$

Then the survival curve is estimated by:

$$\hat{S}(t) = \prod_{j|t_{(j)} \leq t} (1 - \frac{d_j}{r_j}), \tag{10.3}$$

where r_j is the number of units at risk at $t_{(j)}$ and d_j is the number of "failure" or events at $t_{(j)}$. Note that units censored at $t_{(j)}$ are included in r_j.

The variance of this estimator can be estimated by:

$$v\hat{a}r(S(t)) = (\hat{S}(t))^2 \sum_{j|t_{(j)} \leq t} \frac{d_j}{r_j(r_j - d_j)}. \tag{10.4}$$

10.2 Hazard function

We are often interested in which periods have the highest or lowest change of "death" or "failure" or other events. That is, we may be interested in the probability that a state ends between t and $t + \Delta t$ conditional on having reached t in the first place.

The probability is

$$\Pr(t < T \leq t + \Delta t | T \geq t) = \frac{F(t + \Delta t) - F(t)}{S(t)} \tag{10.5}$$

The hazard function is defined by

$$h(t) = \lim_{\Delta t \to 0} \frac{\Pr(t < T \leq t + \Delta t | T \geq t)}{\Delta t} = \frac{f(t)}{S(t)} = \frac{f(t)}{1 - F(t)} \tag{10.6}$$

One of the simplest functional forms is the exponential distribution

$$f(t, \theta) = \theta e^{-\theta t} \tag{10.7}$$

Therefore, the hazard function is

$$h(t) = \theta \tag{10.8}$$

A much more flexible functional form is Weibull, with CDF, survival, and density

$$F(t, \theta, \alpha) = 1 - \exp(-(\theta t)^\alpha), \quad S(t) = \exp(-(\theta t)^\alpha), \quad f(t) = \alpha \theta^\alpha t^{\alpha - 1} \exp(-(\theta t)^\alpha). \tag{10.9}$$

Hazard function can be shown to be

$$h(t) = \alpha \theta^\alpha t^{\alpha - 1} \tag{10.10}$$

However, Weibull distribution does not allow for the possibility that the hazard may first increase then decrease over time. Log-normal distribution does allow this.

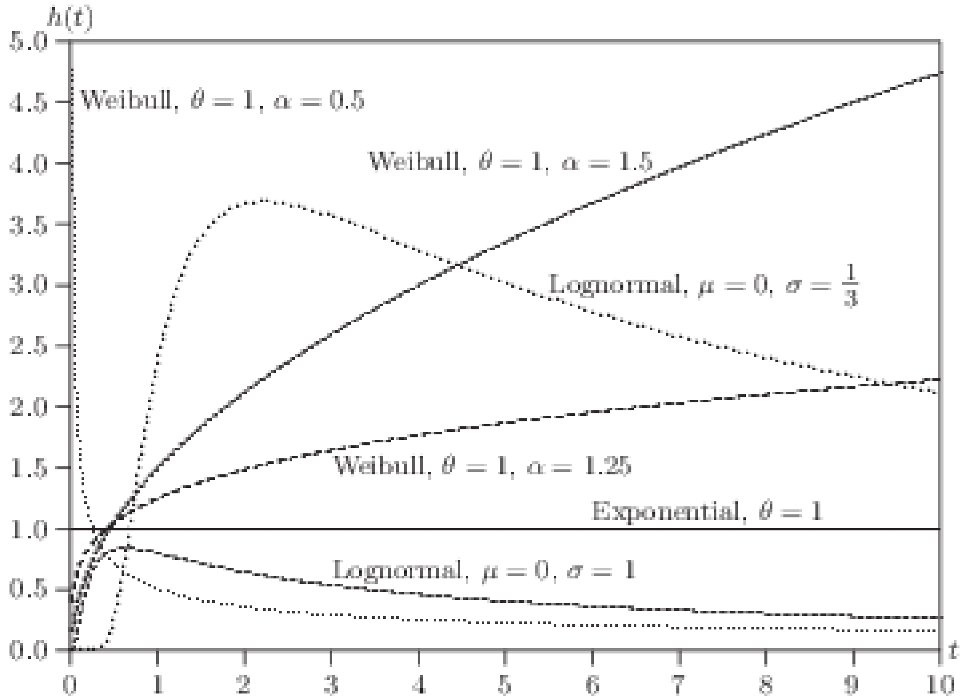


Figure 10.1: Various hazard functions from Davidson and MacKinnon

10.3 Log-likelihood function

Suppose we have n units with a survival function $S(t)$, density $f(t)$, and hazard $h(t)$. If for unit i, the event happens at t_i, it's contribution to the likelihood function is the density at that point:

$$L_i = f(t_i) = S(t_i)h(t_i) \tag{10.11}$$

If no event at t_i, all we know is that the lifetime exceeds t_i. The probability is

$$L_i = S(t_i), \tag{10.12}$$

which is the contribution to the likelihood function from a censored observation.

Therefore, if U denotes the set of uncensored observations, the log-likelihood function for the entire sample can be written as

$$\ell(t, \theta) = \sum_{i \in U} \log h(t_i | \mathbf{X}_i, \theta) + \sum_{i=1}^{n} \log S(t_i | \mathbf{X}_i, \theta) \tag{10.13}$$

10.4 Proportional Hazard Models

10.4.1 Model and likelihood

One class of models that is quite widely used is the proportional hazard models (based on Cox (1972)).

$$h(t|\mathbf{x}_i) = h_0(t)\exp(\mathbf{x}_i'\beta), \tag{10.14}$$

In this model $h_0(t)$ is called baseline hazard rate; it describes the risk for units with $\mathbf{x_i} = \mathbf{0}$. $\exp(\mathbf{x}_i'\beta)$ represents the relative risk, a proportionate increase or decrease in risk, due to $\mathbf{x_i}$. The interpretation of β is that $\exp(\beta_i)$ gives the relative risk change associated with an increase of one unit in x_i, all other explanatory variables remaining constant.

β is estimated by maximizing a partial likelihood function. It is partial likelihood function since the baseline hazard is factored out. Suppose one event happens at time t_j. Conditional on this event the probability that case i dies (meaning this event happened to case i, not other cases in the risk set) is

$$L_i = \frac{h_0(t_j)\exp(x_i'\beta)}{\sum_l I(T_l \geq t_j)h_0(t_j)\exp(x_l'\beta)} = \frac{\exp(x_i'\beta)}{\sum_l I(T_l \geq t_j)\exp(x_l'\beta)} \tag{10.15}$$

We can see here that Cox's model is for continuous time survival data. It is assumed there is no ties at time t_j (in continuous time is not possible for two events to happen at the same time). In reality, we observe ties of event time, because in many situations, we observe grouped data. For example, we observe case m and n have events at the same time j. In that case, in the denominator, whether we include case m in the calculation of the likelihood of case n is ambiguous. In that case, special treatment is needed. The standard approximation here is Breslow's:

$$L_{m\in D(t_j)} \simeq \frac{\prod_{m\in D(t_j)}\exp(x_m'\beta)}{[\sum_{l\in R(t_j)}\exp(x_l'\beta)]^{d_j}} \tag{10.16}$$

where $D(t_j)$ is the set of cases that die at time t_j and d_j denotes the number of cases that die at time t_j. This approximation works well with small number of ties relative to total number of cases at risk. Peto's approximation is a close relative of it. Efron's is a finer adjustment that apportions the risk set among the tied cases, and it is the more accurate of the two when ties are common. The defaults differ by software: R's `coxph` uses Efron, Stata's `stcox` uses Breslow, so the same data can produce slightly different coefficients in the two without either being wrong.

One important feature (disadvantage?) for the proportional hazard model is that for all units, the effect is the same at all time t. To see this:

$$\frac{h_1(t)}{h_2(t)} = \frac{h_0(t)\exp(\mathbf{x}_1'\beta)}{h_0(t)\exp(\mathbf{x}_2'\beta)} = \frac{\exp(\mathbf{x}_1'\beta)}{\exp(\mathbf{x}_2'\beta)}, \tag{10.17}$$

which does not depend on t.

One advantage of Cox's model is that it is considered as a semi-parametric model since it does not have to specify the distribution of the survival time. The estimation process relies only on the order in which events occur, not on the spacing between them. The exact times still matter, because they are what determines who sits in each risk set. What drops out is the baseline hazard, and with it any need to model how far apart the events are. Parameter estimates are derived assuming continuous survival times.

10.4.2 Interpretation

In the equation above, if we would like to calculate the partial effect of x_j on h, we have:

$$\partial h(t|\mathbf{x}_i)/\partial x_j = h_0(t)\exp(\mathbf{x}_i'\beta)\beta_j = \beta_j h(t|\mathbf{x}_i), \tag{10.18}$$

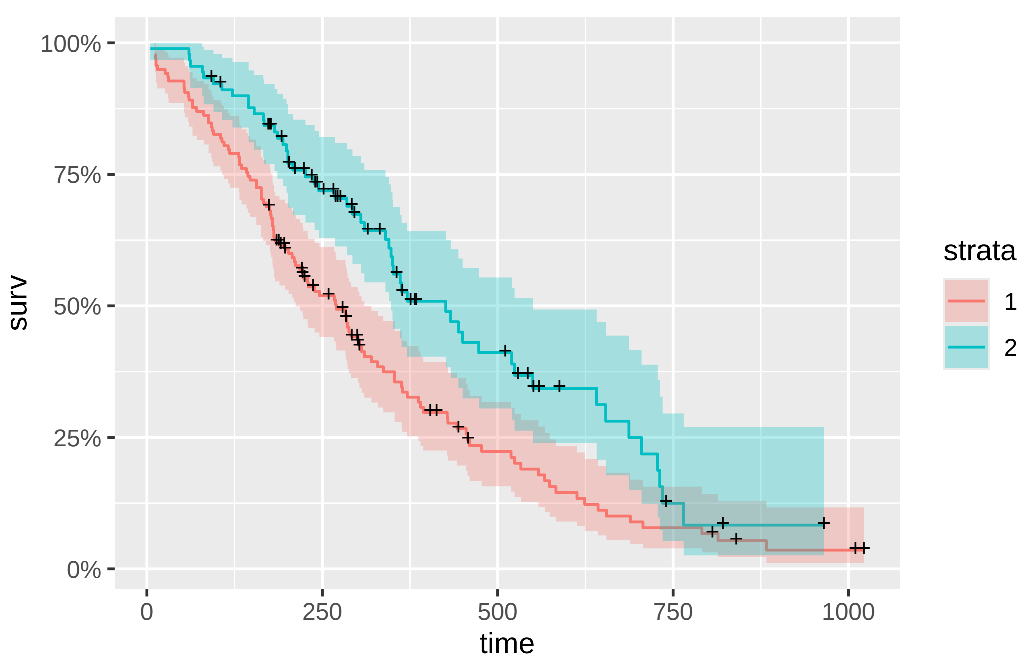
If we do it the other way, plug in $x_j + 1$ will generate

$$h_0(t)\exp(\mathbf{x}_i'\beta + \beta_j) = \exp(\beta_j)h(t|\mathbf{x}_i) \tag{10.19}$$

Therefore, a one unit change in x_j multiplies the hazard by $\exp(\beta_j)$ (a proportionate change of $\exp(\beta_j) - 1$).

10.4.3 Example in R

```r
library(ggfortify)
library(survival)
fit <- survfit(Surv(time, status) ~ sex, data = lung)
autoplot(fit)
```



Censored observations are denoted by crosses. This is the so-called Kaplan-Meier curve.

Now let's do a Cox model example.

```
fit2 <- coxph(Surv(time, status) ~ sex + age, data = lung, method="breslow")
summary(fit2)
```

```
Call:
coxph(formula = Surv(time, status) ~ sex + age, data = lung,
    method = "breslow")

  n= 228, number of events= 165

         coef exp(coef)  se(coef)      z Pr(>|z|)
sex -0.512565  0.598957  0.167462 -3.061  0.00221 **
age  0.017013  1.017158  0.009222  1.845  0.06506 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

    exp(coef) exp(-coef) lower .95 upper .95
sex     0.599     1.6696    0.4314    0.8316
age     1.017     0.9831    0.9989    1.0357

Concordance= 0.603  (se = 0.025 )
Likelihood ratio test= 14.08  on 2 df,   p=9e-04
Wald test            = 13.44  on 2 df,   p=0.001
Score (logrank) test = 13.69  on 2 df,   p=0.001
```

Examples of parametric survival models: exponential, Weibull, or log-logistic:

```
exp_fit <-  survreg(Surv(time, status) ~ sex + age, data = lung, dist="exponential")
summary(exp_fit)
```

```
Call:
survreg(formula = Surv(time, status) ~ sex + age, data = lung,
    dist = "exponential")
               Value Std. Error     z      p
(Intercept)  6.35967    0.63547 10.01 <2e-16
sex          0.48093    0.16709  2.88  0.004
age         -0.01562    0.00911 -1.72  0.086

Scale fixed at 1

Exponential distribution
Loglik(model)= -1156.1   Loglik(intercept only)= -1162.3
```

```
    Chisq= 12.48 on 2 degrees of freedom, p= 0.002
Number of Newton-Raphson Iterations: 4
n= 228
```

```r
weibull <-  survreg(Surv(time, status) ~ sex + age, data = lung, dist="weibull")
summary(weibull)
```

```
Call:
survreg(formula = Surv(time, status) ~ sex + age, data = lung,
    dist = "weibull")
               Value Std. Error     z       p
(Intercept)  6.27485    0.48137 13.04 < 2e-16
sex          0.38209    0.12748  3.00  0.0027
age         -0.01226    0.00696 -1.76  0.0781
Log(scale)  -0.28230    0.06188 -4.56 5.1e-06

Scale= 0.754

Weibull distribution
Loglik(model)= -1147.1   Loglik(intercept only)= -1153.9
    Chisq= 13.59 on 2 degrees of freedom, p= 0.0011
Number of Newton-Raphson Iterations: 5
n= 228
```

```r
loglog <-  survreg(Surv(time, status) ~ sex + age, data = lung, dist="loglogistic")
summary(loglog)
```

```
Call:
survreg(formula = Surv(time, status) ~ sex + age, data = lung,
    dist = "loglogistic")
               Value Std. Error     z       p
(Intercept)  5.92232    0.53269 11.12 < 2e-16
sex          0.47751    0.14036  3.40 0.00067
age         -0.01401    0.00771 -1.82 0.06946
Log(scale)  -0.56991    0.06543 -8.71 < 2e-16

Scale= 0.566

Log logistic distribution
Loglik(model)= -1152.9   Loglik(intercept only)= -1160.9
    Chisq= 16.07 on 2 degrees of freedom, p= 0.00032
Number of Newton-Raphson Iterations: 4
n= 228
```

One caveat when reading these next to the Cox fit above. `survreg` estimates *accelerated failure time* models, parameterized on log survival time, whereas `coxph` estimates proportional hazards. The coefficients therefore carry *opposite* signs for the same underlying effect: a covariate that raises the hazard shortens expected survival time. A sign flip between the two tables is the expected result of the differing parameterizations, not a contradiction.

Expect the reversal of direction; do not expect mirror-image magnitudes. Exponential and Weibull are the families that are both PH and AFT, and there the two parameterizations are related exactly through the shape parameter, so the AFT coefficient scaled by the shape recovers the PH coefficient up to sign. Log-logistic, the third fit above, is an AFT family that is *not* proportional hazards at all, so it has no PH coefficient to be the mirror of. Setting a log-logistic fit beside a Cox fit compares two different models of the same data, and only the direction of the effect should be expected to agree.

10.5 Discrete-time Survival Models

Cox (1972) is the *continuous-time* proportional hazards model; the discrete-time analogue models the odds of an event at time t_j given survival up to that point. This proportional-odds specification for the discrete hazard,

$$\frac{h(t_j|\mathbf{x}_i)}{1-h(t_j|\mathbf{x}_i)} = \frac{h_0(t_j)}{1-h_0(t_j)}\exp(\mathbf{x}_i'\beta), \tag{10.20}$$

is a discrete-time *proportional-odds* hazard model rather than a direct discretization of Cox's continuous-time proportional hazards model. (The complementary log-log link is the discrete-time form that corresponds more directly to a grouped continuous-time proportional hazards model.) If we take logs on both sides, we have

$$\text{logit}(h(t_j|\mathbf{x}_i)) = \alpha_j + \mathbf{x}_i'\beta, \tag{10.21}$$

This looks very similar to a logit model. In fact, we can fit a discrete-time proportional hazard model by running a logistic regression on a set of pseudo observations generated by "filling in" the observations between the start time to the event time or time at the end of the study period (censored). The outcome of this process is 0 unless there is an event, then it turns 1. It is treated as independent Bernoulli observations with probability given by hazard h_{ij} for individual i at time t_j. The likelihood function for the discrete-time survival model coincides with the likelihood of a sequence of independent Bernoulli trials (Singer and Willett, 1993, has more details).

10.5.1 Do it in Stata

To do a discrete-time survival model in Stata, your data set needs to be set up as by unit-time. For example, observations by company month. The event (or failure) variable needs to be set to zero until the event happens. If no event until the end of the study period, then it is censored. A data set by unit-time can be analyzed by logit model, optionally with clustered standard error. To set up a structure like this, user-written programs called *dthaz* and *prsnperd* (means person-period) can be used.

To do a continuous-time survival model in Stata, your data can be either set up as discrete-time case (in which case you can use a time-varying explanatory variable) or a single-record-per-person data set. Stata has built-in command *stset* to set up a survival analysis structure. Then you can run *stcox* for Cox's proportional hazard model.

11 Dynamic Panel Data

11.1 When Is It a Problem

Model setup:

$$y_{i,t} = \gamma y_{i,t-1} + X'_{i,t}\beta + C_i + \epsilon_{i,t} \tag{11.1}$$

Pooled regression is biased and inconsistent. The problem is:

$$Cov(y_{i,t-1}, (C_i + \epsilon_{i,t})) \approx \frac{\sigma_c^2}{(1-\gamma)} \tag{11.2}$$

Random effects model is biased and inconsistent, for the same reason that $y_{i,t-1}$ is necessarily correlated with C_i, which is part of the composite error term in the random effect model.

Fixed effect model setup:

$$y_{i,t} - \bar{y}_i = (X_{i,t} - \bar{X}_i)'\beta + \gamma(y_{i,t-1} - \overline{y_{i,-1}}) + (\epsilon_{i,t} - \bar{\epsilon}_i) \tag{11.3}$$

where $\overline{y_{i,-1}}$ is the unit mean of the *lagged* dependent variable. This is not exactly $\bar{y}_i$ (the mean of current $y_{i,t}$). Over the usable periods the two are $(y_{i2}+\cdots+y_{iT})/(T-1)$ and $(y_{i1}+\cdots+y_{i,T-1})/(T-1)$, so their difference is exactly

$$\frac{y_{iT} - y_{i1}}{T-1}, \tag{11.4}$$

the endpoints and nothing else. Balance does not make that vanish – only T growing does, under stability. The distinction is asymptotically negligible and finite-sample real.

Nickell (1981) shows that (see also Anderson and Hsiao 1981)

$$Cov((y_{i,t-1} - \overline{y_{i,-1}}), (\epsilon_{i,t} - \bar{\epsilon}_i)) \approx -\frac{\sigma_\epsilon^2}{T(1-\gamma)^2}[\frac{(T-1) - T\gamma + \gamma^T}{T}] \tag{11.5}$$

which indicates that the demeaned lagged y is correlated with the demeaned error term. The correlation may be large if T is small, but when T is big, then the correlation goes down to near zero. When T is big, we don't have a problem with fixed effect model.

11.2 How Big Is the Bias

When T is small, which is typically the case in many microeconomic settings, then we say the fixed effect model with a lagged dependent variable is inconsistent. But how bad is it?

The limit of $\hat{\gamma} - \gamma$ is approximately $-\frac{(1+\gamma)}{T-1}$. When $T = 10$ and $\gamma = .5$, the bias is about -0.167 which is 1/3 of the true value.

But a simulation study (Judson and Owen 1999) shows that the bias can be as big as 20% of the true coefficient even when $T = 30$.

11.3 Anderson and Hsiao estimator

Anderson and Hsiao (1981) suggested looking at the first difference estimator:

$$y_{i,t} - y_{i,t-1} = (X_{i,t} - X_{i,t-1})'\beta + \gamma(y_{i,t-1} - y_{i,t-2}) + (\epsilon_{i,t} - \epsilon_{i,t-1}) \tag{11.6}$$

or

$$\Delta y_{i,t} = \Delta X_{i,t}\beta + \gamma\Delta y_{i,t-1} + \Delta\epsilon_{i,t} \tag{11.7}$$

This does not solve the endogeneity problem, since $\Delta y_{i,t-1}$ is still correlated with the error term. AH's idea is to instrument $\Delta y_{i,t-1}$ with the past level $y_{i,t-2}$, or past difference $y_{i,t-2} - y_{i,t-3}$. This estimator is consistent, since neither of these instruments is correlated with $\Delta\epsilon_{i,t}$ – provided the idiosyncratic error is serially uncorrelated. That proviso carries the argument. If $\epsilon_{i,t}$ is itself AR(1), then $y_{i,t-2}$ is correlated with $\Delta\epsilon_{i,t}$, the instrument fails, and one has to reach further back.

The assumption is testable, which is the point of the Arellano-Bond test for serial correlation in the differenced residuals. First-order correlation is expected by construction, since $\Delta\epsilon_{i,t}$ and $\Delta\epsilon_{i,t-1}$ share $\epsilon_{i,t-1}$; it is the *second*-order test that carries information. Rejecting AR(2) says the lag being used as an instrument does not sit far enough back.

The other regressors need classifying too, because that is what decides how they are instrumented. Strictly exogenous X can serve as its own instrument. Predetermined X – correlated with past shocks but not with current or future ones – is instrumented by its own lags from $t-1$ backwards. Endogenous X is treated like the lagged dependent variable, with lags from $t-2$ backwards. Misclassifying a regressor here is one of the commoner ways a difference-GMM specification goes quietly invalid.

11.4 Arellano-Bond estimator

Arellano and Bond (1991) expanded the idea by using additional lags of the dependent variable as instruments. For example, both $y_{i,t-2}$ and $y_{i,t-3}$ can be used as instruments. In fact, as t increases, the number of instruments available also increases. In period 3 only $y_{i,1}$ is available. In period 4 $y_{i,1}$ and $y_{i,2}$ are available. In period 5 $y_{i,1}$ and $y_{i,2}$ and $y_{i,3}$ are available, and so on. In other words, we'll have an instrument matrix with one row for each time period that we are instrumenting:

$$Z_i = \begin{bmatrix} y_{i,1} & 0 & 0 & 0 & 0 & 0 & \cdots & 0 & 0 & 0 \\ 0 & y_{i,1} & y_{i,2} & 0 & 0 & 0 & \cdots & 0 & 0 & 0 \\ 0 & 0 & 0 & y_{i,1} & y_{i,2} & y_{i,3} & \cdots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots & \cdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & 0 & \cdots & 0 & y_{i,1} & \cdots & y_{i,T-2} \end{bmatrix} \tag{11.8}$$

This is the so called difference GMM estimator.

11.5 Blundell and Bond estimator

AB model has a problem: when the instruments are weak, the estimator is not good. That will happen when y follows random walk or near random walk. In that case the past levels won't be a good predictor of the future changes.

BB add a second set of moment conditions in *levels*. The basic idea is that lagged differences $\Delta y_{i,t-1}$ are uncorrelated with the composite error $C_i + \epsilon_{i,t}$ under an additional initial-condition/stationarity restriction, so they can instrument the level equation.

This is implemented by so-called *system GMM*, which stacks two equations and estimates them jointly:

- the **differenced equation** ($\Delta y_{i,t}$), instrumented by lagged *levels* $y_{i,t-2}, y_{i,t-3}, \ldots$ (the Arellano–Bond moments); and
- the **level equation** ($y_{i,t}$ in levels, keeping C_i), instrumented by lagged *differences* $\Delta y_{i,t-1}$.

The level-equation moments are valid only under the extra assumption that the initial deviations from the long-run mean are uncorrelated with C_i (mean stationarity of the process). System GMM is more efficient than difference GMM when y is persistent (near a random walk), and it allows time-invariant regressors. A practical warning: the instrument count grows with T^2, so *instrument proliferation* can overfit the endogenous regressors and weaken the Hansen J test; lags should be collapsed or capped in practice.

11.6 Diagnostics in practice

The warning about instrument proliferation is easy to state and easy to ignore. It is worth seeing what it does to real estimates, and what the standard diagnostics look like when they are working.

The data are Arellano and Bond's own: an unbalanced panel of 140 UK firms observed between 1976 and 1984, shipped with `plm` as `EmplUK`. The specification is theirs too – log employment on its own first two lags, current and lagged log wages, three lags of log capital, current and lagged log output, with firm and year effects removed and two-step weighting. Only the instrument set changes across the three fits: every available lag of log employment from $t-2$ back, the same capped at three lags, and then the system-GMM version that adds the level-equation moments.

```r
library(plm)
data("EmplUK", package = "plm")

dif_full <- pgmm(log(emp) ~ lag(log(emp), 1:2) + lag(log(wage), 0:1) +
                   lag(log(capital), 0:2) + lag(log(output), 0:1) |
                   lag(log(emp), 2:99),
                 data = EmplUK, effect = "twoways", model = "twosteps")

dif_cap  <- pgmm(log(emp) ~ lag(log(emp), 1:2) + lag(log(wage), 0:1) +
                   lag(log(capital), 0:2) + lag(log(output), 0:1) |
                   lag(log(emp), 2:4),
                 data = EmplUK, effect = "twoways", model = "twosteps")

sys      <- pgmm(log(emp) ~ lag(log(emp), 1:2) + lag(log(wage), 0:1) +
                   lag(log(capital), 0:2) + lag(log(output), 0:1) |
                   lag(log(emp), 2:99),
                 data = EmplUK, effect = "twoways", model = "twosteps",
                 transformation = "ld")

diag_row <- function(fit, label) {
  co <- summary(fit, robust = TRUE)$coefficients
  sg <- sargan(fit)
  data.frame(specification = label,
             moments  = ncol(fit$W[[1]]),
             lag1     = round(co[1, 1], 3),
             se       = round(co[1, 2], 3),
             sargan_p = round(sg$p.value, 3),
             AR1_p    = round(mtest(fit, order = 1)$p.value, 3),
             AR2_p    = round(mtest(fit, order = 2)$p.value, 3))
}

rbind(diag_row(dif_full, "difference GMM, lags 2+"),
      diag_row(dif_cap,  "difference GMM, lags 2-4"),
      diag_row(sys,      "system GMM"))
```

```
                  specification moments  lag1    se sargan_p AR1_p AR2_p
chisq   difference GMM, lags 2+      40 0.645 0.194    0.162 0.002 0.596
chisq1 difference GMM, lags 2-4      30 0.437 0.359    0.166 0.065 0.848
chisq2               system GMM      55 1.114 0.051    0.121 0.028 0.689
```

Read the AR columns first, because they are the ones that can invalidate everything else. First-order serial correlation in the differenced residuals is expected: $\Delta\epsilon_{i,t}$ and $\Delta\epsilon_{i,t-1}$ both contain $\epsilon_{i,t-1}$, so a small AR(1) p-value is the normal state of affairs and says nothing against the specification. The AR(2) test is the informative one, and none of the three fits shows evidence that the instrumenting lag sits too close.

The Sargan column passes in all three cases, which is exactly where the instrument-count warning earns its keep. The full difference-GMM fit imposes 40 moment conditions on 15 parameters across 140 firms; the system-GMM fit imposes 55. A Sargan or Hansen test with many moments and few groups has little power, so a comfortable p-value is weak reassurance rather than strong evidence. The test is being asked to detect a violation using an instrument set flexible enough to fit almost anything.

The coefficients show what that costs. Capping the instruments at three lags cuts the moment conditions from 40 to 30, moves the coefficient on lagged log employment from about 0.65 to about 0.44, and nearly doubles its standard error. That is the honest trade: fewer, better-behaved instruments buy less precision. System GMM goes the other way and returns a coefficient above 1 with a standard error near 0.05, which should stop the reader rather than reassure them. A persistence parameter above one implies an explosive employment process, and the tight interval around it comes from 55 moment conditions on 140 firms. Every diagnostic in the table passes. The estimate is still not credible.

So what to report: the AR(2) test, the Sargan or Hansen statistic with its degrees of freedom, the instrument count next to the number of groups, and the estimate under at least one narrower instrument set. The last of those does most of the work, because it is the one thing in the list that a flexible instrument set cannot make look good.

12 Missing Data

Data set like:

y	x	z	w	q
1	4	2	?	?
4	?	1	2	?
2	?	?	?	?
?	2	1	?	?
?	?	?	?	?

12.1 Different cases under different assumptions

1. Missing completely at Random (MCAR): the occurrence of missing data is not related to the missing value, the values of any other variables, or the pattern of missingness in other variables. Too good to be true situation.
2. Missing at Random (MAR): the occurrence of missing values for a variable is random, contingent on the value or missingness of observable variables. Or, the missingness can be modeled.
3. Missing Not at Random (MNAR): the occurrence of missing values is systematically related to the unobserved value itself, even after conditioning on observed data. This is the hardest case: there is no model-free fix, but it is not hopeless. It can be addressed with selection models, pattern-mixture models, or sensitivity analysis, all of which rely on untestable assumptions about the missingness mechanism.

The MAR case is the one we are interested in.

For MCAR: you lose efficiency if you simply drop the data. For MAR: if you don't model it, you generally suffer bias as well as efficiency.

"Generally" is doing work in that sentence, because the consequence depends on the analysis and not only on the mechanism. Complete-case regression of y on X stays consistent for β when missingness depends only on X – which is MAR, not MCAR – because dropping rows on the basis of X leaves the conditional distribution of y given X untouched. What breaks consistency is missingness that depends on y itself, or on something unobserved, after conditioning on X. So MAR is not automatically fatal and MCAR is not the only safe case. Which it is depends on what is missing and on what you are trying to estimate.

If we model missingness, for MCAR, you gain efficiency. For MAR, you correct bias and gain efficiency.

12.2 Old methods

1. Listwise deletion
2. Mean imputation.

Problems: * understates variability in the imputed variable * does not recover associations between variables.

So standard errors are in general under-estimated.

3. Regression-based imputation

Use a model such as $x = \alpha_0 + \alpha_1 z + \alpha_2 y$. But that introduce extra noise that are not accounted for (similar to generated variable problem).

4. Interpolation of panel data

That is, to use the observation from last period or a linear interpolation for the same unit.

12.3 Modern methods

- Account for uncertainty in imputed variable
- Use a model to predict missing observation.
- Instead of picking one, pick many
- uncertainty is represented by VCV matrix of the coefficients used to predict missing values.

12.3.1 Basic ideas

Imputation model: $x = \alpha_0 + \alpha_1 z + \alpha_2 y$

Main model: $y = \beta_0 + \beta_1 x + \beta_2 z$

1. Pick M values of α out of the asymptotic distribution, the multivariate normal $N(\hat{\alpha}, \hat{\Sigma})$, using the estimate $\hat{\alpha}$ and its VCV $\hat{\Sigma}$ as the mean and VCV of the distribution.
2. For each of the M draws, predict the missing values *and add a random residual* drawn from the imputation model's estimated error distribution, creating M data sets. Both draws are needed: the draw of α carries parameter uncertainty, and the residual draw carries the irreducible scatter around the prediction. Dropping the residual draw gives what is called "improper" imputation, and reintroduces precisely the understated-variability problem that sank mean imputation and single regression imputation above.

A fully proper procedure draws the residual variance too. Step 1 as written holds $\hat{\Sigma}$, and with it $\hat{\sigma}^2$, fixed at its estimate, so it propagates uncertainty about α but not about σ^2. The Bayesian version draws $\sigma^{2(m)}$ from its posterior – for the normal linear model a scaled inverse-χ^2 – then draws $\alpha^{(m)}$ from $N(\hat{\alpha}, \sigma^{2(m)}(\mathbf{X}'\mathbf{X})^{-1})$, and finally the residual from $N(0, \sigma^{2(m)})$. In large samples the difference is slight. In small ones, fixing σ^2 understates the between-imputation variance.

3. calculate M new estimates and combine them by averaging, $\tilde{\beta} = \frac{1}{M}\sum_{m=1}^{M}\tilde{\beta}_m$, using each of the imputed data sets. then calculate the variance of $\tilde{\beta}$:

$V_\beta = W + (1 + 1/M)B$

where $W = \frac{1}{M}\sum_{m=1}^{M} s_m^2$, $B = \frac{1}{M-1}\sum_{m=1}^{M}(\tilde{\beta}_m - \tilde{\beta})^2$, in other words, the within-imputation and between-imputation variation. The standard error is $\sqrt{V_\beta}$. These formulas (Rubin's rules) are written here for a single scalar coefficient β; for the full coefficient vector, W is the average of the within-imputation covariance matrices, B is the between-imputation covariance matrix $\frac{1}{M-1}\sum_m(\tilde{\beta}_m - \tilde{\beta})(\tilde{\beta}_m - \tilde{\beta})'$, and $V_\beta = W + (1 + 1/M)B$ is the total covariance matrix.

12.3.2 MI through Chained Equations (MICE) (by Buuren)

1. Discard observations with all missing.
2. Fill in the missing data with random draws from the observed values.
3. Move through the columns and perform single-variable imputation using some method.
4. Replace the original replacements with the fitted replacements. Repeat 3 for a large number of times, or with a convergence criteria.
5. Do 1-4 m times to create m imputed data sets.

Many ways to implement MICE.

- Regression (linear, logit, multinomial), get $\hat{w}$ or $f(\hat{w})$ sample.
- The default is Predictive Mean Matching (PMM)
  a. create predicted value
  b. pick a small set of donor cases that have the closest predicted values (the number of donors is software-specific; e.g., three or five).
  c. randomly choose one of those donors' observed values to impute.

12.3.3 Bayesian Data Augmentation

- MICE is Markov Chain
- Build a missing data model into a Bayesian model, treating missing values as another parameter to estimate by drawing out of its posterior distribution.

$$p(\beta, y_{mis} \mid y_{obs}) \propto p(y_{obs}, y_{mis} \mid \beta)\, p(\beta)$$

The complete-data model appears once, applied to the observed and missing components together, and the prior is over β alone. If the missingness mechanism is modelled, it enters as its own factor with its own parameters rather than being folded into the prior.

- We integrate out the missing values by sampling from the total distribution, then averaging out the beta distributions over the space of missing data points.
- Ignorability is what allows the missingness mechanism to be left out of that expression, and it asks for more than MAR. It needs MAR *and* distinct parameters – those of the missingness mechanism separate from β – and, in the Bayesian version, priors that factor accordingly. MAR on its own is not sufficient.

12.3.4 FIML

If there are no missing data, the likelihood function is

$L(\mu, \Sigma) = \prod_{i=1} f(y_i|\mu, \Sigma)$

If there are missing values, each unit contributes the density of the part of y_i that was actually observed:

$$L(\mu, \Sigma) = \prod_i f(y_{i,obs}\,;\,\mu_{i,obs},\,\Sigma_{i,obs,obs})$$

where $y_{i,obs}$ is unit i's observed subvector, and $\mu_{i,obs}$ and $\Sigma_{i,obs,obs}$ are the matching subvector of μ and submatrix of Σ. The parameters are the same μ and Σ for everyone; what varies by unit is which rows and columns get selected. Written this way the mechanism is visible – each contribution is the marginal density obtained by integrating out that unit's missing components, and no imputation happens anywhere.

12.3.5 mi in stata

```
log using mi_10_model2.log, replace

clear
set more off

* set matsize is obsolete since Stata 16 (matrix sizes are now dynamic)
* and can be safely omitted on modern Stata; kept here only because the
* original script targeted an older Stata version.
set matsize 4000

use "CROSSED_ContestUserActivity FEB 2014 temp.dta", clear
*sample 10

mi set flong
*local dummies "x1 x2"
local continuous "CulturalDist_KS5D_ctr tight_ctr targ_tight_ctr user_country_opennessValue_ct

mi register imputed `continuous'
mi impute chained  (regress) `continuous' , add(10) rseed(1)
```

```
mi estimate, cmdok: heckprob has_won c.CulturalDist_KS5D_ctr##c.tight_ctr targ_tight_ctr gende
```

This is a sample code. In stata, you have to do “mi set”, then “mi register” to register variables you need to impute. Then “mi impute chained” if you have multiple variables to impute. Then the imputation will take all variables you registered (imputed or regular) in the prediction model unless you specify otherwise. Then do “mi estimate”.

13 Interpreting OLS: Effect Weights

An OLS regression returns one number, $\hat{\beta}$. What that number estimates when effects are heterogeneous, and how each observation contributes to it, depends on the geometry of the sample and on what else is in the model. This chapter derives the implicit weights OLS uses, and follows them across cross-sectional and panel settings.

Two companion notes work through related applications in causal inference: Weights in OLS in causal inference and OLS cannot be ATE.

13.1 OLS as a weighted average

Start with a simple regression of y on x with an intercept:

$$y_i = \alpha + \beta x_i + e_i \tag{13.1}$$

The OLS slope is the familiar ratio of a covariance to a variance:

$$\hat{\beta} = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2} \tag{13.2}$$

Now give each observation its own slope through the sample centroid $(\bar{x}, \bar{y})$:

$$s_i \equiv \frac{y_i - \bar{y}}{x_i - \bar{x}} \tag{13.3}$$

Write $d_i \equiv x_i - \bar{x}$ for the horizontal distance from the centroid, and $w_i \equiv d_i^2$. Multiply and divide each term of Equation 13.2 by d_i and it rearranges into

$$\hat{\beta} = \frac{\sum_{i=1}^{n} d_i^2 s_i}{\sum_{i=1}^{n} d_i^2} = \frac{\sum_{i=1}^{n} w_i s_i}{\sum_{i=1}^{n} w_i} \tag{13.4}$$

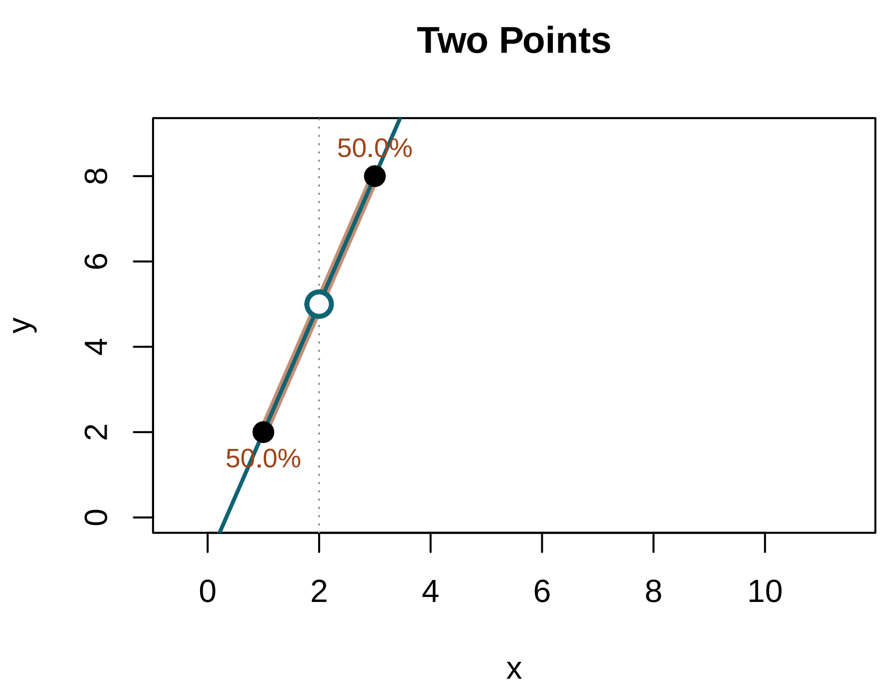
So $\hat{\beta}$ is an exact weighted average of the deviation slopes s_i, and the weight each observation carries is its squared distance from the mean of x.

13.1.1 What the weights look like

The weights are highly concentrated. Because they grow with the *square* of distance from $\bar{x}$, a few points far from the mean can dominate the estimate while most points near the mean contribute almost nothing. The examples below show how adding or moving a single observation redistributes the weights across the entire sample.

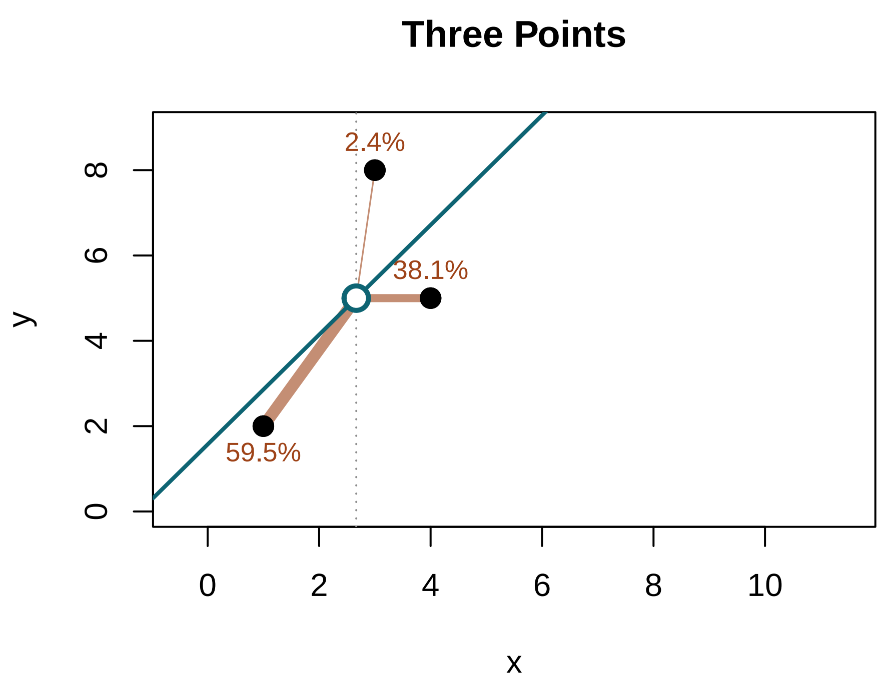
The figures draw the weights w_i of Equation 13.4 as the thickness of each ray from the centroid.

```
show <- function(x, y, main = "") {
  xb <- mean(x); yb <- mean(y); d <- x - xb; s <- (y - yb) / d; w <- d^2
  plot(x, y, pch = 16, cex = 1.5, xlim = c(-0.5, 11.5), ylim = c(0, 9),
       xlab = "x", ylab = "y", main = main)
  abline(v = xb, lty = 3, col = "grey55")
  segments(xb, yb, x, y, lwd = 0.6 + 9 * w / sum(w), col = "#9E431899")
  abline(lm(y ~ x), lwd = 2, col = "#0E6473")
  points(x, y, pch = 16, cex = 1.5)
  points(xb, yb, pch = 21, bg = "white", col = "#0E6473", lwd = 2.5, cex = 1.7)
  text(x, y, labels = sprintf("%.1f%%", 100 * w / sum(w)),
       pos = ifelse(y >= yb, 3, 1), cex = .85, col = "#9E4318")
}
show(c(1, 3), c(2, 8), "Two Points")
```



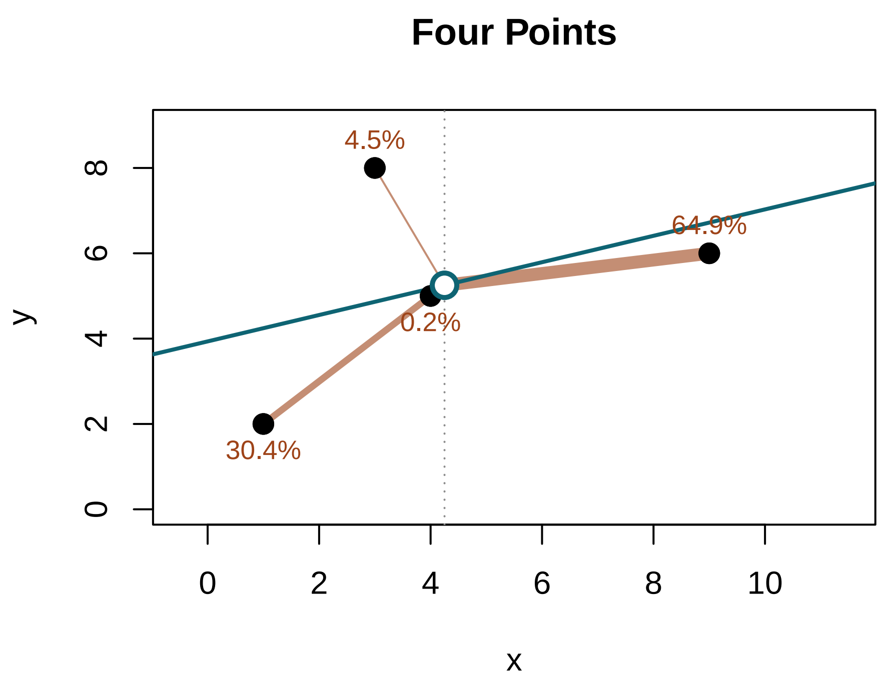
With two points the weights are equal, 50% each, and both s_i are just the line through the pair.

```
show(c(1, 3, 4), c(2, 8, 5), "Three Points")
```



Add a third point at $x = 4$ and $\bar{x}$ moves to 2.67. The point at $x = 3$ now sits almost on top of $\bar{x}$, so it takes only 2.4% of the weight even though its deviation slope is the largest of the three ($s_2 = 9.0$).

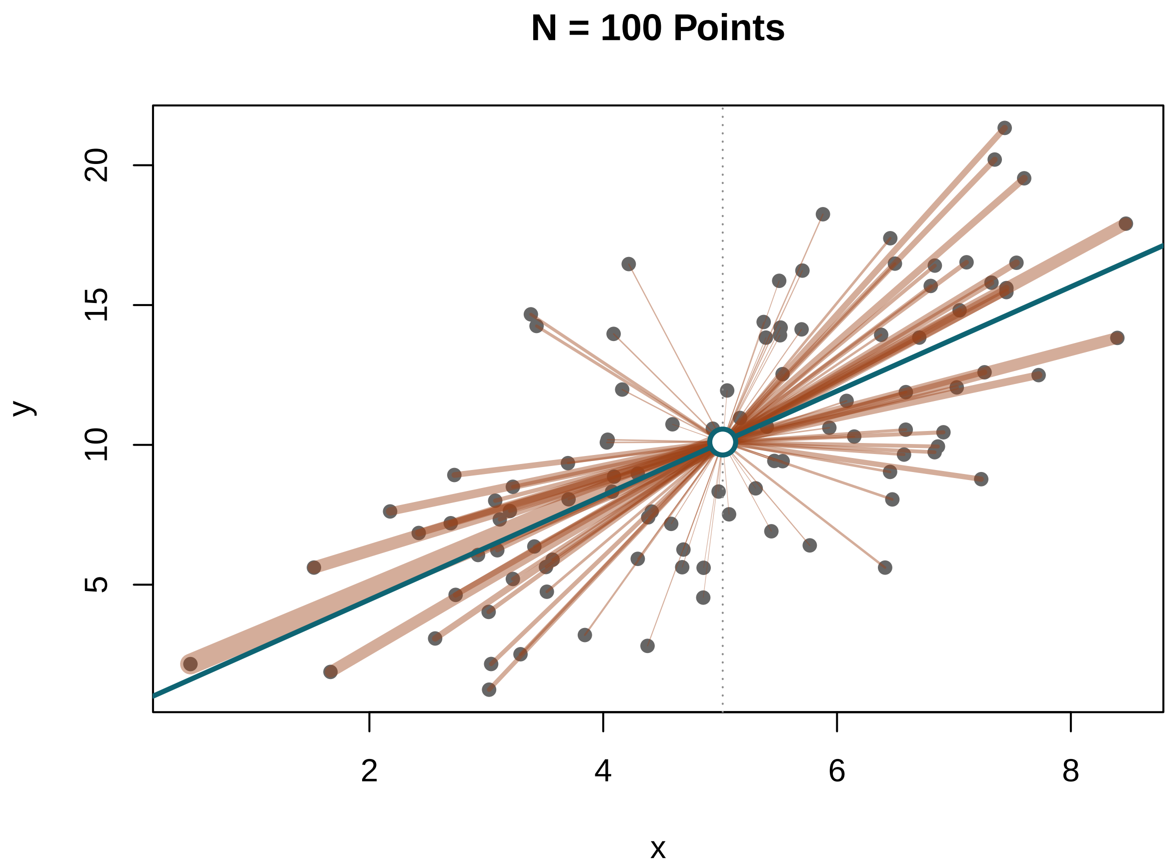
```
show(c(1, 3, 4, 9), c(2, 8, 5, 6), "Four Points")
```



Add an extreme point at $x = 9$ and $\bar{x}$ moves to 4.25. That one observation now carries 65% of the total weight, while $x = 4$ falls to 0.18% because it has landed near the new centroid. The weights are relative to the whole sample: add data anywhere and the weights everywhere change.

Simulation — a 100-point sample. The concentration does not wash out in larger samples. We draw $n = 100$ points, $x_i \sim N(5, 2^2)$ and $y_i = 2x_i + \varepsilon_i$ with $\varepsilon_i \sim N(0, 3^2)$, and compare the weight held by the ten points nearest $\bar{x}$ against the ten furthest.

```
set.seed(3)
n <- 100
x <- rnorm(n, 5, 2)
y <- 2 * x + rnorm(n, 0, 3)

xb <- mean(x); yb <- mean(y); w <- (x - xb)^2
o <- order(w, decreasing = TRUE)

plot(x, y, pch = 16, cex = 1, col = "grey40", xlab = "x", ylab = "y", main = "N = 100 Points")
abline(v = xb, lty = 3, col = "grey55")
segments(xb, yb, x, y, lwd = 0.3 + 10 * w / max(w), col = "#9E431870")
abline(lm(y ~ x), lwd = 2.5, col = "#0E6473")
points(xb, yb, pch = 21, bg = "white", col = "#0E6473", lwd = 2.5, cex = 1.8)
```




```
vtab(c(`Weight 10 Nearest Mean (%)` = 100 * sum(w[o[91:100]]) / sum(w),
       `Weight 10 Furthest (%)`     = 100 * sum(w[o[1:10]]) / sum(w),
       `OLS Estimate`                = coef(lm(y ~ x))[["x"]]), 3)
```

	value
Weight 10 Nearest Mean (%)	0.138
Weight 10 Furthest (%)	35.360
OLS Estimate	1.865

The 10% of points furthest from $\bar{x}$ hold 35% of the weight; the 10% closest hold 0.14%. A point sitting exactly at $\bar{x}$ has an undefined s_i and zero weight: the wild value arrives paired with a weight that annihilates it.

13.2 Deviation slopes s_i versus true effects b_i

Now let each unit have its own structural slope b_i:

$$y_i = \alpha + b_i x_i + e_i \tag{13.5}$$

b_i is a parameter we never see. s_i from Equation 13.3 is computed from the data. The two are not the same object, we observe s_i, but are interested in b_i, or function of b_i.

13.2.1 Decomposing s_i

Let $\bar{b} = n^{-1}\sum_i b_i$ and $C = n^{-1}\sum_i (b_i - \bar{b})(x_i - \bar{x})$, the sample covariance between b and x. Substituting $x_i = d_i + \bar{x}$ into Equation 13.5,

$$s_i = b_i + \frac{\bar{x}(b_i - \bar{b}) - C}{d_i} + \frac{e_i - \bar{e}}{d_i} \tag{13.6}$$

The middle term is the trouble. When effects are heterogeneous ($b_i \neq \bar{b}$) it does not vanish, and dividing it by d_i makes s_i blow up for points near $\bar{x}$ — before any noise enters. Taking expectations over e_i,

$$E[s_i \mid x, b] = b_i + \frac{\bar{x}(b_i - \bar{b}) - C}{d_i} \neq b_i \tag{13.7}$$

So s_i is biased for b_i, observation by observation.

13.2.2 Aggregating up to $\hat{\beta}_{OLS}$

Take the d_i^2-weighted expectation of s_i, combining Equation 13.4 and Equation 13.7:

$$E[\hat{\beta}_{OLS} \mid x, b] = \frac{\sum_i d_i^2 E[s_i \mid x, b]}{\sum_i d_i^2} = \frac{\sum_i d_i^2 b_i + \bar{x}\sum_i d_i(b_i - \bar{b}) - C\sum_i d_i}{\sum_i d_i^2} \tag{13.8}$$

Because $\sum_i d_i = 0$ and $\sum_i d_i(b_i - \bar{b}) = nC$, the numerator collapses:

$$\sum_i d_i^2 E[s_i \mid x, b] = \sum_i d_i^2 b_i + n\bar{x}C = \sum_i b_i d_i(d_i + \bar{x}) = \sum_i b_i d_i x_i \tag{13.9}$$

And since $\sum_i d_i^2 = \sum_i d_i x_i$, we get the exact expectation:

$$E[\hat{\beta}_{OLS} \mid x, b] = \frac{\sum_{i=1}^n d_i x_i b_i}{\sum_{i=1}^n d_i x_i} = \frac{\sum_{i=1}^n (x_i - \bar{x}) x_i b_i}{\sum_{i=1}^n (x_i - \bar{x})^2} \tag{13.10}$$

So the OLS estimator is unbiased for a weighted average of b_i's.

13.2.3 So what?

Equation 13.10 has three consequences for practice.

1. **OLS is not the ATE under heterogeneity.** $\hat{\beta}_{OLS}$ is not the plain average effect $\bar{b} = E[b_i]$. It is a weighted parameter whose weights are $d_i x_i = (x_i - \bar{x})x_i$ — signed, not squared — and it equals the ATE only when the b_i are uncorrelated with those weights.
2. **Leverage annihilates the individual bias.** Equation 13.7 shows s_i is a hopeless estimate of b_i precisely where $d_i \approx 0$. The weight $w_i = d_i^2$ zeroes out exactly those unstable points, turning biased observation-level slopes into the exact parameter of Equation 13.10.
3. **The same mechanism runs through the causal estimators.** With a binary regressor it puts the most weight where treatment is near a coin flip (Angrist 1998); grouped by treatment status it reverses the propensity weights, favouring the smaller group (Słoczyński 2022). Both are Equation 13.10 read at different granularities, as the rest of the chapter shows.

Simulation — deviation slopes versus true effects. We check the algebra directly, drawing $n = 2000$ with heterogeneous slopes $b_i \sim N(1, 0.5^2)$, regressor $x_i \sim N(0, 2^2)$, and $y_i = b_i x_i + \varepsilon_i$, $\varepsilon_i \sim N(0, 2^2)$.

```
set.seed(6)
n <- 2000
b <- rnorm(n, 1, 0.5)
x <- rnorm(n, 0, 2)
y <- b * x + rnorm(n, 0, 2)
xb <- mean(x); d <- x - xb; w <- d^2
s  <- (y - mean(y)) / d

mtab(cbind(value = c(
  `$\\mathrm{cor}(s_i, b_i)$, Pearson`    = cor(s, b),
  `$\\mathrm{cor}(s_i, b_i)$, Spearman`   = cor(s, b, method = "spearman"),
  `$\\mathrm{sd}(s_i)$`                   = sd(s),
  `weighted average of $s_i$`       = sum(w * s) / sum(w),
  `OLS Estimate`                = unname(coef(lm(y ~ x))[2]),
  `$E[\\beta \\mid x, b]$ (eq-exact)` = sum(d * x * b) / sum(d * x))), 3)
```

	value
$\mathrm{cor}(s_i, b_i)$, Pearson	0.019
$\mathrm{cor}(s_i, b_i)$, Spearman	0.258
$\mathrm{sd}(s_i)$	117.420
weighted average of s_i	0.950
OLS Estimate	0.950
$E[\beta \mid x, b]$ (eq-exact)	0.991

The s_i are wildly dispersed ($\mathrm{sd}(s_i) > 100$) and barely correlated with the true b_i — Pearson 0.02, though the rank correlation is 0.26, higher once the explosive tails near $\bar{x}$ are set aside. Their leverage-weighted average equals $\hat{\beta}_{OLS}$ exactly; that is Equation 13.4, an identity. The same leverage applied to the true b_i gives 0.99, the value $\hat{\beta}$ estimates *in expectation* (Equation 13.10) — and this sample's $\hat{\beta} = 0.95$ sits about two standard errors below it. The gap between 0.95 and 0.99 is ordinary sampling variation, not a failure of the weighting identity. Equation 13.10 pins down what $\hat{\beta}$ converges to across repeated samples; any single draw fluctuates around that target, and 0.95 versus 0.99 is well within the range one would expect. The structural relationship between the weights and the true slopes is exact and permanent; it is the realisation, not the identity, that is noisy.

Noise is not what distorts s_i. Turn it off entirely, set $y_i = b_i x_i$, and ask how well each s_i tracks its own b_i:

```
y0 <- b * x
s0 <- (y0 - mean(y0)) / (x - xb)
vtab(c(`$\\mathrm{cor}(s_i, b_i)$, no noise` = cor(s0, b),
       `weighted average of $s_i$`     = sum(w * s0) / sum(w),
       `weighted average of $b_i$` = sum(w * b) / sum(w),
       `OLS, noiseless`          = unname(coef(lm(y0 ~ x))[2])), 4)
```

	value
$\mathrm{cor}(s_i, b_i)$, no noise	0.1436
weighted average of s_i	0.9914
weighted average of b_i	0.9919
OLS, noiseless	0.9914

The correlation stays at 0.14 with no noise in the data at all. The reason is visible in Equation 13.6: even without the error term, s_i contains $\bar{x}(b_i - \bar{b})/d_i$, which depends on every unit's slope through $\bar{b}$ and on every unit's position through $\bar{x}$. A ray from the centroid is anchored at $(\bar{x}, \bar{y})$, and $\bar{y}$ shifts when any other unit's slope changes. So s_i is a function of the entire sample, not of unit i alone, and heterogeneity in the b_i makes it impossible for any single ray to isolate its own unit's effect. Adding noise inflates the variance of s_i further through the $(e_i - \bar{e})/d_i$ term, but the low correlation is already present without it.

13.2.4 Under homogeneity, the weights buy precision

If $b_i = b$ for all i, Equation 13.6 reduces to $s_i = b + (e_i - \bar{e})/d_i$. With homoskedastic errors $\mathrm{Var}(e_i) = \sigma^2$,

$$E[s_i \mid X] = b, \qquad \mathrm{Var}(s_i \mid X) = \frac{\sigma^2(1 - 1/n)}{d_i^2} \propto \frac{1}{d_i^2} \tag{13.11}$$

Now $w_i = d_i^2$ is exactly the inverse-variance weight. Any weighting of the s_i — equal, d_i^2, anything — is unbiased for b; what the weights buy is precision, not correctness, and d_i^2 is the choice that minimises variance. That is OLS as BLUE. The argument is conditional on X, where d_i is a known constant and the weights are functions of x alone.

13.3 The general rule: FWL

The panel chapter met two weightings — by T_i when a panel is unbalanced, and by variance when a regression mixes within and between variation.

Regress y on a regressor of interest V and a control matrix X, and let $\tilde{V}$ be the residual from projecting V on X. By the Frisch–Waugh–Lovell theorem,

$$\hat{\beta} = \frac{\sum_{i=1}^{n} \tilde{V}_i y_i}{\sum_{i=1}^{n} \tilde{V}_i^2} \tag{13.12}$$

Be careful what $\tilde{V}_i$ is. A single observation has no variation of its own; variation belongs to a collection. What it has is a *deviation* — how far its regressor sits from what the rest of the model predicts for it. With only an intercept, that is distance from the grand mean. With unit fixed effects, it is distance from the unit's own mean. Which mean gets subtracted is the most consequential detail in this section, and we come back to it.

Equation 13.12 reads two ways. This chapter takes the first; the second opens the companion chapter Interpreting OLS: Outcome Weights.

Weights on effects. Under a heterogeneous model $y_i = \beta_i V_i + X_i\gamma + e_i$, substituting into Equation 13.12 gives

$$\hat{\beta} = \frac{\sum_{i=1}^{n} w_i \beta_i}{\sum_{i=1}^{n} w_i} + \frac{\sum_{i=1}^{n} \tilde{V}_i e_i}{\sum_{i=1}^{n} \tilde{V}_i^2}, \qquad w_i = \tilde{V}_i V_i \tag{13.13}$$

The control term drops out because $\tilde{V}$ is orthogonal to X by construction. The error term does not. It vanishes in conditional expectation when $E[e_i \mid V, X, \beta] = 0$, and in probability under the usual orthogonality and large-sample conditions. That is when the weighted-average reading is the right one. The outcome-weight identity in the companion chapter carries no such remainder — it is exact in every sample.

The denominator $\sum_i w_i = \sum_i \tilde{V}_i V_i = \sum_i \tilde{V}_i^2 \geq 0$ always, by OLS orthogonality. Individual weights $w_i = \tilde{V}_i V_i$ can be negative — they are whenever $\tilde{V}_i$ and V_i have opposite signs.

13.3.1 Where the square comes from

Substitute $y_i = \beta_i V_i + X_i\gamma + e_i$ into the numerator of Equation 13.12 and it becomes $\sum_i \beta_i \tilde{V}_i V_i + \sum_i \tilde{V}_i e_i$; setting the error term aside as above, over the whole sample,

$$\sum_{i=1}^{n} \tilde{V}_i V_i = \sum_{i=1}^{n} \tilde{V}_i^2 \tag{13.14}$$

because $\tilde{V}$ is orthogonal to the fitted part of V.

In a panel, several rows share a unit, and it is natural to ask whether the same identity holds *within a single unit* — summed over that unit's own t's — and not only over the whole sample. Write V_{it} for the observation and $\bar{V}_i$ for the unit's own mean. With unit fixed effects $\tilde{V}_{it} = V_{it} - \bar{V}_i$, and expanding the product gives the within-unit version directly,

$$\sum_{t=1}^{T}(V_{it} - \bar{V}_i)V_{it} = \sum_{t=1}^{T} V_{it}^2 - T\bar{V}_i^{\,2} = \sum_{t=1}^{T}(V_{it} - \bar{V}_i)^2 \tag{13.15}$$

13.4 Two causal decompositions: Angrist and Słoczyński

Equation 13.13 says the coefficient is a weighted average of effects with weights $w_i = \tilde{V}_i V_i$; everything below only evaluates those weights. Angrist reads them at the stratum level. Słoczyński reads the same weights one level up, grouped by treatment status, where they come out reversed relative to the natural shares of the ATE.

Let $D \in \{0, 1\}$ be a binary treatment and X discrete strata $x \in \{1, \dots, K\}$, with the controls saturated — one indicator per stratum, so no functional form is imposed within cells. Discreteness is essential: saturated controls require a finite number of cells, so continuous X cannot be saturated. With continuous controls the linear projection $\hat{D}(X)$ is not the cell proportion — it is a linear approximation that can fall outside $[0, 1]$ — and the stratum-by-stratum decomposition below does not go through. With continuous X one could bin it, but the result then holds for the chosen bins, not for X itself.

13.4.1 Angrist (1998): weighting by stratum

Residualising D_i on a full set of stratum indicators gives $\tilde{D}_i = D_i - \hat{D}(x)$, where $\hat{D}(x) = E[D \mid X = x]$. Saturation is what makes this the true cell proportion: the OLS projection on a complete set of dummies recovers the conditional mean exactly. Sum the effect weights $w_i = \tilde{D}_i D_i$ within a stratum:

$$w_x = \sum_{i \in x} \tilde{D}_i D_i = \sum_{i \in x} D_i^2 - \hat{D}(x) \sum_{i \in x} D_i = n_x \hat{D}(x) - n_x \hat{D}(x)^2 = n_x \hat{D}(x)(1 - \hat{D}(x)) \tag{13.16}$$

Nothing new has been added: it is $w_i = \tilde{V}_i V_i$ from Equation 13.13 with a binary V, where $D_i^2 = D_i$ makes the sum collapse. The maximum at $\hat{D}(x) = 1/2$ is just the fact that a binary variable has most variance when it is a coin flip.

Note what plays the role of s_i here. A single observation with a binary regressor has no slope; you need one treated and one control unit to form a difference. The smallest unit carrying an effect is therefore the *stratum*, and s_i becomes the stratum difference in means.

A word on the labels before going further. $\hat{\delta}_x$ is a difference in *observed* means within a stratum. Calling the aggregate an ATT or an ATE needs the usual identification conditions: conditional exchangeability, so that treatment is as good as randomly assigned within a stratum, and overlap, so that each stratum holds both treated and control units. Saturation delivers the algebra, not the interpretation. Without those conditions the formulas below still hold exactly, but they describe weighted averages of observed contrasts rather than causal effects. The weighting is what OLS does; whether the thing being weighted is an effect is settled outside the regression.

Let $\hat{\delta}_x$ be the stratum difference-in-means. Substituting Equation 13.16 into Equation 13.13,

$$\hat{\beta}_{OLS} = \frac{\sum_{x=1}^{K} n_x \hat{D}(x)(1-\hat{D}(x))\,\hat{\delta}_x}{\sum_{x=1}^{K} n_x \hat{D}(x)(1-\hat{D}(x))} \tag{13.17}$$

so strata with propensity near $\hat{D}(x) = 0.5$ get the most weight per observation. A stratum that contains only treated or only control units has $\hat{D}(x) = 1$ or 0, so $w_x = 0$: it drops out entirely, because there is no within-stratum comparison to make. A linear model with continuous controls would extrapolate into those strata instead of dropping them.

13.4.2 Słoczyński (2022): aggregating to two groups

Angrist sorts by covariate stratum. Słoczyński sorts by treatment status instead, collapsing everything into two numbers — the effect on the treated and the effect on the untreated. Aggregating the stratum effects $\hat{\delta}_x$ of Equation 13.17 with each stratum's count of treated or untreated units gives

$$\text{ATT} = \frac{\sum_x n_x \hat{D}(x)\,\hat{\delta}_x}{\sum_x n_x \hat{D}(x)}, \qquad \text{ATU} = \frac{\sum_x n_x (1-\hat{D}(x))\,\hat{\delta}_x}{\sum_x n_x (1-\hat{D}(x))} \tag{13.18}$$

and the ATE combines them with their natural shares, $\rho = P(D = 1)$. To see this, write ATE $= E[\tau_i]$ and condition on treatment status:

$$E[\tau_i] = P(D=1)\,E[\tau_i \mid D=1] + P(D=0)\,E[\tau_i \mid D=0] = \rho\,\text{ATT} + (1-\rho)\,\text{ATU}.$$

In the finite-sample, stratum-weighted version the same decomposition holds because $\sum_x n_x \hat{D}(x) = n_1$ and $\sum_x n_x (1-\hat{D}(x)) = n_0$, so

$$\text{ATE} = \rho\,\text{ATT} + (1-\rho)\,\text{ATU} \tag{13.19}$$

The OLS weights of Equation 13.17 go as the propensity *variance* $\hat{D}(x)(1-\hat{D}(x))$; the group weights of Equation 13.18 go as the propensity itself or its complement. The relation between the two is Słoczyński's result. Write $p = \hat{D}(X)$ for the cell propensity. Because the design is saturated — the controls are a full set of stratum dummies — residualising D on the strata leaves exactly $\tilde{D} = D - p$, so the coefficient is $E[\tilde{D}\,Y]/E[p(1-p)]$. Since p is constant within a stratum, Y enters only through the group means $m_j(p) = E[Y \mid p, D = j]$, and, using $\tilde{D} = 1 - p$ for treated and $-p$ for untreated units,

$$\beta_{OLS} = \frac{\rho\,E[(1-p)\,m_1(p) \mid D=1] - (1-\rho)\,E[p\,m_0(p) \mid D=0]}{E[p(1-p)]} \tag{13.20}$$

Substitute the within-group linear projections $m_j(p) = \alpha_j + \gamma_j p + \varepsilon_j(p)$. The residual term $E[p(1-p)(\varepsilon_1 - \varepsilon_0)]/E[p(1-p)]$ vanishes when the group means are linear in p, leaving

$$\beta_{OLS} = (\alpha_1 - \alpha_0) + (\gamma_1 - \gamma_0)\,\frac{E[p^2(1-p)]}{E[p(1-p)]} \tag{13.21}$$

The last ratio is moment algebra — an identity in the first three moments of p, exact for any distribution. With $\mu_j = E[p \mid D = j]$, $V_j = \mathrm{Var}(p \mid D = j)$, and w_1, w_0 the weights defined in Equation 13.23 below,

$$\frac{E[p^2(1-p)]}{E[p(1-p)]} = w_1\mu_1 + w_0\mu_0 \tag{13.22}$$

holds identically in the distribution of p. Under linearity $\mathrm{ATT} = (\alpha_1 - \alpha_0) + (\gamma_1 - \gamma_0)\mu_1$ and $\mathrm{ATU} = (\alpha_1 - \alpha_0) + (\gamma_1 - \gamma_0)\mu_0$, so Equation 13.20 through Equation 13.22 combine into the main result,

$$\hat{\beta}_{OLS} = w_1\,\mathrm{ATT} + w_0\,\mathrm{ATU}, \qquad w_1 = \frac{(1-\rho)V_0}{\rho V_1 + (1-\rho)V_0}, \quad w_0 = 1 - w_1 \tag{13.23}$$

where V_j is the variance of the propensity score within group j. This is exact for the effects of the within-group linear projections; for ATT and ATU it is exact when the group means are linear in p and an approximation otherwise.

Two things about Equation 13.23 matter. First, the weight w_1 on ATT falls as the treated share ρ rises: the larger the treated group, the smaller its say. That is the whole of "smaller groups get more weight." Second, when the propensity has equal variance in the two groups, $V_1 = V_0$, the weights collapse to the reversed shares,

$$\hat{\beta}_{OLS} \approx (1-\rho)\,\mathrm{ATT} + \rho\,\mathrm{ATU} \tag{13.24}$$

the same two quantities as the ATE in Equation 13.19, with the weights **swapped**. At $\rho = 0.1$ the coefficient puts 0.9 on the effect for the treated tenth; the ATE puts 0.1 there.

One caveat. Equation 13.24 is exact under equal propensity variance across groups — not "no covariates," though no covariates is one trivial way to get there. Słoczyński calls the condition unlikely to hold exactly and offers ρ as a rule of thumb, cleanest near $\rho = 0\%$, 50%, or 100%.

Simulation — Angrist and Słoczyński weights. We draw $n = 400{,}000$ units in five covariate strata ($X \in \{1, \ldots, 5\}$), each with its own treatment effect $\tau_x = (0, 1, 2, 3, 4)$, so the plain ATE is 2. The outcome is $y = \tau_x D + X + \varepsilon$. We run the simulation twice with different propensity schedules: once with the treated share near half ($\rho \approx 0.5$, propensities ranging from 0.08 to 0.93 across strata) and once with few treated ($\rho \approx 0.09$, propensities from 0.02 to 0.20). The point is to show that the OLS coefficient moves with the treated share even though the stratum effects do not change.

```
set.seed(21)
n <- 400000
X <- sample(1:5, n, TRUE)

compare <- function(px) {
  d  <- rbinom(n, 1, px[X]); ta <- c(0, 1, 2, 3, 4)[X]
  yy <- ta * d + X + rnorm(n)
  dt <- resid(lm(d ~ factor(X)))
  nx <- as.vector(table(X)); dh <- as.vector(tapply(d, X, mean))
  DIM <- as.vector(tapply(yy[d == 1], X[d == 1], mean) - tapply(yy[d == 0], X[d == 0], mean))
  rho <- mean(d); ATT <- weighted.mean(ta, d); ATU <- weighted.mean(ta, 1 - d)
  c(`$\\rho$`                        = rho,
    `$\\max\\lvert\\text{weight diff}\\rvert$`         = max(abs(as.vector(tapply(dt * d, X, su
    `OLS Saturated`             = unname(coef(lm(yy ~ d + factor(X)))[["d"]]),
    `Angrist Weighted`          = sum(nx*dh*(1-dh) * DIM) / sum(nx*dh*(1-dh)),
    `Sloczynski Approx`         = (1 - rho) * ATT + rho * ATU,
    `ATE`                       = mean(ta))
}
res <- cbind(`$\\rho \\approx 0.5$`  = compare(c(.08, .20, .50, .75, .93)),
             `$\\rho \\approx 0.09$` = compare(c(.02, .04, .08, .12, .20)),
             `$\\rho \\approx 0.91$` = compare(c(.80, .88, .92, .96, .98)))
mtab(res, 4)
```

	$\rho \approx 0.5$	$\rho \approx 0.09$	$\rho \approx 0.91$
ρ	0.4913	0.0909	0.9069
$\max\lvert\text{weight diff}\rvert$	0.0000	0.0000	0.0000
OLS Saturated	2.0121	2.8740	1.1262
Angrist Weighted	2.0121	2.8740	1.1262
Sloczynski Approx	2.0293	2.8577	1.1403
ATE	1.9978	1.9978	1.9978

The two weight vectors — the stratum sums $\sum \tilde{D}_i D_i$ and Angrist's $n_x \hat{D}(x)(1-\hat{D}(x))$ — are the same, below 10^{-9} (the `$\\max\\lvert\\text{weight diff}\\rvert$` row, 0 to four decimals), and the Angrist-weighted average reproduces the OLS coefficient exactly. With $\rho \approx 0.5$ the coefficient is 2.012 against an ATE of 1.998: close, because near-equal treated and control shares are where the weighting does least. Push the treated share to $\rho \approx 0.09$ and the coefficient jumps to 2.874; push it the other way to $\rho \approx 0.91$ and it falls. The two extremes are symmetric: at $\rho = 0.09$ OLS overweights ATT, at $\rho = 0.91$ it overweights ATU. Nothing about the effects changed. Only the treated share did.

Why does OLS overweight the treated when they are few? The FWL residual is $\tilde{D}_i = D_i - \hat{D}(x)$. When ρ is small, $\hat{D}(x)$ is small in most strata: a treated unit has $\tilde{D}_i = 1 - \hat{D}(x) \approx 1$, a control unit has $\tilde{D}_i = -\hat{D}(x) \approx 0$. Treated units are the unusual ones — they sit far from the stratum mean of D — so they carry most of the leverage, the same way a point far from $\bar{x}$ dominates the simple regression. At $\rho = 0.09$ the reversed-share approximation puts weight $1 - \rho = 0.91$ on ATT and only 0.09 on ATU, while the ATE puts 0.09 on ATT and 0.91 on ATU. In observational data where the treated group is small — a common setting — the OLS coefficient can be far from the population average effect even under unconfoundedness, purely because of the weighting.

The reversed-share Słoczyński approximation $(1-\rho)\,\text{ATT} + \rho\,\text{ATU}$ gives 2.029 against the exact 2.012, and 2.858 against 2.874 — within about a percent, but the exact weights of Equation 13.23 are what to report.

13.4.3 The three, side by side

Decomposition	Sorted by	Smallest effect unit	Exact when	Reference
Observation rule	nothing	s_i	always (identity)	Equation 13.4
Angrist (1998)	covariate stratum x	stratum difference $\hat{\delta}_x$	saturated controls	Equation 13.17
Słoczyński (2022)	treatment status D	group averages ATT, ATU	equal propensity variance	Equation 13.24

Read down the last column. The observation identity always holds and says little. Angrist's needs saturation and gives an interpretable weight. Słoczyński's needs more and gives the sharpest statement of the three: the coefficient and the ATE are the same two numbers with the weights reversed. Underneath all of them, Hazlett and Shinkre (2024) show these weights are exactly the bias from fitting a constant effect when effects vary, and argue for estimators that tolerate heterogeneity rather than for diagnosing the weights after the fact.

13.5 Within versus between, in panels

Pooled OLS mixes two sources of variation: within-unit, the deviations from a unit's own mean, and between-unit, the differences across unit means. Which one dominates is not chosen; it follows from the same rule. Take the within side first.

Nothing new happens here — it is the opening leverage rule with a different mean subtracted. Equation 13.12 weights every observation by its squared deviation $\tilde{V}_i^2$ from whatever the controls predict for it; Equation 13.13 weights each effect by $w_i = \tilde{V}_i V_i$. Simple OLS residualises against an intercept alone, so the deviation is from the grand mean and $w_i = (x_i - \bar{x})^2$, the leverage of Equation 13.4. Panel fixed effects residualises against unit dummies, so the deviation is from the unit's *own* mean and $w_i = \sum_t (x_{it} - \bar{x}_i)^2$ by Equation 13.15. Pooled OLS drops the dummies and is back to distance from the grand mean — which is exactly why the between comparison creeps back in. So the whole section is Equation 13.4 read with the unit mean in place of the grand mean.

13.5.1 Within-unit weighting: an extreme case

A fixed-effects estimate is identified by units whose x changes over time. Units whose x barely moves get near-zero weight regardless of their true effect. So the FE coefficient estimates the effect for the movers, not the population average. When the movers and the stayers have different effects, the two diverge.

The simulation below constructs the extreme version. Half the units barely move and have a true slope of zero; the other half move a lot and have a true slope of two. The average slope across all units is one. The DGP draws $N = 1000$ units for $T = 6$ periods, with $y_{it} = b_i x_{it} + \alpha_i + \varepsilon_{it}$, $\alpha_i \sim N(0, 1)$, $\varepsilon_{it} \sim N(0, 1)$. Both groups have $E[x_{it}] = 0$, so they differ only in how much x moves within a unit, not in its level.

Group	Units	sd(x_{it})	True slope b_i
Steady	1–500	0.2	0
Movers	501–1000	3	2

```r
set.seed(1)
half <- N / 2
g <- data.frame(id   = rep(1:N, each = T),
                move = rep(c(rep(0.2, half), rep(3, half)), each = T),
                b    = rep(c(rep(0,   half), rep(2, half)), each = T))
g$x <- rnorm(N * T, 0, g$move)
g$y <- g$b * g$x + rep(rnorm(N), each = T) + rnorm(N * T)

wg <- tapply(resid(feols(x ~ 1 | id, g))^2, g$id, sum)
mtab(cbind(value = c(
  `average of the two true slopes`      = mean(c(0, 2)),
  `FE estimate`                         = coef(feols(y ~ x | id, g))[["x"]],
  `Pooled OLS`                          = unname(coef(lm(y ~ x, g))[["x"]]),
  `weight held by the steady half (%)`  = 100 * sum(wg[1:half]) / sum(wg))), 3)
```

	value
average of the two true slopes	1.000
FE estimate	1.984
Pooled OLS	1.983
weight held by the steady half (%)	0.473

Both estimates are near 2, not the simple average of 1. The steady units hold under half of one percent of the FE weight: their six observations sit almost on top of their own unit mean, and squaring those tiny deviations puts them nowhere in the sum. Pooled OLS agrees because both groups have $E[x_{it}] = 0$ — there is no between-group difference in x levels, so $V_{between} = 0$ and pooled OLS collapses to β_{within}. Both estimators weight by variance, and the movers contribute $\sigma^2 = 9$ against the steady group's 0.04, so neither comes close to the simple average. The two estimators diverge only when the group means of x differ, creating a between channel with its own slope. The next subsection shifts the steady group's mean x away from zero and shows FE holding steady while pooled OLS collapses toward the between-slope.

13.5.2 Splitting the two channels

That result was about the *within-unit* deviation. Plain pooled OLS uses distance from the grand mean instead, and the two coincided above only because both halves were built around zero. Separate them and the algebra says exactly what happens.

Let G index groups — any number of them, any sizes — with shares p_g, group means $\mu_g = E[x \mid G = g]$ and $\bar{y}_g = E[y \mid G = g]$, within-group variances σ_g^2, and true slopes β_g. The pooled slope is plim $\hat{\beta}_{OLS} = \mathrm{Cov}(x, y)/\mathrm{Var}(x)$, and the whole derivation is one move: condition on G and split numerator and denominator each into a within piece and a between piece. The laws of total covariance and total variance do this exactly,

$$\mathrm{Cov}(x, y) = \underbrace{E[\mathrm{Cov}(x, y \mid G)]}_{\text{within}} + \underbrace{\mathrm{Cov}(E[x \mid G],\, E[y \mid G])}_{\text{between}} \tag{13.25}$$

$$\mathrm{Var}(x) = \underbrace{E[\mathrm{Var}(x \mid G)]}_{\text{within}} + \underbrace{\mathrm{Var}(E[x \mid G])}_{\text{between}} \tag{13.26}$$

Name each variance, and define each slope as its own covariance over its own variance:

$$V_{within} = E[\mathrm{Var}(x \mid G)], \qquad \beta_{within} = \frac{E[\mathrm{Cov}(x, y \mid G)]}{E[\mathrm{Var}(x \mid G)]} \tag{13.27}$$

$$V_{between} = \mathrm{Var}(E[x \mid G]), \qquad \beta_{between} = \frac{\mathrm{Cov}(E[x \mid G],\, E[y \mid G])}{\mathrm{Var}(E[x \mid G])} \tag{13.28}$$

With those definitions the within numerator of Equation 13.25 *is* $V_{within}\beta_{within}$ and the between numerator *is* $V_{between}\beta_{between}$ — no algebra, just the definitions. Dividing Equation 13.25 by Equation 13.26 therefore gives the pooled slope as a convex combination,

$$\mathrm{plim}\ \hat{\beta}_{OLS} = \frac{V_{within}\,\beta_{within} + V_{between}\,\beta_{between}}{V_{within} + V_{between}} \tag{13.29}$$

Nothing here used how many groups there are or that they are balanced; Equation 13.29 holds for any grouping — including $G =$ the panel unit, with N groups of sizes T_i. The only modelling input is $\mathrm{Cov}(x, y \mid G = g) = \beta_g \sigma_g^2$, that within a group the error is uncorrelated with x. It makes β_{within} an explicit share- and variance-weighted average of the group slopes, and leaves $\beta_{between}$ as the across-group regression of the group y-means on the group x-means:

$$\beta_{within} = \frac{\sum_g p_g \beta_g \sigma_g^2}{\sum_g p_g \sigma_g^2}, \qquad \beta_{between} = \frac{\sum_g p_g (\mu_g - \bar{\mu})(\bar{y}_g - \bar{\mu}_y)}{\sum_g p_g (\mu_g - \bar{\mu})^2} \tag{13.30}$$

with grand means $\bar{\mu} = \sum_g p_g \mu_g$ and $\bar{\mu}_y = \sum_g p_g \bar{y}_g$.

The two-equal-group case in the simulation below is where these collapse to closed forms. Equal shares turn the averages into $\frac{1}{2}$'s. The two group means then form a fair two-point variable, and such a variable taking values a_1, a_2 has variance $\frac{1}{4}(a_1 - a_2)^2$ — that is where the $\frac{1}{4}$ comes from — so

$$V_{within} = \tfrac{1}{2}(\sigma_1^2 + \sigma_2^2), \quad \beta_{within} = \frac{\beta_1\sigma_1^2 + \beta_2\sigma_2^2}{\sigma_1^2 + \sigma_2^2}, \quad V_{between} = \tfrac{1}{4}\Delta\mu^2, \quad \beta_{between} = \frac{\Delta\bar{y}}{\Delta\mu} \tag{13.31}$$

where $\Delta\mu = \mu_1 - \mu_2$ and $\Delta\bar{y} = \bar{y}_1 - \bar{y}_2$: with only two groups $\beta_{between}$ is a single rise over run between the centroids.

β_{within} is a variance-weighted average of the group slopes — the same σ^2 weighting as everywhere else in this chapter, one level up. And $\beta_{between}$ depends only on where the group centroids sit, so it can take any value at all, regardless of the within-group slopes.

The closed forms are special to two equal groups. With two *unequal* groups every $\frac{1}{2}$ becomes a share p_g and the $\frac{1}{4}$ becomes $p(1-p)$ — Angrist's weight again; with *more than two* groups $\beta_{between}$ is the genuine across-group regression slope of Equation 13.30, not a two-point ratio.

Fixed effects removes $V_{between}$ by demeaning and isolates β_{within}. As the groups pull apart, $\Delta\mu \to \infty$, $V_{between}$ swamps V_{within} and drags $\hat{\beta}_{OLS}$ toward $\beta_{between}$.

Simulation — pulling the groups apart. The DGP is the same as the within-unit weighting simulation: 500 steady units ($b_i = 0$, $\mathrm{sd}(x) = 0.2$) and 500 movers ($b_i = 2$, $\mathrm{sd}(x) = 3$). The one change is that the steady group's mean x is shifted from 0 to 10 or 100, while the movers stay centred at zero. This opens a gap $\Delta\mu$ between the two group centroids without changing any within-unit slope. The between-slope $\beta_{between} = \Delta\bar{y}/\Delta\mu$ is near zero, because the steady group's slope is zero and the movers' mean y is also near zero. So as $\Delta\mu$ grows, pooled OLS is dragged toward zero while FE, which removes the between channel by demeaning, stays at $\beta_{within} \approx 2$.

```
b_within <- (0 * 0.2^2 + 2 * 3^2) / (0.2^2 + 3^2)
V_within <- 0.5 * (0.2^2 + 3^2)
predicted <- function(shift) V_within * b_within / (V_within + 0.25 * shift^2)

park <- function(shift) {
  set.seed(1)
  half <- N / 2
  move <- c(rep(0.2, half), rep(3, half))
  b    <- c(rep(0,   half), rep(2, half))
  x    <- rnorm(N * T, 0, rep(move, each = T)) + rep(c(rep(shift, half), rep(0, half)), each =
  y    <- rep(b, each = T) * x + rep(rnorm(N), each = T) + rnorm(N * T)
  d    <- data.frame(id = rep(1:N, each = T), x = x, y = y)
  c(`FE estimate`       = coef(feols(y ~ x | id, d))[["x"]],
    `OLS Predicted`     = predicted(shift),
    `OLS Actual`        = coef(lm(y ~ x, d))[["x"]])
}

mtab(t(sapply(c(`shift 0` = 0, `shift 10` = 10, `shift 100` = 100), park)), 3)
```

	FE estimate	OLS Predicted	OLS Actual
shift 0	1.984	1.991	1.983
shift 10	1.984	0.305	0.311
shift 100	1.984	0.004	0.004

Fixed effects stays flat near the theoretical $\beta_{within} = 1.991$ wherever the steady group sits. Plain OLS falls from 1.983 to 0.004 on identical true slopes: as $\Delta\mu$ grows, the between weight $\frac{1}{4}\Delta\mu^2$ swamps the fixed within weight 4.52, and the coefficient is dragged toward $\beta_{between} = 0$ like $9/(4.52 + \Delta\mu^2/4)$, the numerator $\frac{1}{2}(\beta_1\sigma_1^2 + \beta_2\sigma_2^2) = 9$.

The figure below draws this directly. Two groups of 200 points each share the same within-group slopes: movers have $y = 2x+\varepsilon$ with $x \sim N(0, 3^2)$, the steady group has $y = \varepsilon$ with $x \sim N(\text{shift}, 0.2^2)$. Across the three panels the shift increases from 0 to 10 to 100, opening a gap $\Delta\mu$ between the group centroids without changing any within-group slope. The dashed line connects the two centroids (the between-slope), and the solid line is the pooled OLS fit. As the centroids separate, the between channel dominates and the pooled fit rotates from $\beta_{within} \approx 2$ toward $\beta_{between} \approx 0$.

```
set.seed(5)
m <- 200
draw <- function(shift, main) {
  xw <- rnorm(m, 0, 3);       yw <- 2 * xw + rnorm(m)
  xs <- rnorm(m, shift, 0.2); ys <- rnorm(m)
  x <- c(xw, xs); y <- c(yw, ys)
  plot(x, y, pch = 16, cex = .5, col = c(rep("#9E431855", m), rep("#0E6473", m)),
       xlab = "x", ylab = "y", main = main)
  abline(v = mean(x), lty = 3, col = "grey50")
  segments(mean(xw), mean(yw), mean(xs), mean(ys), lty = 2, lwd = 2, col = "grey30")
  points(c(mean(xw), mean(xs)), c(mean(yw), mean(ys)), pch = 21, bg = "white", cex = 1.4)
  abline(lm(y ~ x), lwd = 2, col = "#0E6473")
  dmu <- mean(xs) - mean(xw)
  Vb  <- 0.25 * dmu^2
  Vw  <- 0.5 * (var(xs) + var(xw))
  c(`Within Slope (movers)` = unname(coef(lm(yw ~ xw))[2]),
    `Within Slope (steady)` = unname(coef(lm(ys ~ xs))[2]),
    `Between Slope`       = unname((mean(ys) - mean(yw)) / dmu),
    `Between Weight (%)` = 100 * Vb / (Vb + Vw),
    `OLS Estimate`        = unname(coef(lm(y ~ x))[2]))
}
op <- par(mfrow = c(1, 3), mar = c(4, 3.5, 2.5, 0.5))
a <- draw(0,   expression(Delta*mu == 0))
b <- draw(10,  expression(Delta*mu == 10))
c_ <- draw(100, expression(Delta*mu == 100))
```

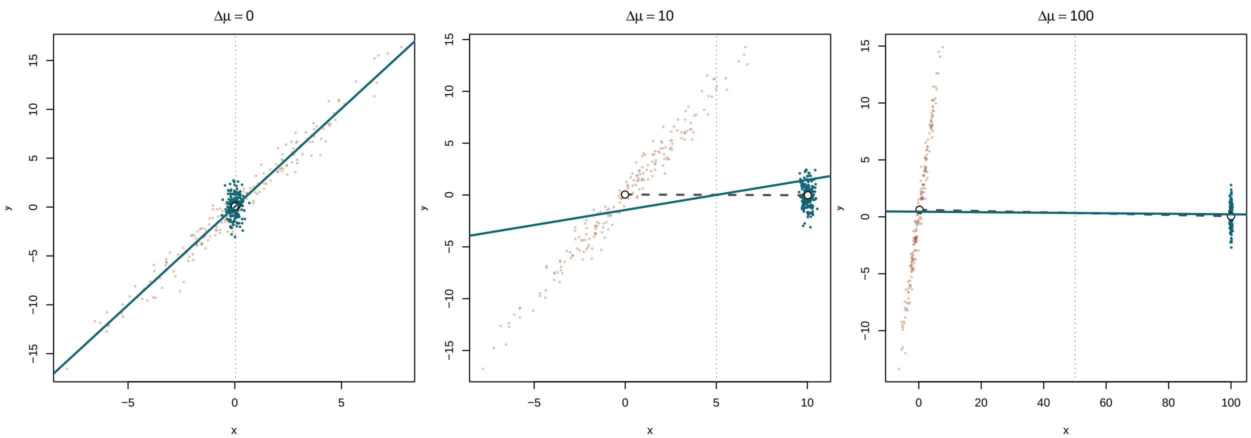


Figure 13.1: Pooled OLS as a within/between blend (Equation 13.29). Points are the two groups, open circles their centroids. The dashed line is the between-slope $\beta_{between} = \Delta\bar{y}/\Delta\mu$ through the centroids; the solid line is the pooled fit $\hat{\beta}_{OLS}$. The within slopes are identical across panels; widening $\Delta\mu$ (left to right) inflates $V_{between}$ and rotates $\hat{\beta}_{OLS}$ toward $\beta_{between}$.

```r
par(op)
mtab(cbind(`$\\Delta\\mu = 0$` = a, `$\\Delta\\mu = 10$` = b, `$\\Delta\\mu = 100$` = c_), 3)
```

	$\Delta\mu = 0$	$\Delta\mu = 10$	$\Delta\mu = 100$
Within Slope (movers)	2.019	2.009	2.002
Within Slope (steady)	0.007	-0.288	-0.136
Between Slope	0.882	-0.007	-0.006
Between Weight (%)	0.026	85.113	99.833
OLS Estimate	2.010	0.291	-0.002

Pooled OLS as a within/between blend (Equation 13.29). Points are the two groups, open circles their centroids. The dashed line is the between-slope $\beta_{between} = \Delta\bar{y}/\Delta\mu$ through the centroids; the solid line is the pooled fit $\hat{\beta}_{OLS}$. The within slopes are identical across panels; widening $\Delta\mu$ (left to right) inflates $V_{between}$ and rotates $\hat{\beta}_{OLS}$ toward $\beta_{between}$.

Read across the three columns. The within slopes barely change — about 2 for the movers, indistinguishable from zero for the steady strip. What changes is the between weight: 0.03% when the centres coincide, 85% at $\Delta\mu = 10$, and 99.6% at $\Delta\mu = 100$. So the OLS fit falls from 2.01 to 0.29 to 0.02 on identical true slopes.

The between slope reported in the first column is meaningless — and harmlessly so: with the centres nearly on top of each other $\Delta\mu \approx 0$, an unstable ratio is multiplied by a vanishing weight and contributes nothing.

The strip has influence in the second and third panels despite barely moving. Influence is distance from the mean of x, and the strip sits $\Delta\mu$ units from it; what it contributes is not its slope but its position — it anchors one end of the dashed line.

A group dummy removes that channel. Demeaning subtracts each cloud's *own* mean, sliding the shifted strip back to the centre, so every panel collapses onto the first — centroids coinciding. Run OLS on that and the solid fit swings back up to the within slope. That is all fixed effects is, and why its slope matches the leftmost panel's regardless of how far the strip had been pushed.

So "the weight is how much x moves" is the demeaned case only. It is exact once unit effects are removed, and false in general, because with the halves at different levels of x part of the fitted line is a comparison *between* them rather than a weighting of within-unit slopes.

13.5.3 When the weighting changes nothing

The previous two subsections showed variance weighting at its most consequential: FE estimated the movers' effect rather than the population average, and pooled OLS was dragged toward a between-slope of zero. The reader might conclude that unequal weighting is always a problem. It is not. The variance-weighted average equals the plain average whenever effects are unrelated to the weights. The weighting tilts the estimate only when units whose x moves more have systematically different slopes — that is, when b_i covaries with w_i.

The following two designs illustrate the distinction. Both draw $N = 1000$ units, $T = 6$ periods, with $\text{sd}_i(x) \sim U(0.2, 3)$ and $y_{it} = b_i x_{it} + \alpha_i + \varepsilon_{it}$. Both share the same x. What differs is how the true slopes b_i relate to regressor volatility:

- **(A) Unrelated:** $b_i \sim N(1, 1)$, independent of $\text{sd}_i(x)$.
- **(B) Aligned:** $b_i = 1 + 0.8(\text{sd}_i - 1.6)$, so the slope rises with volatility.

Both designs average to $\bar{b} \approx 1$. The question is whether the FE estimate — a variance-weighted average of the b_i — differs from that plain average.

```
set.seed(7)
sdX <- runif(N, 0.2, 3)

panel <- function(b) {
  h   <- data.frame(id = rep(1:N, each = T),
                    x  = rnorm(N * T, 0, rep(sdX, each = T)))
  h$y <- rep(b, each = T) * h$x + rep(rnorm(N), each = T) + rnorm(N * T)
  h
}
compareAB <- function(h, b) {
  w <- tapply(resid(feols(x ~ 1 | id, h))^2, h$id, sum)
  c(`FE estimate`             = coef(feols(y ~ x | id, h))[["x"]],
    `true slopes, plain avg` = mean(b),
    `true slopes, weighted`  = sum(w * b) / sum(w))
}

b1 <- rnorm(N, 1, 1)
b2 <- 1 + 0.8 * (sdX - 1.6)
h1 <- panel(b1); h2 <- panel(b2)
```

```r
mtab(cbind(`(A) unrelated` = compareAB(h1, b1),
           `(B) aligned`   = compareAB(h2, b2)), 4)
```

	(A) unrelated	(B) aligned
FE estimate	0.9813	1.5338
true slopes, plain avg	0.9716	1.0140
true slopes, weighted	0.9816	1.5422

In (A) all three numbers agree to within sampling noise: the weighting is there but costs nothing, because the slopes are independent of the weights. In (B) the FE estimate is 1.53 while the plain average of the true slopes is 1.01, and the weighted average, 1.54, confirms what the estimator was tracking all along — it overweights the high-slope units precisely because they are the high-variance units.

FE still removes confounding from the unit effects α_i — that is its job, and it does it in both designs. What FE does not do is change the within-variance weighting. Even after demeaning, OLS leans on units whose x moves more, and in design (B) those units happen to have higher slopes. The gap between the FE estimate and the plain average is not a failure of FE; it is what OLS does by construction, and it matters only when b_i covaries with $\mathrm{Var}_i(x)$. Without knowing the true slopes, the problem is undetectable from the data alone — the FE estimate looks well-behaved, standard errors are fine, it is just consistently estimating a different parameter than the simple average.

Reweighting to recover the simple average. The fix is weighted least squares. The FE weight on unit i is proportional to its within-variance $\hat{\sigma}_i^2 = \sum_t (x_{it} - \bar{x}_i)^2$. Running WLS-FE with weight $1/\hat{\sigma}_i^2$ gives each unit equal influence, so the estimate converges to the simple average of slopes rather than the variance-weighted average. This is the logic behind Gibbons, Suárez Serrato, and Urbancic (2018), who formalize the reweighting to recover what they call the "unit-average treatment effect," and it is the continuous analog of Słoczyński's reweighting in the binary case.

The trade-off is efficiency. The high-variance units are the ones whose slope is best estimated — their x moves a lot, so there is real information there. Downweighting them inflates the standard error. We are giving up precision to change which parameter we estimate.

The table below adds the reweighted FE estimate to both designs.

```r
compareAB2 <- function(h, b) {
  wvar <- tapply(resid(feols(x ~ 1 | id, h))^2, h$id, sum)
  inv_w <- 1 / tapply(resid(feols(x ~ 1 | id, h))^2, h$id, mean)
  h$rw <- rep(inv_w, each = T)
  c(`FE estimate`             = coef(feols(y ~ x | id, h))[["x"]],
    `Reweighted FE`           = coef(feols(y ~ x | id, h, weights = ~rw))[["x"]],
    `true slopes, plain avg` = mean(b),
    `true slopes, weighted`  = sum(wvar * b) / sum(wvar))
}

mtab(cbind(`(A) unrelated` = compareAB2(h1, b1),
           `(B) aligned`   = compareAB2(h2, b2)), 4)
```

	(A) unrelated	(B) aligned
FE estimate	0.9813	1.5338
Reweighted FE	0.9781	1.0240
true slopes, plain avg	0.9716	1.0140
true slopes, weighted	0.9816	1.5422

In (A) the reweighted estimate agrees with the standard FE estimate — reweighting changes nothing when effects are independent of variance. In (B) the reweighted estimate falls from 1.53 back to near 1.01, recovering the simple average. The price is a larger standard error, because we have deliberately downweighted the most informative units.

13.6 Summary

1. **Leverage weighting.** OLS weights each observation by regressor variance $\tilde{V}_i^2 = (x_i - \bar{x})^2$ (Equation 13.4); units that vary more dominate the slope.
2. **What is being estimated.** Under heterogeneous slopes b_i, OLS estimates $\sum_i d_i x_i b_i / \sum_i d_i x_i$ with weights $d_i x_i = (x_i - \bar{x}) x_i$ (Equation 13.10); it equals the ATE only when the b_i are uncorrelated with those weights. Keep the two weightings apart: d_i^2 weights the observed deviation slopes s_i, while $d_i x_i$ weights the structural b_i. They share a total, since $\sum_i d_i x_i = \sum_i d_i^2$, but individually differ by $d_i \bar{x}$, and $d_i x_i$ can be negative where a square cannot. Centering the treatment, $y_i = \alpha + b_i(x_i - \bar{x}) + e_i$, is what makes them coincide.
3. **Between-group bias in pooled OLS.** Without fixed effects, group-mean differences $\Delta\mu$ enter through $V_{between}\beta_{between}$ (Equation 13.29) and distort the within-unit parameter; demeaning removes them.
4. **When it matters.** The weighting always happens. It changes the answer only when effects are heterogeneous and the heterogeneity lines up with the weight. With homogeneous effects it does no harm; with heterogeneous effects the question is not whether OLS is weighting but whether we know what it weighted.

Everything here is the *effect* reading — weights on the unobservable β_i, answering what $\hat{\beta}$ estimates. The same FWL identity also weights the *observed* outcomes, a signed and fully computable combination that answers which rows drive the number and reaches, unchanged, into estimators far beyond OLS. That is the companion chapter, Interpreting OLS: Outcome Weights.

14 Interpreting OLS: Outcome Weights

The companion chapter Interpreting OLS: Effect Weights read the OLS coefficient as a weighting of unobservable effects — it answered *whose* effect $\hat{\beta}$ estimates. The same Frisch–Waugh–Lovell identity reads a second way, as a weighting of the *observed* outcomes. This reading answers a different question — which rows drive $\hat{\beta}$, and in which direction — and, unlike the first, it is computable and generalises past OLS. This chapter follows it.

14.1 The other reading of FWL

Residualise the regressor of interest V on the controls X, write $\tilde{V}_i$ for the residual, and recall the residual-on-residual form of the coefficient, Equation 13.12:

$$\hat{\beta} = \frac{\sum_{i=1}^{n} \tilde{V}_i y_i}{\sum_{i=1}^{n} \tilde{V}_i^2}. \tag{14.1}$$

The previous chapter grouped each $\tilde{V}_i$ with V_i and read $\hat{\beta}$ as a weighting $w_i = \tilde{V}_i V_i$ of effects (Equation 13.13). Group $\tilde{V}_i$ with y_i instead — pairing each residual with its own outcome in the numerator — and the *same* expression is a linear combination of the observed outcomes:

$$\hat{\beta} = \sum_{i=1}^{n} \omega_i y_i, \qquad \omega_i = \frac{\tilde{V}_i}{\sum_{j=1}^{n} \tilde{V}_j^2}. \tag{14.2}$$

Equation 14.2 is the same algebra rearranged, but a different object from the effect weights.

	Weights on outcomes ω_i	Weights on effects w_i
Weights what	the observed y_i	that observation's own effect β_i
Sign	signed, sums to zero	non-negative (under saturation)
Computable	yes, y is observed	no, the β_i are never seen
Holds when	always, it is algebra	in expectation; per-unit needs saturation
Answers	which rows drive $\hat{\beta}$, any wrong sign	is this the effect I meant, whose effect

Two constraints pin ω_i down, both from Equation 14.2. An intercept among the controls makes $\sum_i \tilde{V}_i = 0$, so the weights **sum to zero**. And $\sum_i \tilde{V}_i V_i = \sum_i \tilde{V}_i^2$ makes the weighted regressor **sum to one**, $\sum_i \omega_i V_i = 1$.

Use this reading when overlap is in doubt and a row may enter with the wrong sign — which the non-negative w_i cannot show. Keep the effect reading for questions about the estimand.

14.2 What the outcome weights look like

Take the simple regression of y on x with an intercept, so $\tilde{V}_i = x_i - \bar{x} = d_i$ and

$$\omega_i = \frac{d_i}{\sum_j d_j^2}. \tag{14.3}$$

Unlike the effect weights $w_i = d_i^2$ — non-negative, growing with the *square* of distance — this is **linear in d_i, and so signed**: a point left of $\bar{x}$ enters $\hat{\beta}$ with a negative coefficient on its outcome, a point right of it positive. It is not a slope and does not involve $\bar{y}$; it is simply a signed weight on the observed y_i.

Simulation — simple regression. To see it, we simulate $n = 100$ points — regressor $x_i \sim N(5, 2^2)$ and outcome $y_i = 2x_i + \varepsilon_i$ with $\varepsilon_i \sim N(0, 3^2)$ — then size each point by $|\omega_i|$ and colour it by sign.

```
set.seed(3)
n <- 100; x <- rnorm(n, 5, 2); y <- 2 * x + rnorm(n, 0, 3)
xb <- mean(x); d <- x - xb; om <- d / sum(d^2); col <- ifelse(om >= 0, teal, rust)
R <- d > 0; L <- d < 0
xR <- sum(abs(d[R]) * x[R]) / sum(abs(d[R])); yR <- sum(abs(d[R]) * y[R]) / sum(abs(d[R]))
xL <- sum(abs(d[L]) * x[L]) / sum(abs(d[L])); yL <- sum(abs(d[L]) * y[L]) / sum(abs(d[L]))
plot(x, y, pch = 16, cex = 0.35 + 2 * abs(om) / max(abs(om)),
     col = paste0(col, "99"), xlab = "x", ylab = "y", main = "N = 100 Points")
abline(v = xb, lty = 3, col = "grey55")
abline(lm(y ~ x), lwd = 1.3, lty = 2, col = "grey55")
segments(xL, yL, xR, yR, lwd = 2.6, col = "grey15")
points(c(xL, xR), c(yL, yR), pch = 21, bg = c(rust, teal), col = "white", cex = 2.3, lwd = 2)
```

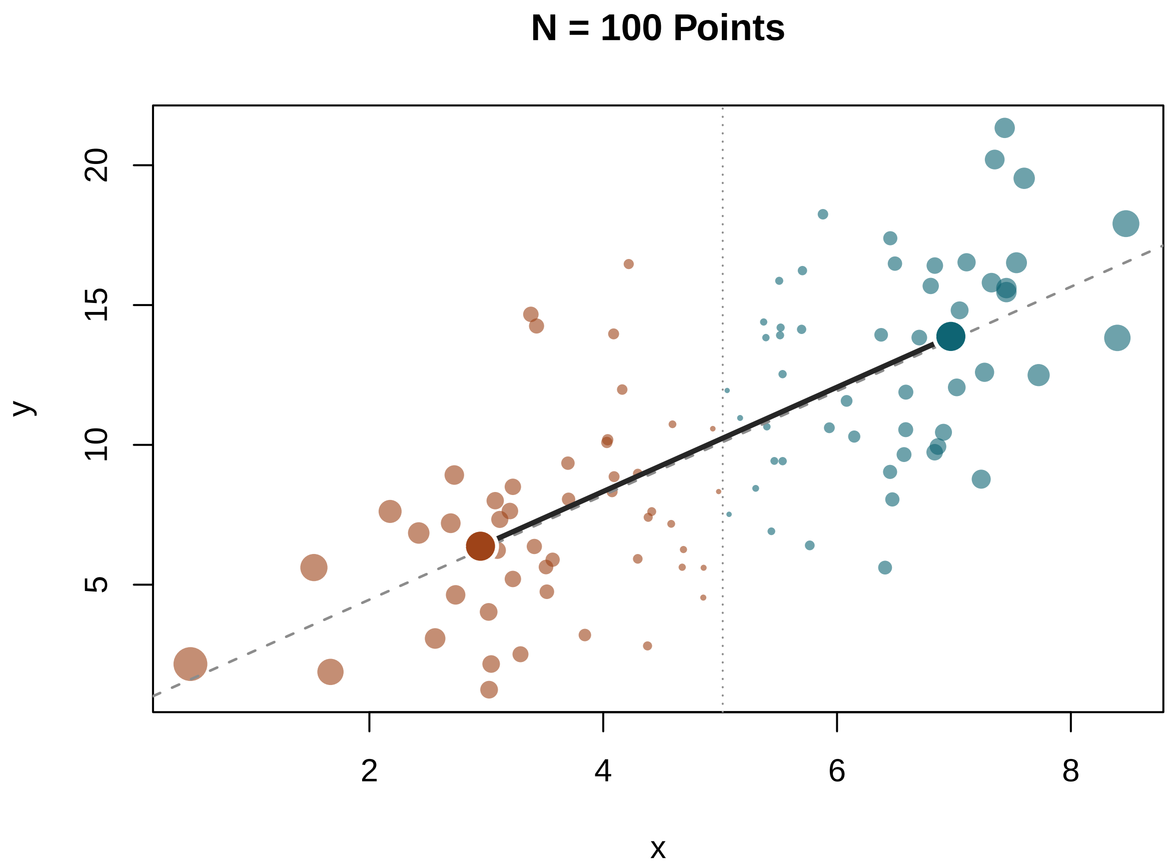


Figure 14.1: Outcome weights in a 100-point sample. Each small point's **size scales with its weight** $|\omega_i|$, coloured by sign — teal ($\omega_i > 0$) pushes $\hat{\beta}$ up, rust ($\omega_i < 0$) pulls it down; the dotted line is the fulcrum $x = \bar{x}$. The two large dots are the $|d|$-weighted centroids of the points left and right of $\bar{x}$; the solid segment joining them has slope $\hat{\beta}$, and the dashed line is the OLS fit (same slope, shifted through the grand mean).

```
vtab(c(`$\\sum_i \\omega_i$`   = sum(om),      `$\\sum_i \\omega_i x_i$` = sum(om * x),
       `positive mass` = sum(om[om > 0]), `negative mass` = sum(om[om < 0]),
       `OLS`          = coef(lm(y ~ x))[[2]], `$\\sum_i \\omega_i y_i$` = sum(om * y)), 4)
```

	value
$\sum_i \omega_i$	0.0000
$\sum_i \omega_i x_i$	1.0000
positive mass	0.2486
negative mass	-0.2486
OLS	1.8648
$\sum_i \omega_i y_i$	1.8648

Outcome weights in a 100-point sample. Each small point's **size scales with its weight** $|\omega_i|$, coloured by sign — teal ($\omega_i > 0$) pushes $\hat{\beta}$ up, rust ($\omega_i < 0$) pulls it down; the dotted line is the fulcrum $x = \bar{x}$. The two large dots are the $|d|$-weighted centroids of the points left and right of $\bar{x}$; the solid segment joining them has slope $\hat{\beta}$, and the dashed line is the OLS fit (same slope, shifted through the grand mean).

The printed sums confirm Equation 14.2: the weights split at $\bar{x}$ into a positive mass ($\approx +0.25$) and an equal negative one (≈ -0.25), and $\sum_i \omega_i x_i = 1$.

But the figure shows more. Split the points at $\bar{x}$, weight each side by $|d_i|$, and take its centroid — the two large dots. **The segment joining them has slope $\hat{\beta}$:** the OLS slope is the rise over run between the low-x and high-x weighted centres.

The dashed fit shares that slope, shifted through the grand mean, so the two coincide only when the sides balance. This collapse to two points falls out of the slope formula.

14.2.1 Why it collapses to two points

Write $d_i = x_i - \bar{x}$. Because $\sum_i d_i = 0$, the positive and negative sides carry equal total weight $W = \sum_{i:d_i>0} d_i$. An inner product with d is therefore a between-group contrast: for any variable v,

$$d \cdot v = W(\bar{v}_R - \bar{v}_L), \tag{14.4}$$

where $\bar{v}_R$ and $\bar{v}_L$ are the $|d_i|$-weighted means on each side of $\bar{x}$. The slope $\hat{\beta} = d \cdot y \,/\, d \cdot x$ is the ratio of two such contrasts; W cancels:

$$\hat{\beta} = \frac{\bar{y}_R - \bar{y}_L}{\bar{x}_R - \bar{x}_L}. \tag{14.5}$$

That is the rise over run between the two $|d|$-weighted centroids — the two dots in the figure.

14.3 When the sign flips

In simple regression the sign of ω_i is just the side of the mean: half the weights are negative, structurally, and a low-x point *should* count against a high-x one. With controls $\tilde{V}_i$ becomes a *residual*, and a row can flip to the sign its raw position denies — a hidden extrapolation the non-negative w_i can never show.

With a binary treatment D and covariate x, the weight $\omega_i \propto \tilde{V}_i = D_i - \hat{D}_i$, with $\hat{D}$ the linear projection of D on $(1, x)$. Regressed on D alone, every treated unit is positive and every control negative — the *structural* pattern, producing $\hat{\beta} = \bar{y}_1 - \bar{y}_0$. Add x and the fitted $\hat{D}_i$ can leave $[0, 1]$ where treatment and control barely overlap: a treated unit with $\hat{D}_i > 1$ gets $\tilde{V}_i < 0$, so its outcome is subtracted from $\hat{\beta}$ — a *wrong-sign* weight that the non-negative $\tilde{V}_i^2$ hides. The test is one line: **fit the treatment on the controls; a treated unit with $\hat{D}_i > 1$, or a control with $\hat{D}_i < 0$, is an extrapolated, wrong-sign unit** — a point where the groups fail to overlap. Out-of-range the other way is a different thing: a treated unit with $\hat{D}_i < 0$ still has $\tilde{V}_i > 0$, and a control with

$\hat{D}_i > 1$ still has $\tilde{V}_i < 0$, so both keep their conventional sign even though the prediction is out of bounds. (With a continuous regressor there is no bounded range to leave; overlap is read from leverage, not sign.)

A wrong-sign weight is extrapolation: the model reaches outside common support, and $\hat{\beta}$ can leave the convex hull of the observed outcomes. Whether it matters depends on the data — a sensitivity check is one trim away.

Simulation — thin overlap on a curved baseline. The DGP: 150 controls near $x = 2.5$ and 150 treated near $x = 4.5$ (both sd 1.1), with $y_i = D_i + 0.8\,(x_i - 3.5)^2 + 0.3\,x_i + \varepsilon_i$, $\varepsilon_i \sim N(0, 0.5^2)$ — a constant true effect of 1 sitting on a convex baseline.

```
set.seed(7)
n <- 150
x <- c(rnorm(n, 2.5, 1.1), rnorm(n, 4.5, 1.1))
d <- c(rep(0, n), rep(1, n))
y <- 1.0 * d + 0.8 * (x - 3.5)^2 + 0.3 * x + rnorm(2 * n, 0, 0.5)
df <- data.frame(y, d, x)
X <- cbind(1, x); dh <- as.vector(X %*% solve(crossprod(X)) %*% t(X) %*% d)
flag <- (d == 1 & dh > 1) | (d == 0 & dh < 0)
pch <- ifelse(d == 1, 17, 16); col <- ifelse(d == 1, teal, rust)
plot(x, y, type = "n", xlab = "x", ylab = "y")
xs <- seq(min(x), max(x), length = 200)
mc <- function(z) 0.8 * (z - 3.5)^2 + 0.3 * z            # true control mean E[y | x, D = 0]
lines(xs, mc(xs),     col = rust, lwd = 2.4)             # control mean
lines(xs, mc(xs) + 1, col = teal, lwd = 2.4, lty = 3)    # treated mean = control + true effe
points(x, y, pch = pch, col = paste0(col, "88"), cex = 0.85)              # all points, by gro
points(x[flag], y[flag], pch = 1, col = "grey15", cex = 1.7, lwd = 1.4)    # ring the flagged u
arrows(3.5, mc(3.5), 3.5, mc(3.5) + 1, code = 3, length = 0.05, angle = 90, lwd = 2.4, col = "g
segments(3.5, mc(3.5) + 0.5, 2.05, 13.3, col = "grey55", lwd = 1)
text(2.0, 14, "gap = true effect = 1", pos = 4, cex = 0.82, col = "grey10")
legend("bottomright", bty = "n", cex = 0.82,
       legend = c("control", "treated", "flagged (trimmed)", "control mean", "treated mean"),
       pch = c(16, 17, 1, NA, NA), lty = c(NA, NA, NA, 1, 3),
       col = c(rust, teal, "grey15", rust, teal))
```

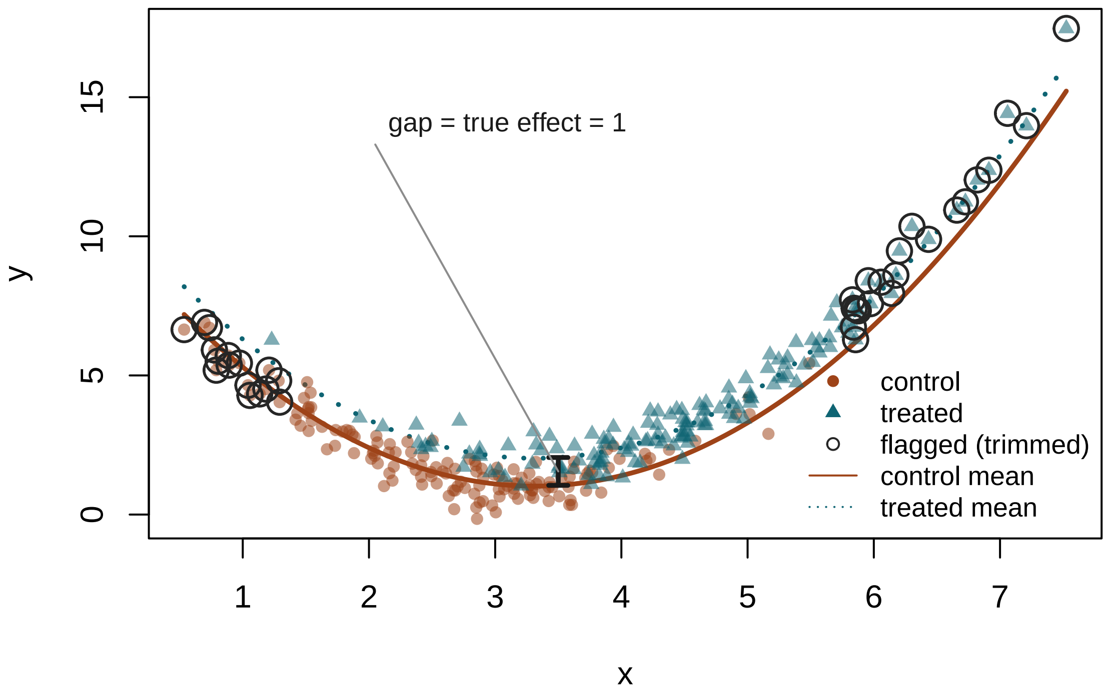


Figure 14.2: Thin overlap on a convex baseline. The two curves are the true means — control (rust, solid) and treated (teal, dotted), a constant 1 apart (arrow). Controls sit low on the curve and treated high; the groups share only the middle. Ringed points are wrong-sign units — $\hat{D}_i \notin [0, 1]$ — where the groups fail to overlap. The straight $y \sim D + x$ reports $\hat{\beta} = 0.69$; dropping the flagged units, or reweighting to the overlap, restores ≈ 1.

```
w_ato <- WeightIt::weightit(d ~ x, data = df, method = "glm", estimand = "ATO")
vtab(c(`flagged units`      = sum(flag),
        `$\\hat\\beta$ (full)`        = coef(lm(y ~ d + x))[["d"]],
        `$\\hat\\beta$ (trimmed)`     = coef(lm(y ~ d + x, data = df[!flag, ]))[["d"]],
        `$\\hat\\beta$ (overlap-wtd)` = coef(lm(y ~ d + x, weights = w_ato$weights))[["d"]]), 3
```

	value
flagged units	37.000
$\hat{\beta}$ (full)	0.688
$\hat{\beta}$ (trimmed)	0.991
$\hat{\beta}$ (overlap-wtd)	0.979

Thin overlap on a convex baseline. The two curves are the true means — control (rust, solid) and treated (teal, dotted), a constant 1 apart (arrow). Controls sit low on the curve and treated high;

the groups share only the middle. Ringed points are wrong-sign units — $\widehat{D}_i \notin [0,1]$ — where the groups fail to overlap. The straight $y \sim D + x$ reports $\hat{\beta} = 0.69$; dropping the flagged units, or reweighting to the overlap, restores ≈ 1.

Thirty-seven units carry the wrong sign, all in the non-overlapping tails. Trim them and $\hat{\beta}$ jumps from 0.69 to 0.99; the overlap weights (`"ATO"`, Li, Morgan, and Zaslavsky 2018) give 0.98. Either fix confirms the estimate leaned on the extrapolation, read off the regression itself with no separate model. Had the outcome been linear, the same trim would barely move $\hat{\beta}$: the wrong-sign weights are still there, but they are harmless. The check tells you which case you are in.

This is the implied-weights diagnostic of Chattopadhyay and Zubizarreta (2023); their `lmw` package computes the weights and the balance, extrapolation, and influence diagnostics built on them, and `OutcomeWeights` (Knaus 2024) carries it to machine-learning estimators. The mainstream overlap check takes the other route, through the propensity score (`WeightIt` for estimation, `cobalt` for balance, and Crump et al. 2009 for trimming); both routes reach the same conclusion.

14.4 Out of OLS: the same weights beyond the linear model

The outcome reading has one advantage the effect reading lacks: it survives leaving OLS. Any estimator that can be written $\hat{\tau} = \sum_i \omega_i Y_i = \omega' Y$ — a linear combination of the observed outcomes — *has* outcome weights, whatever machinery produced them.

Knaus (2024) makes this the organising idea, and his `OutcomeWeights` package computes ω for AIPW, double machine learning, and the causal forest.

14.4.1 The weights, derived

The OLS outcome weights (Equation 14.2) followed from one fact: in the FWL numerator $\sum_i \tilde{V}_i y_i$, the outcome y_i appears once, linearly, so $\hat{\beta} = \sum_i \omega_i y_i$ with $\omega_i = \tilde{V}_i / \sum_j \tilde{V}_j^2$. The same fact holds whenever the outcome model is a *smoother* — a map $\hat{m} = SY$ that is linear in the training outcomes. Substituting SY for $\hat{m}$ keeps Y linear in the numerator, and reading off its coefficient gives ω. We carry this out for PLR and AIPW.

PLR. Take the partially-linear estimator (the R-learner behind the causal forest). It is FWL with the projections done by a flexible learner: residualise treatment and outcome on the covariates, then regress one residual on the other,

$$\hat{\tau} = \frac{\sum_i (D_i - \hat{e}_i)(Y_i - \hat{m}_i)}{\sum_i (D_i - \hat{e}_i)^2}, \tag{14.6}$$

with $\hat{e} = \hat{E}[D \mid X]$ and $\hat{m} = \hat{E}[Y \mid X]$ the fitted nuisances.

The outcome nuisance is a *smoother*: its fitted values are a linear map of the training outcomes, $\hat{m} = SY$ — true for OLS, ridge, kernels, and honest forests alike. The word *honest* carries weight there. If the learner's structure is itself chosen using the outcomes — splits grown on the same Y being predicted, a lasso whose active set is outcome-selected, boosting — then S depends on Y and the map stops being linear in it. What follows then holds *conditional on the fitted learner*, which is the weaker claim: the weights describe the estimate you got rather than a globally linear estimator. Honesty and cross-fitting are what restore the clean reading, by fitting the structure on data other than the outcomes being weighted. So the outcome residual is $Y - \hat{m} = (\mathbf{I} - S)Y$. Writing $\tilde{D} = D - \hat{e}$ for the treatment residual and substituting,

$$\hat{\tau} = \frac{\tilde{D}'(\mathbf{I} - S)\,Y}{\tilde{D}'\tilde{D}}. \tag{14.7}$$

Now Y appears once, linearly, in the numerator. Read off its coefficient and the weights are done:

$$\hat{\tau} = \omega' Y, \qquad \omega' = \frac{\tilde{D}'(\mathbf{I} - S)}{\tilde{D}'\tilde{D}}. \tag{14.8}$$

That is Equation 14.2 — the OLS weights — with the projection annihilator M_X replaced by the smoother annihilator $\mathbf{I} - S$. OLS is the case of a linear learner, $S = P_X$: there S is a projection, $\mathbf{I} - S = M_X$, and Equation 14.8 is Equation 14.2 exactly. Component by component, the weight on observation i is

$$\omega_i = \frac{\tilde{D}_i - \sum_j S_{ji}\,\tilde{D}_j}{\sum_k \tilde{D}_k^2}: \tag{14.9}$$

the treatment residual of unit i, minus how much other units' treatment residuals leak to i through the outcome smoother, normalised by the total treatment-residual variance.

The outcome weights tell which rows drive the estimate; the *effect* weights tell whose effect it estimates. Substitute $Y_i = \tau_i D_i + m(X_i) + \varepsilon_i$ into the PLR numerator. With consistent nuisances $\hat{m} \approx m$, the noise vanishes in expectation and the numerator reduces to $\sum_i \tilde{D}_i D_i \tau_i$, so the effect weight on τ_i is $\tilde{D}_i D_i = (D_i - e_i) D_i$. Since D is binary ($D_i^2 = D_i$),

$$E[\tilde{D}_i D_i \mid X_i] = E[D_i \mid X_i] - e_i\, E[D_i \mid X_i] = e_i - e_i^2 = e_i(1 - e_i). \tag{14.10}$$

The denominator $\tilde{D}'\tilde{D} = \sum_i \tilde{D}_i^2$ has the same expectation, $\sum_i e_i(1 - e_i)$, since $\mathrm{Var}(D_i \mid X_i) = e_i(1 - e_i)$. Both numerator and denominator weight by propensity variance, and the ratio is the variance-weighted average treatment effect,

$$\tau_{\mathrm{VWATE}} = \frac{\sum_i e(X_i)(1 - e(X_i))\,\tau(X_i)}{\sum_i e(X_i)(1 - e(X_i))}, \tag{14.11}$$

not the plain ATE $\bar{\tau} = n^{-1} \sum_i \tau_i$. The two coincide when effects are constant, or when treatment is randomly assigned so that $e(1 - e)$ is the same everywhere. Neither condition is necessary. All that is required is that $\tau(X)$ be uncorrelated with the weight $e(X)(1 - e(X))$: heterogeneity that happens not to line up with propensity variance leaves the two equal. The weighting is the same $e(1 - e)$ that appeared in the previous chapter's Equation 13.17, where Angrist (1998) showed OLS with saturated strata weights each stratum by $n_x \hat{D}(x)(1 - \hat{D}(x))$. That result required discrete strata and saturated controls; Equation 14.11 is the same propensity-variance weighting with continuous covariates and a flexible learner. The individual $\tau(X_i)$ are never observed or estimated — they are the structural unit-level effects in $Y_i = \tau(X_i)D_i + m(X_i) + \varepsilon_i$. PLR returns one number, $\hat{\tau}$, that converges to their $e(1-e)$-weighted average; to estimate $\tau(X_i)$ itself one needs a CATE estimator (a causal forest, say).

Three moves did it: write the estimator, substitute the smoother $\hat{m} = SY$, read off the coefficient on Y. They work for any estimator built from smoothers; only the operator on Y changes. AIPW is worth carrying through, because its operator is more than $\mathbf{I} - S$.

AIPW. With outcome smoothers $\hat{\mu}_1 = S_1 Y$, $\hat{\mu}_0 = S_0 Y$ and propensity $\hat{e}$, the score of unit i is

$$\psi_i = \hat{\mu}_{1i} - \hat{\mu}_{0i} + \frac{D_i}{\hat{e}_i}(Y_i - \hat{\mu}_{1i}) - \frac{1-D_i}{1-\hat{e}_i}(Y_i - \hat{\mu}_{0i}), \qquad \hat{\tau} = \frac{1}{n}\sum_i \psi_i. \tag{14.12}$$

Each $\hat{\mu}_{ji}$ is $(S_j Y)_i$, so Y again enters only linearly; collecting its coefficient gives $\hat{\tau} = \omega' Y$ with

$$T = S_1 - S_0 + \operatorname{diag}\!\left(\tfrac{D}{\hat{e}}\right)(\mathbf{I} - S_1) - \operatorname{diag}\!\left(\tfrac{1-D}{1-\hat{e}}\right)(\mathbf{I} - S_0), \qquad \omega' = \frac{\mathbf{1}'T}{n}. \tag{14.13}$$

That is, ω_j is the j-th column-sum of T divided by n: observation j's outcome enters the ATE through its direct IPW term and through its indirect contribution via the two outcome smoothers S_1, S_0 to every other unit's augmentation correction.

Comparing the two:

	PLR (Equation 14.8)	AIPW
Matrix form	$\omega' = \tilde{D}'(\mathbf{I} - S) / \tilde{D}'\tilde{D}$	$\omega' = \mathbf{1}'T / n$
Per unit	$\omega_i = (\tilde{D}_i - \sum_j S_{ji}\tilde{D}_j) / \sum_k \tilde{D}_k^2$	$\omega_j = \frac{1}{n}\sum_i T_{ij}$
Smoother	one S (outcome on X)	two: S_1 (treated), S_0 (control)
Normalisation	treatment-residual variance	sample size n

Both follow the same recipe — substitute the smoother $\hat{m} = SY$, read off Y's coefficient — and differ only in the operator acting on Y.

Simulation — heterogeneous effects (DML). Under a heterogeneous effect the two estimators diverge, and the weights show why. We simulate $n = 2000$ units: five standard-normal covariates X, selection $W_i \sim \text{Bernoulli}(\Lambda(-1 + 1.2\,X_{i1}))$ with Λ the logistic function (treatment is scarce and tied to X_1), and outcome $Y_i = \tau_i W_i + X_{i1} + 0.5\,X_{i2} + \varepsilon_i$ with $\varepsilon_i \sim N(0,1)$ and an effect $\tau_i = 1 + X_{i1}$ that varies across units — a $W \times X_1$ interaction. The average effect is $E[\tau] = 1$, but units with larger X_1 have larger effects. `dml_with_smoother` returns a partially-linear (PLR) and an AIPW estimator; `get_outcome_weights` extracts the weight matrix ω, one row per estimator.

```
library(OutcomeWeights)
set.seed(1)
n <- 2000; p <- 5
X <- matrix(rnorm(n * p), n, p)
etrue <- plogis(-1 + 1.2 * X[, 1])                    # asymmetric selection on X1
W <- rbinom(n, 1, etrue)
```

```
tau <- 1 + X[, 1]                                        # heterogeneous effect: W x X1
Y <- tau * W + X[, 1] + 0.5 * X[, 2] + rnorm(n)
dml <- dml_with_smoother(Y, W, X, estimators = c("PLR", "AIPW_ATE"))
om  <- get_outcome_weights(dml)                          # om$omega: 2 x n, om$treat = W
invisible(capture.output(s <- summary(dml)))
W1 <- om$treat == 1
tab <- t(sapply(rownames(om$omega), function(nm) {
  w <- om$omega[nm, ]
  c(estimate = s[nm, 1], `$\\boldsymbol\\omega'\\boldsymbol Y$` = sum(w * Y),
    treated = sum(w[W1]), control = sum(w[!W1]))
}))
ov <- etrue * (1 - etrue)                                # e(1-e): PLR's per-unit weight
# ATE = plain mean of tau; overlap-ATE = e(1-e)-weighted mean of tau (PLR's target)
cat(sprintf("true ATE = %.2f   true overlap-ATE = %.2f\n", mean(tau), sum(ov * tau) / sum(ov))
```

```
true ATE = 0.99   true overlap-ATE = 1.31
```

```
mtab(tab, 3)
```

	estimate	$\omega' Y$	treated	control
PLR	1.287	1.287	0.997	-0.997
AIPW-ATE	1.074	1.074	1.000	-1.000

Two things stand out. First, the estimators now *disagree* — PLR reports 1.29, AIPW 1.07 — and neither is wrong; they target different populations. PLR targets the VWATE (Equation 14.11), not the ATE. Here treatment selection loads on X_1 through $e = \Lambda(-1 + 1.2\, X_1)$, and $\tau = 1 + X_1$ also rises in X_1, so the units PLR overweights — those with e near $\frac{1}{2}$, around $X_1 \approx 0.8$ — carry above-average effects, pushing PLR's estimate above the true ATE. AIPW instead reweights by $1/e$ and $1/(1-e)$ to undo selection, returning the plain average, the ATE (≈ 1). Same outcomes Y, weighted two ways, two population-averages of τ — and the weights are what make the difference visible.

Second, the weights still reproduce each estimate exactly and sum to zero overall. They part on the other constraint from Equation 14.2, $\sum_i \omega_i V_i = 1$: for a binary treatment that constraint is the treated weights summing to $+1$ (and, with the zero total, controls to -1). AIPW keeps it exactly; the partially-linear estimator sums to ± 0.997. That shortfall has a direct reading. If the treated weights sum to a and the controls to $-a$, then $\hat{\tau} = a\,(\bar{y}^{\omega}_{\text{treated}} - \bar{y}^{\omega}_{\text{control}})$, so the estimate is the weighted contrast scaled by a: at $a = 0.997$ the two differ by 0.3%. Whether that is an understatement depends on treating the normalized contrast as the target, which nothing here establishes — better read as a normalization discrepancy, and as a measure of how far a cross-fitted residual maker sits from a projection. The reason OLS and AIPW hit 1 exactly is that their residual maker is a projection, giving $\tilde{V}' V = \tilde{V}' \tilde{V}$ and hence a ratio of one. A cross-fitted smoother is not idempotent, so the identity only holds approximately, and the sum reports how far off it is.

That divergence shows what the representation is for. In Knaus's (2024) framing ω does three things: it puts OLS, IPW, AIPW, DML, and the causal forest on **one footing** ($\omega' Y$); it **characterises the target population** each reports — the PLR-versus-AIPW gap above; and it **checks covariate balance** via `cobalt`, flagging when weights break a property usually assumed — the sum-to-± 1 within treated and control groups that PLR violates here.

The formulas take a few lines to check against the package:

```
n  <- length(Y)
S  <- dml$NuPa.hat$smoothers$S[1, , ]
S1 <- dml$NuPa.hat$smoothers$S.d1[1, , ]
S0 <- dml$NuPa.hat$smoothers$S.d0[1, , ]
e  <- dml$NuPa.hat$predictions$D.hat[, 1]
Wtil   <- W - e
om_plr <- as.numeric((Wtil - crossprod(S, Wtil)) / sum(Wtil^2))
Imat    <- diag(n)
Tm      <- S1 - S0 + (W / e) * (Imat - S1) - ((1 - W) / (1 - e)) * (Imat - S0)
om_aipw <- colSums(Tm) / n
vtab(c(`PLR: largest gap vs package` = max(abs(om_plr  - om$omega["PLR", ])),
       `AIPW: largest gap vs package` = max(abs(om_aipw - om$omega["AIPW-ATE", ])),
       `PLR: $\\boldsymbol\\omega'\\boldsymbol Y$`                = sum(om_plr  * Y),
       `AIPW: $\\boldsymbol\\omega'\\boldsymbol Y$`                = sum(om_aipw * Y)), 4)
```

	value
PLR: largest gap vs package	0.0000
AIPW: largest gap vs package	0.0000
PLR: $\omega' Y$	1.2873
AIPW: $\omega' Y$	1.0738

14.4.2 Two uses: balance and extrapolation

Having ω, two checks run directly on it. Does the estimator's implied weighting balance the covariates? Hand ω to `cobalt` and read the largest standardised mean difference (SMD), the between-group difference in a covariate's weighted mean divided by its standard deviation. And is it extrapolating? Count the wrong-sign weights — a control with $\omega_i > 0$ or a treated unit with $\omega_i < 0$, the signature of a comparison reaching off common support.

```
library(cobalt)
uses <- t(sapply(rownames(om$omega), function(nm) {
  w <- om$omega[nm, ]
  c(`$\\max\\lvert\\mathrm{SMD}\\rvert$`         = max(abs(bal.tab(as.data.frame(X), treat = W,
    `wrong-sign units` = sum((om$treat == 1 & w < 0) | (om$treat == 0 & w > 0)))
}))
mtab(uses, 3)
```

	max|SMD|	wrong-sign units
PLR	0.025	21
AIPW-ATE	0.091	0

Raw, the groups are badly imbalanced — the largest covariate SMD is 1.11, since treatment was tied to X_1. Both weightings pull it down to near zero (0.03 for PLR, 0.09 for AIPW): each balances the covariates, which for a causal forest you could not otherwise check.

They differ on extrapolation. PLR carries 21 wrong-sign controls — it leans slightly off common support — while AIPW carries none. Neither is disqualifying here; the point is that balance and extrapolation are now *checkable*, on estimators that never exposed a weighting.

14.5 The DiD setting

When the regressor is a treatment indicator, every unit in a group gets the same weight, so $\hat{\beta}$ reduces to a contrast of group means. Regressed on the treatment alone it is the difference in means between treated and control. Add a time period and its interaction with treatment and it becomes a difference-in-differences across four group-by-period cells. We take these in turn, then bring in covariates.

Simulation — treatment alone. Regress y on the treatment D alone — here 30 treated drawn $y \sim N(5, 1)$ and 50 controls $y \sim N(2, 1)$. The residual $\tilde{V}_i = D_i - \bar{D}$ is constant within each group, so the weight is uniform within a group and $\hat{\beta}$ reduces to the two group means: $\hat{\beta} = \bar{y}_1 - \bar{y}_0$, the difference in means. Every treated unit then carries positive weight, every control negative — the structural pattern of the previous section, with the wrong-sign off-diagonal empty until covariates enter.

```
set.seed(7)
n1 <- 30; n0 <- 50
d <- c(rep(1, n1), rep(0, n0)); y <- c(rnorm(n1, 5), rnorm(n0, 2))
dev <- d - mean(d); om <- dev / sum(dev^2); col <- ifelse(om >= 0, teal, rust)
y1 <- mean(y[d == 1]); y0 <- mean(y[d == 0])
plot(jitter(d, 0.4), y, pch = 16, col = paste0(col, "55"), cex = 0.9,
     xlim = c(-0.3, 1.3), xaxt = "n", xlab = "", ylab = "y", main = "Regression on treatment")
axis(1, at = c(0, 1), labels = c("control", "treated"))
segments(0, y0, 1, y1, lwd = 2.6, col = "grey15")
points(c(0, 1), c(y0, y1), pch = 21, bg = c(rust, teal), col = "white", cex = 2.5, lwd = 2)
```

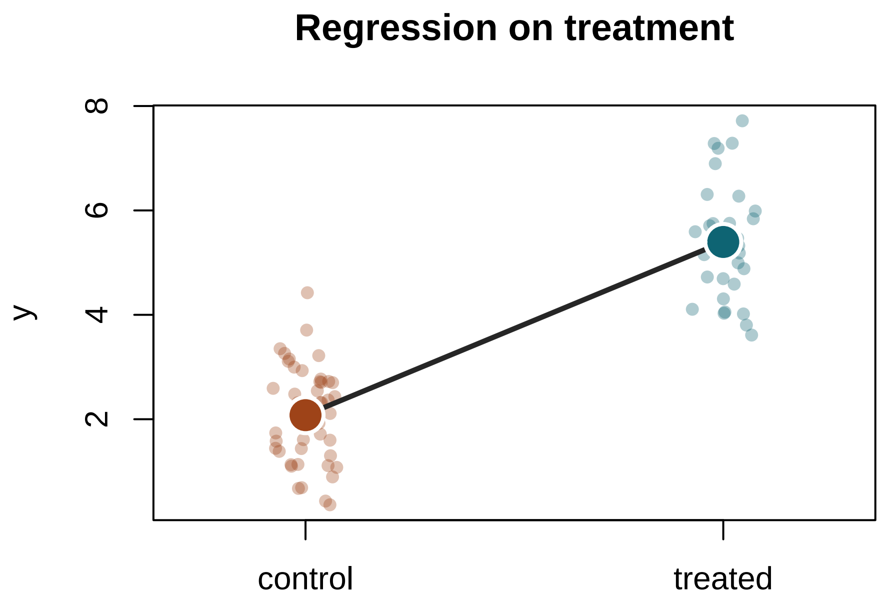


Figure 14.3: Regression on the treatment alone. Points are the two groups (rust control, teal treated); the large dots are the group means, and the segment joining them has slope $\hat\beta = \bar y_1 - \bar y_0$, the difference in means.

```
vtab(c(`$\\hat\\beta$ (lm)` = coef(lm(y ~ d))[["d"]], `$\\bar y_1 - \\bar y_0$` = y1 - y0,
       `off-diagonal weights` = sum((d == 1 & om < 0) | (d == 0 & om > 0))), 3)
```

	value
$\hat\beta$ (lm)	3.318
$\bar y_1 - \bar y_0$	3.318
off-diagonal weights	0.000

Regression on the treatment alone. Points are the two groups (rust control, teal treated); the large dots are the group means, and the segment joining them has slope $\hat\beta = \bar y_1 - \bar y_0$, the difference in means.

The estimate equals $\bar y_1 - \bar y_0$, and no weight carries the wrong sign — the off-diagonal count is zero.

Simulation — 2×2 difference-in-differences. Add a period and interact it with the group, and the design has four cells. Each collapses to its mean, and the interaction coefficient is their double contrast,

$$\widehat{\text{DiD}} = (\bar y_{T,\text{post}} - \bar y_{T,\text{pre}}) - (\bar y_{C,\text{post}} - \bar y_{C,\text{pre}}). \tag{14.14}$$

Now there are four cell means, and with them four slopes — the two group trends over time and the two cross-sectional gaps. The DiD is the difference between either parallel pair, trend minus trend or gap minus gap. We simulate 200 units in each group×period cell from $y = 3 + 2g + p + 2\,(g \cdot p) + \varepsilon$ with $\varepsilon \sim N(0, 1)$, so the true DiD (the interaction) is 2:

```
set.seed(3)
g <- rep(c(1, 1, 0, 0), each = 200); p <- rep(c(0, 1, 0, 1), each = 200)
y <- 3 + 2 * g + 1 * p + 2 * (g * p) + rnorm(800)
m <- tapply(y, list(g, p), mean)
Tpre <- m["1", "0"]; Tpost <- m["1", "1"]; Cpre <- m["0", "0"]; Cpost <- m["0", "1"]
cf <- Tpre + (Cpost - Cpre)
plot(NA, xlim = c(-0.15, 1.25), ylim = range(y), xaxt = "n", xlab = "", ylab = "y",
     main = "Difference-in-differences")
axis(1, at = c(0, 1), labels = c("pre", "post"))
segments(0, Tpre, 1, Tpost, lwd = 2.4, col = teal)
segments(0, Cpre, 1, Cpost, lwd = 2.4, col = rust)
segments(0, Tpre, 1, cf, lty = 2, lwd = 1.6, col = "grey55")
segments(1, cf, 1, Tpost, lwd = 4, col = "grey15")
points(c(0, 1, 0, 1), c(Tpre, Tpost, Cpre, Cpost),
       pch = 21, bg = c(teal, teal, rust, rust), col = "white", cex = 2.3, lwd = 2)
text(1.02, (cf + Tpost) / 2, "DiD", pos = 4, cex = 0.9, col = "grey15")
```

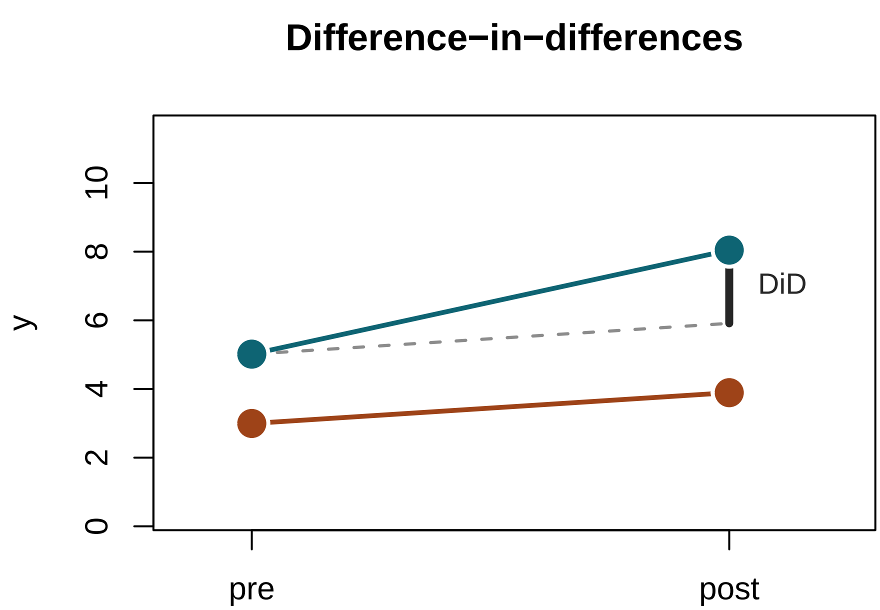


Figure 14.4: The 2×2 difference-in-differences: four cell means (dots), the treated (teal) and control (rust) trends, the parallel-trends counterfactual (dashed), and the DiD (black bar) as the gap between the treated post-mean and that counterfactual.

```r
vtab(c(`treated trend` = Tpost - Tpre, `control trend` = Cpost - Cpre,
       `pre gap` = Tpre - Cpre, `post gap` = Tpost - Cpost,
       `DiD (interaction)` = coef(lm(y ~ g * p))[["g:p"]]), 3)
```

	value
treated trend	3.029
control trend	0.897
pre gap	2.019
post gap	4.151
DiD (interaction)	2.132

The 2×2 difference-in-differences: four cell means (dots), the treated (teal) and control (rust) trends, the parallel-trends counterfactual (dashed), and the DiD (black bar) as the gap between the treated post-mean and that counterfactual.

The estimated interaction recovers the true DiD of 2, and the four cell-mean weights carry the structural signs $(+, -, -, +)$ as long as every cell is populated.

The clean 2×2 has trivial weights — four cell means, each ± 1. Real difference-in-differences carries covariates: parallel trends holds only *conditional* on X, and the comparison has to be modelled. That model is a weighting of outcomes too.

The doubly-robust DiD of Sant'Anna and Zhao (2020) is AIPW read on the outcome *change* $\Delta Y = Y_1 - Y_0$: a propensity for treatment, a control-group regression of ΔY on X, combined. By the recipe of the last section its estimate is $\widehat{\text{ATT}} = \omega' \Delta Y$, with $\omega' = a'(\mathbf{I} - S)$ — the control-trend smoother S and the group weights $a = w_{\text{treat}} - w_{\text{cont}}$.

Simulation — doubly-robust DiD. Two periods and $n = 2000$, with covariates $x_1, x_2 \sim N(0, 1)$. Treatment is selected on x_1 with thin overlap, $\Pr(D = 1) = \Lambda(-0.5 + 2x_1)$. The outcomes are $y_{i0} = 1 + x_{1i} + 0.5\, x_{2i} + \varepsilon_{i0}$ and $y_{i1} = y_{i0} + (0.5 + 0.4\, x_{1i}) + D_i + \varepsilon_{i1}$, with both errors $N(0, 1)$. The control trend $0.5 + 0.4x_1$ depends on x_1, so trends are parallel only given X, and the true ATT is 1. We fit DR-DiD with `DRDID`, rebuild its weights by hand, and check.

```r
library(DRDID)
set.seed(2)
n  <- 2000
x1 <- rnorm(n); x2 <- rnorm(n); Xcov <- cbind(1, x1, x2)
D  <- rbinom(n, 1, plogis(-0.5 + 2 * x1))        # selection into treatment on x1 (thin overlap
y0 <- 1 + x1 + 0.5 * x2 + rnorm(n)               # pre-period outcome
y1 <- y0 + (0.5 + 0.4 * x1) + 1 * D + rnorm(n)   # control trend depends on x1; true ATT = 1
dY <- y1 - y0
att <- drdid_panel(y1 = y1, y0 = y0, D = D, covariates = Xcov, boot = FALSE)$ATT

# rebuild the DR-DiD weights: omega = (I - S)' a, acting on the outcome change dY
e    <- pmin(as.vector(glm(D ~ x1 + x2, family = binomial)$fitted), 1 - 1e-6)
keep <- rep(TRUE, n); keep[D == 0] <- e[D == 0] < 0.995     # DRDID's control trimming
Xc   <- Xcov[D == 0, ]
```

```
S     <- matrix(0, n, n); S[, D == 0] <- Xcov %*% solve(crossprod(Xc)) %*% t(Xc)
w.t   <- keep * D
w.c   <- keep * e * (1 - D) / (1 - e)
a     <- w.t / sum(w.t) - w.c / sum(w.c)          # group weights, treated minus reweighted con
omega <- as.vector(a - crossprod(S, a))

Xk <- cbind(x1, x2); sds <- apply(Xk, 2, sd)
vtab(c(`DRDID ATT`                    = att,
       `$\\boldsymbol\\omega'\\Delta\\boldsymbol Y$`                = sum(omega * dY),
       `wrong-sign weights`        = sum((D == 1 & omega < 0) | (D == 0 & omega > 0)),
       `covariate imbalance, raw` = max(abs((colMeans(Xk[D == 1, ]) - colMeans(Xk[D == 0, ])) /
       `covariate imbalance, wtd` = max(abs(as.vector(a %*% Xk) / sds))), 3)
```

	value
DRDID ATT	0.980
$\omega' \Delta Y$	0.980
wrong-sign weights	151.000
covariate imbalance, raw	1.198
covariate imbalance, wtd	0.086

$\omega' \Delta Y$ reproduces the `DRDID` estimate exactly — the difference-in-differences is a signed weighting of observed outcome *changes*, and the diagnostics of this chapter apply to it. The weighting balances the covariate that drove selection (its standardised imbalance falls from 1.20 to 0.09). And 151 units carry a wrong-sign weight — a treated unit subtracted, or a control added — the signature of a comparison reaching past common support.

14.5.1 Staggered designs: TWFE at the observation level

Negative weights in staggered designs are usually described at the level of two-by-two comparisons: TWFE averages many such comparisons, and some enter with a negative weight — the forbidden comparisons that use already-treated units as controls (de Chaisemartin and D'Haultfœuille 2020; Goodman-Bacon 2021). That is the effect-weight reading. The outcome-weight reading goes one level down, to the individual y_{it}.

TWFE is still FWL: residualise D_{it} on unit and time fixed effects to get $\tilde{D}_{it}$, and $\hat{\beta} = \sum_{it} \omega_{it} y_{it}$ with $\omega_{it} = \tilde{D}_{it} / \sum_{jt} \tilde{D}_{jt}^2$. A treated observation with $\omega_{it} < 0$ is a cell whose outcome is subtracted from the estimated effect — used as a control even though it is treated. An untreated observation with $\omega_{it} > 0$ enters as though it were treated. Both are the same sign flip seen earlier under thin overlap, caused here by the unit-and-time residualisation pushing $\tilde{D}_{it}$ past zero: treated cells in late periods of early cohorts, where the fixed-effect prediction of D_{it} exceeds 1, and untreated cells in early periods of never-treated units, where it falls below 0.

Simulation — staggered panel. $N = 300$ units over $T = 8$ periods, in three cohorts: never-treated (20%), early-treated at period 3 with effect $\tau = 3$ (40%), and late-treated at period 6 with effect $\tau = 1$ (40%). Outcomes are $y_{it} = \alpha_i + \gamma_t + \tau_g D_{it} + \varepsilon_{it}$, with unit effects $\alpha_i \sim N(0, 1)$, a time effect γ_t rising linearly from 0 to 1.4, and $\varepsilon_{it} \sim N(0, 0.5^2)$. The high treated share makes the forbidden comparison visible.

```
library(fixest)
set.seed(5)
N <- 300; TT <- 8
g <- sample(c(0, 3, 6), N, TRUE, prob = c(.2, .4, .4))
panel <- data.frame(id = rep(1:N, each = TT), t = rep(1:TT, N))
panel$g <- g[panel$id]
panel$D <- as.integer(panel$g > 0 & panel$t >= panel$g)
tau <- ifelse(panel$g == 3, 3, ifelse(panel$g == 6, 1, 0))
alpha <- rnorm(N); gamma <- seq(0, 1.4, length.out = TT)
panel$y <- alpha[panel$id] + gamma[panel$t] + tau * panel$D + rnorm(nrow(panel), 0, 0.5)

Dtil <- resid(feols(D ~ 1 | id + t, data = panel))
omega_tw <- Dtil / sum(Dtil^2)
n1 <- sum(panel$D)
vtab(c(`true ATT`                 = mean(tau[panel$D == 1]),
       `TWFE`                     = coef(feols(y ~ D | id + t, panel))[["D"]],
       `$\\boldsymbol\\omega'\\boldsymbol y$`              = sum(omega_tw * panel$y),
       `treated, $\\omega_i < 0$`    = sum(panel$D == 1 & omega_tw < 0),
       `untreated, $\\omega_i > 0$`  = sum(panel$D == 0 & omega_tw > 0)), 3)
```

	value
true ATT	2.426
TWFE	1.915
$\omega' y$	1.915
treated, $\omega_i < 0$	384.000
untreated, $\omega_i > 0$	551.000

The TWFE coefficient sits well below the true ATT, and $\sum \omega_{it} y_{it}$ reproduces it exactly. Hundreds of treated cells carry a negative weight: these are the early cohort's cells in periods 6–8, the ones where the unit-and-time prediction of D exceeds 1 and the residual $\tilde{D}_{it}$ flips sign. Their outcomes are subtracted from $\hat{\beta}$ — the observation-level signature of the Bacon forbidden comparison.

The wrong-sign diagnostic in a panel is by observation (i, t), not by unit. In the cross-sectional case earlier, the fix was to trim wrong-sign observations and refit. In a panel, trimming specific (i, t) cells removes observations that anchor that unit's fixed effect and distorts the estimates for its remaining cells. The response in the staggered setting is to switch estimators — imputation (Borusyak, Jaravel, and Spiess 2024), Callaway and Sant'Anna, or Sun and Abraham — that avoid the forbidden comparison by construction. The imputation estimator, for example, fits unit and time fixed effects on untreated observations alone and imputes each treated cell's counterfactual; because its smoother draws only on untreated outcomes, the weight on every treated cell reduces to $1/n_1$ and no treated observation can count as a control.

14.5.2 Synthetic difference-in-differences: a rank-one weighting

Synthetic difference-in-differences is a weighted two-way fixed effects regression, from Arkhangelsky, Athey, Hirshberg, Imbens, and Wager (2021). The estimators so far imply their weights. This one chooses them:

$$(\hat{\tau}, \hat{\mu}, \hat{\alpha}, \hat{\beta}) = \arg\min \sum_i \sum_t \left(Y_{it} - \mu - \alpha_i - \beta_t - W_{it}\tau\right)^2 \omega_i \lambda_t. \tag{14.15}$$

The recipe: fit ω_i on the control units so the reweighted control path tracks the treated path over the pre-period, and fit λ_t on the pre-periods so the reweighted pre-period resembles the post-period. Treated units enter at $1/N_1$, post-periods at $1/T_1$. Both fitted sets lie on the simplex, non-negative and summing to one.

Equation 14.15 is weighted least squares, so the FWL reading of Equation 14.2 goes through with the weights carried along. Then $\hat{\tau}$ is again a weighting of outcomes. Let u_i be $1/N_1$ for a treated unit and $-\omega_i$ for a control. Let v_t be $1/T_1$ for a post-period and $-\lambda_t$ for a pre-period. Then

$$\hat{\tau} = \sum_i \sum_t \omega_{it} Y_{it}, \qquad \omega_{it} = u_i v_t. \tag{14.16}$$

In words, the weight on a cell is a unit part times a time part. Collect the weights into an $N \times T$ matrix Ω, one entry per cell. Then Equation 14.16 reads $\Omega = uv'$, an outer product, and a matrix of that form has **rank one**. Every row is a multiple of every other row. So each unit carries the same weight profile over time, scaled by its own u_i, and each period weights the units in the same pattern, scaled by its own v_t. There is no unit-by-period interaction in the weights.

That product structure is what makes this a difference-in-differences. A double difference contrasts units and contrasts periods, and multiplying the two contrasts gives a rank-one weighting. We come back to the rank below, where it separates the block designs from the staggered ones.

Multiply Equation 14.16 out and we get the 2×2 of Equation 14.14 with two of its four averages reweighted, the control average by ω and the pre-period average by λ:

$$\hat{\tau} = \Big(\bar{Y}_{T,\text{post}} - \sum_{t \in \text{pre}} \lambda_t \bar{Y}_{T,t}\Big) - \Big(\sum_{i \in C} \omega_i \bar{Y}_{i,\text{post}} - \sum_{i \in C} \sum_{t \in \text{pre}} \omega_i \lambda_t Y_{it}\Big). \tag{14.17}$$

The four blocks keep the signs $(+, -, -, +)$ of Equation 14.14. Only the averages inside two of them change, and they are no longer uniform.

Simulation — the outcome weights of synthetic DiD. A panel of 40 control and 10 treated units over 12 pre- and 6 post-periods. Untreated outcomes are

$$Y_{it}(0) = \alpha_i + \beta_t + f_i' g_t + \varepsilon_{it}, \tag{14.18}$$

with a unit effect $\alpha_i \sim N(0, 0.5^2)$, a time effect β_t rising linearly from 0 to 1, two factors in g_t (one linear in t, one a sine), loadings $f_i \sim N(0, \mathbf{I})$ shifted by 0.8 for the treated units, and $\varepsilon_{it} \sim N(0, 0.3^2)$. The spread in f_i makes the control trends differ from each other, and the shift makes the treated trend differ from all of them. The true ATT is 2. We fit with `synthdid`, then rebuild Equation 14.16 by hand from the fitted ω and λ.

```r
library(synthdid)
set.seed(11)
N0 <- 40; N1 <- 10; N <- N0 + N1; T0 <- 12; T1 <- 6; Tt <- T0 + T1
f <- cbind(rnorm(N), rnorm(N)); f[(N0 + 1):N, ] <- f[(N0 + 1):N, ] + 0.8
g <- cbind(seq(0, 2, length = Tt), sin(seq(0, 3, length = Tt)))
Y <- outer(rnorm(N, 0, .5), rep(1, Tt)) + outer(rep(1, N), seq(0, 1, length = Tt)) +
     f %*% t(g) + matrix(rnorm(N * Tt, 0, .3), N, Tt)
W <- matrix(0, N, Tt); W[(N0 + 1):N, (T0 + 1):Tt] <- 1
Y <- Y + 2 * W                                                 # true ATT = 2

est <- synthdid_estimate(Y, N0, T0)
wt  <- attr(est, "weights")
u <- c(-wt$omega,  rep(1 / N1, N1))                            # unit part
v <- c(-wt$lambda, rep(1 / T1, T1))                            # time part
Omega <- outer(u, v)                                           # outcome weight on cell (i, t)
```

Three routes should give the same number: what **synthdid** reports, the weighting Equation 14.16 applies to the observed outcomes, and the weighted regression of Equation 14.15.

```r
cw <- outer(c(wt$omega, rep(1 / N1, N1)), c(wt$lambda, rep(1 / T1, T1)))
dd <- data.frame(y = as.vector(Y), Wt = as.vector(W), i = factor(rep(1:N, Tt)),
                 t = factor(rep(1:Tt, each = N)), cw = as.vector(cw))
tau3 <- c(`synthdid_estimate()`              = as.numeric(est),
          `$\\sum_{it} \\omega_{it} Y_{it}$, by hand` = sum(Omega * Y),
          `weighted TWFE regression`        = coef(lm(y ~ Wt + i + t, dd,
                                                 weights = dd$cw))[["Wt"]])
vtab(c(tau3, `largest gap between them` = max(abs(tau3 - tau3[1]))), 6)
```

	value
synthdid_estimate()	2.205444
$\sum_{it} \omega_{it} Y_{it}$, by hand	2.205444
weighted TWFE regression	2.205444
largest gap between them	0.000000

The gap between them is zero to machine precision. The outer-product form is what **synthdid** computes.

Writing the weights this way puts three constraints on them. First, the rank is one, since a cell's weight is a unit part times a time part. Second, each margin sums to zero on its own: ω and λ each sum to one, and the treated and post-period blocks each contribute an offsetting one. Third, no cell carries a wrong sign: no treated post-period cell is subtracted, and no control pre-period cell is added. We check all three.

```r
chk <- rbind(`rank of $\\boldsymbol\\Omega$`      = c(sum(svd(Omega)$d > 1e-12), 1),
             `$\\sum_i u_i$`      = c(sum(u), 0),
             `$\\sum_t v_t$`    = c(sum(v), 0),
             `treated post-cells, $\\omega_{it} < 0$` = c(sum(Omega[(N0+1):N, (T0+1):Tt] < 0),
             `control pre-cells, $\\omega_{it} < 0$`  = c(sum(Omega[1:N0, 1:T0] < 0), 0))
colnames(chk) <- c("found", "should be")
mtab(chk, 4)
```

	found	should be
rank of Ω	1	1
$\sum_i u_i$	0	0
$\sum_t v_t$	0	0
treated post-cells, $\omega_{it} < 0$	0	0
control pre-cells, $\omega_{it} < 0$	0	0

All three hold. Note that the double zero-sum is stronger than the single sum-to-zero of Equation 14.2, which constrains the weights only in total. Here they vanish along each margin. The signs are fixed by construction too: a treated post-period cell gets $1/(N_1 T_1) > 0$, and a control pre-period cell gets $\omega_i \lambda_t > 0$, because the simplex rules out a negative ω_i. The doubly-robust fit above produced 151 wrong-sign units. Synthetic DiD cannot produce any.

The remaining question is how much of the panel the comparison uses. Counting the non-zero weights is one answer, but it treats a weight of 0.4 and a weight of 0.001 alike. The usual summary that does not is the **effective sample size**, and it is worth deriving rather than asserting. Take m observations with equal variance σ^2. An unweighted mean of them has variance σ^2/m. A weighted mean with weights w summing to one has variance $\sigma^2 \sum w^2$. Ask which m makes the two equal:

$$\frac{\sigma^2}{m} = \sigma^2 \sum_j w_j^2 \qquad \Longrightarrow \qquad m = \frac{1}{\sum_j w_j^2}. \tag{14.19}$$

So Equation 14.19 is the number of equally weighted observations that would deliver the same precision as the weighting actually used. It equals the full count when every weight is equal, and 1 when one observation carries everything. We apply it to each margin separately: to the control weights as $1/\sum_i \omega_i^2$, and to the pre-period weights as $1/\sum_t \lambda_t^2$. Both are sparse here, so both fall below the nominal 40 controls and 12 pre-periods.

```r
dsg <- c(sum(wt$omega > 1e-8), 1 / sum(wt$omega^2),
         sum(wt$lambda > 1e-8), 1 / sum(wt$lambda^2), wt$lambda[T0])
names(dsg) <- c(sprintf("controls with non-zero weight (of %d)", N0),
                sprintf("effective controls, $1/\\sum_i \\omega_i^2$ (of %d)", N0),
                sprintf("pre-periods with non-zero weight (of %d)", T0),
                sprintf("effective pre-periods, $1/\\sum_t \\lambda_t^2$ (of %d)", T0),
                "weight on the last pre-period")
vtab(dsg)
```

	value
controls with non-zero weight (of 40)	23.000
effective controls, $1/\sum_i \omega_i^2$ (of 40)	14.557
pre-periods with non-zero weight (of 12)	2.000
effective pre-periods, $1/\sum_t \lambda_t^2$ (of 12)	1.091
weight on the last pre-period	0.957

Only 23 of the 40 controls get any weight, and the effective count is 14.6. Three reference points make that number readable. Weighting all 40 equally would give 40, and that is plain difference-in-differences. Weighting the 23 active controls equally, at 1/23 each, would give 23. The comparison actually delivers 14.6, so the drop from 23 to 14.6 measures nothing but the inequality among those 23 weights: a few carry most of the mass and the rest contribute little.

A larger effective size is not automatically better, and it is worth seeing why. By Cauchy–Schwarz, $\sum_i \omega_i^2 \geq 1/N_0$ with equality only when every weight equals $1/N_0$, so the effective size reaches N_0 exactly when the weighting is uniform — that is, when no reweighting has happened. Matching the treated path requires tilting toward the controls that resemble it, and any tilt lowers the effective size. The two things we want pull against each other, and the effective count is the price paid for the match rather than a score to be maximised. The one case where both arrive together is when the treated path already equals the unweighted control average, and then there was nothing to reweight for.

Synthetic difference-in-differences prices that trade-off explicitly, which is why the effective count is the number to read. Its weights minimise not the pre-period fit alone but

$$\text{(pre-period fit error)} \; + \; \zeta^2 \sum_i \omega_i^2, \tag{14.20}$$

and by Equation 14.19 the penalty $\zeta^2 \sum_i \omega_i^2$ is exactly ζ^2 divided by the effective sample size. So the estimator charges itself for concentration at a rate ζ, which by default rises with the estimated noise in the panel: noisier data is shrunk harder toward uniform, because in noisy data an apparent gain from concentrating is more likely to be fitted noise. The 14.6 above is where that trade-off settled on this panel, and it is computable whether or not we know what generated the data. The time margin is sparser: an effective 1.09 pre-periods, with 0.957 of the mass on the last one. The counterfactual level is anchored on the single period before treatment, so the estimate inherits that period's noise. We can read all of this off the weights. None of it is visible in $\hat{\tau}$.

We plot the two margins to see where the mass sits. The left panel sorts the fitted control weights ω_i from largest to smallest, which shows how many controls the comparison uses. The right panel keeps the pre-period weights λ_t in time order, which shows which periods the counterfactual is built from.

```
par(mfrow = c(1, 2), mar = c(4.2, 4.2, 2.6, 1))
os <- sort(wt$omega, decreasing = TRUE)
barplot(os, col = ifelse(os > 1e-8, teal, "grey85"), border = NA,
        xlab = "control units, sorted", ylab = expression(omega[i]),
        main = "Unit weights")
```

```
barplot(wt$lambda, names.arg = 1:T0, border = NA,
        col = ifelse(wt$lambda > 1e-8, rust, "grey85"),
        xlab = "pre-period", ylab = expression(lambda[t]), main = "Time weights")
```

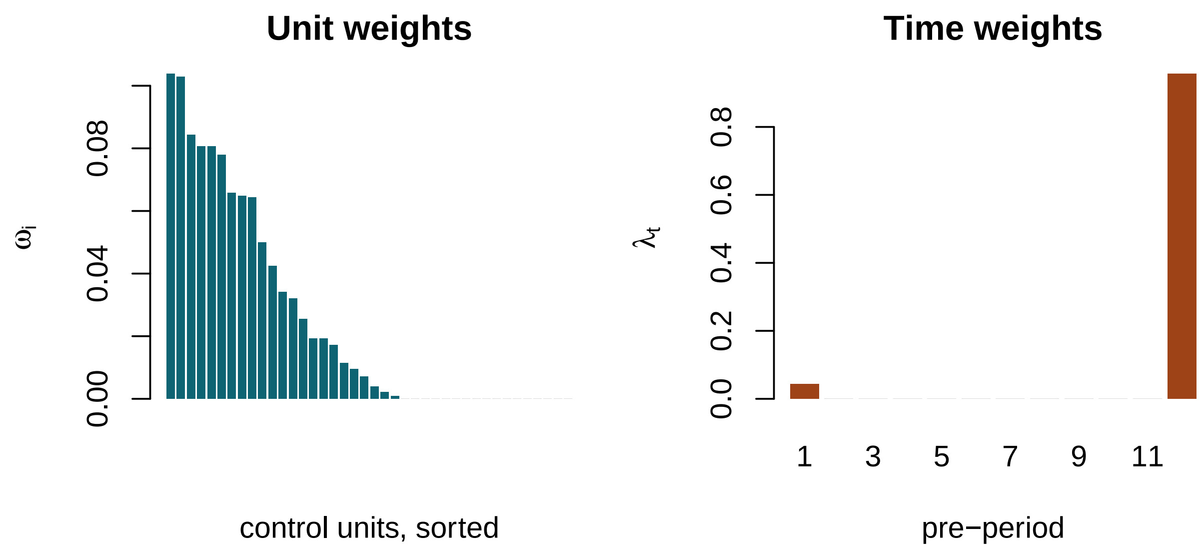


Figure 14.5: The two margins of the synthetic DiD weight matrix. Left: fitted control-unit weights ω_i, sorted. 23 of 40 are non-zero, and the 17 the simplex sets to zero lie flat against the axis. Right: fitted pre-period weights λ_t. Almost all the mass is on the last pre-period, so the counterfactual level is anchored just before treatment.

14.5.3 The rank of the weight matrix

Synthetic difference-in-differences has a rank-one Ω by construction. So does the plain 2×2: set u_i and v_t to the uniform $\pm 1/N_g$ and $\pm 1/T_g$ and Equation 14.14 reappears. Staggered adoption gives a higher rank, bounded by the number of cohorts.

Write the staggered treatment matrix as a sum over adoption cohorts, $W = \sum_{k=1}^{K} c_k s_k'$. Here c_k indicates the units of cohort k, and s_k the periods from its adoption date on. That is a sum of K rank-one terms. Residualising on unit and time dummies is a double centring, $(\mathbf{I} - P)\, W\, (\mathbf{I} - Q)$, and double centring cannot raise a rank. So $\operatorname{rank}(\Omega) \le K$, with equality generically.

We check the bound directly. Each panel has 60 units over 20 periods, with one to five adoption dates plus a never-treated group. For each we build the staggered TWFE weights, then count the non-zero singular values of Ω.

```
set.seed(4)
Ns <- 60; Ts <- 20
rk <- sapply(list(10, c(8, 14), c(6, 10, 14), c(6, 9, 12, 15), c(5, 8, 11, 14, 17)),
  function(dates) {
    coh <- sample(c(dates, Inf), Ns, replace = TRUE)     # Inf = never treated
    Ws  <- matrix(0, Ns, Ts)
```

```
    for (i in 1:Ns) if (is.finite(coh[i])) Ws[i, coh[i]:Ts] <- 1
    ds  <- data.frame(Wt = as.vector(Ws), i = factor(rep(1:Ns, Ts)),
                      t = factor(rep(1:Ts, each = Ns)))
    Wtil <- residuals(lm(Wt ~ i + t, ds))
    Om   <- matrix(Wtil / sum(Wtil^2), Ns, Ts)
    c(`treated cohorts` = length(dates),
      `rank of $\\boldsymbol\\Omega$`   = sum(svd(Om)$d > 1e-9 * max(svd(Om)$d)))
  })
colnames(rk) <- c("1 date", "2 dates", "3 dates", "4 dates", "5 dates")
mtab(rk)
```

	1 date	2 dates	3 dates	4 dates	5 dates
treated cohorts	1	2	3	4	5
rank of $\boldsymbol\Omega$	1	2	3	4	5

The rank is at most the number of treated cohorts, and equals it generically. This ties the chapter's difference-in-differences material together. A block design has one adoption date, so its weight matrix is rank one. The plain 2×2 and synthetic difference-in-differences are the same object at two settings of the margins, uniform in the first and fitted in the second. Staggered adoption raises the rank. Once a cell's weight no longer factors into a unit part times a time part, a treated cell can enter negatively. In the block case it cannot, since every treated post-period cell gets $u_i v_t = 1/(N_1 T_1) > 0$. The forbidden comparisons need the extra rank.

14.5.4 Which margin does the work

The weight matrix has two margins, u over units and v over periods, and Equation 14.16 says the whole weighting is their product. So a natural question about the structure is which margin carries the correction when the control trends differ from the treated group's.

It is worth saying first what such a correction can and cannot be. Parallel trends is a statement about two group means, $E[\Delta Y(0) \mid D = 1] = E[\Delta Y(0) \mid D = 0]$, and no single observation either has it or lacks it, so in the model behind Equation 14.14 there is nothing at the unit level for a weight to act on. Weighting bites only under a model where trends vary across units: unit-specific slopes, or a factor structure. Then each unit has its own trend, and reweighting controls toward those whose loadings resemble the treated group's is well defined. That is what synthetic difference-in-differences does, and what the wider family of trajectory-balancing and history-matching estimators does too (Hazlett and Xu 2018; Imai, Kim, and Wang 2023; Doudchenko and Imbens 2016).

We can then hold one margin uniform and fit the other, and read off how much of the correction each supplies.

Simulation — which margin does the work. Two panels of 40 controls and 10 treated over 12 pre- and 6 post-periods, with a true ATT of 2. Both carry a unit effect $\alpha_i \sim N(0, 1)$, shifted up by 1 for the treated units, a time effect β_t rising linearly from 0 to 1.5, and a serially correlated shock. The shock is a stationary AR(1),

$$\varepsilon_{it} = \rho\,\varepsilon_{i,t-1} + \nu_{it}, \qquad \nu_{it} \sim N(0, \sigma^2), \qquad \rho = 0.5, \tag{14.21}$$

started from its stationary distribution, so ε_{it} has standard deviation $\sigma/\sqrt{1-\rho^2}$. We set $\sigma = 0.4$, which makes the shock's own standard deviation 0.46. Note the distinction: σ is the standard deviation of the innovation ν_{it} that arrives each period, not of the shock ε_{it} that accumulates.

The shock is serially correlated rather than independent for a reason, and it is not decoration. The weights are fitted on the pre-period path. Under independent noise, averaging across twelve pre-periods removes most of it, so the weights would recover the systematic path almost exactly and there would be little left to study. Serial correlation means twelve periods carry well under twelve periods' worth of independent information, and that is what leaves the fitted weights short. It is also the realistic case: panel outcomes persist. AR(1) is the smallest model with a persistence parameter, nesting independent noise at $\rho = 0$ and a random walk as $\rho \to 1$.

These three terms are common to almost any panel, and none of them breaks parallel trends: α_i differences away and β_t is shared by both groups, so a difference-in-differences removes them whatever the weights are.

Parallel trends fails only when the treated and control groups have *different* untreated trends, so a term that varies by unit and moves with t is what we have to put in. Without one there is nothing for the weights to fix. The two panels are two forms of it. The `factor` panel uses the $f_i' g_t$ of Equation 14.18. The `slopes` panel replaces it with a unit-specific linear trend,

$$Y_{it}(0) = \alpha_i + \beta_t + s_i\,t + \varepsilon_{it}, \qquad s_i \sim N(0,\, 0.08^2), \tag{14.22}$$

with s_i shifted up by 0.12 for the treated units. So each unit has its own growth rate, and the treated units grow faster on average, by about one and a half standard deviations of the control slopes. Note that $s_i t$ is itself a one-factor model, with loading s_i and factor $g_t = t$. It is the simplest member of the class Equation 14.18 describes, which is the class synthetic difference-in-differences is built for, so this is a leading case and not a special one. We run synthetic DiD four ways: both margins uniform, which is plain difference-in-differences; unit weights fitted with time weights uniform; time weights fitted with unit weights uniform; and both fitted. 400 replications.

```r
library(parallel)
panel <- function(kind, N0 = 40, N1 = 10, T0 = 12, T1 = 6, sig = .4, rho = .5) {
  N <- N0 + N1; Tt <- T0 + T1
  a <- rnorm(N, 0, 1); a[(N0 + 1):N] <- a[(N0 + 1):N] + 1
  e <- matrix(0, N, Tt); e[, 1] <- rnorm(N, 0, sig / sqrt(1 - rho^2))
  for (t in 2:Tt) e[, t] <- rho * e[, t - 1] + rnorm(N, 0, sig)
  Y0 <- outer(a, rep(1, Tt)) + outer(rep(1, N), seq(0, 1.5, length = Tt)) + e
  if (kind == "factor") {
    f <- cbind(rnorm(N), rnorm(N)); f[(N0 + 1):N, ] <- f[(N0 + 1):N, ] + 0.8
    g <- cbind(seq(0, 2, length = Tt), sin(seq(0, 3, length = Tt)))
    Y0 <- Y0 + f %*% t(g)
  } else {
    s <- rnorm(N, 0, .08); s[(N0 + 1):N] <- s[(N0 + 1):N] + .12   # unit slopes
    Y0 <- Y0 + outer(s, 1:Tt)
```

```
  }
  W <- matrix(0, N, Tt); W[(N0 + 1):N, (T0 + 1):Tt] <- 1
  list(Y = Y0 + 2 * W, N0 = N0, N1 = N1, T0 = T0, T1 = T1)
}
margins <- function(r, kind) {
  set.seed(5000 + r); p <- panel(kind); N0 <- p$N0; T0 <- p$T0
  fx <- function(om, lam) as.numeric(synthdid_estimate(p$Y, N0, T0,
          weights = list(omega = om, lambda = lam)))     # NULL = fit this margin
  unif.o <- rep(1 / N0, N0); unif.l <- rep(1 / T0, T0)
  c(`both uniform (DiD)` = fx(unif.o, unif.l), `unit weights only` = fx(NULL, unif.l),
    `time weights only`  = fx(unif.o, NULL),   `both fitted (SDID)` = fx(NULL, NULL))
}
res <- lapply(c("factor", "slopes"), function(kind) {
  # portable core count: mclapply falls back to serial on Windows
  n_cores <- if (.Platform$OS.type == "windows") 1L else
               max(1L, min(6L, parallel::detectCores() - 1L))
  M <- simplify2array(mclapply(1:400, margins, kind = kind, mc.cores = n_cores))
  cbind(bias = rowMeans(M) - 2, sd = apply(M, 1, sd), rmse = sqrt(rowMeans((M - 2)^2)))
})
out <- cbind(res[[1]], res[[2]])
colnames(out) <- paste(rep(c("factor", "slopes"), each = 3), colnames(res[[1]]))
mtab(out)
```

	factor bias	factor sd	factor rmse	slopes bias	slopes sd	slopes rmse
both uniform (DiD)	0.729	0.403	0.833	1.083	0.284	1.120
unit weights only	0.173	0.337	0.379	0.603	0.231	0.645
time weights only	0.082	0.285	0.296	0.536	0.203	0.573
both fitted (SDID)	0.044	0.299	0.302	0.403	0.208	0.453

Reweighting helps, and by a lot. Under the factor structure the bias falls from 0.73 to 0.04 with both margins fitted. Each margin helps alone: 0.17 from the unit weights, 0.08 from the time weights. Which margin matters is not fixed in advance. Here the time weights do slightly more, because the factor paths are smooth, so anchoring on the periods just before treatment removes most of the divergence. The fitted λ above showed the same thing when it put 0.957 on the last pre-period.

The `slopes` panel gains less: 1.08 down to 0.40. The reason is that the weights are fitted on the observed pre-period path, which carries the shock of Equation 14.21 as well as the trend, so they match a noisy version of what they are aiming at and fall short. The size of that shortfall depends on the pre-period length and the noise level rather than on anything intrinsic to reweighting — with a much cleaner pre-period the same estimator gets the bias under 0.05 — so the direction is the finding here and the magnitude is not.

Two limits are worth stating even though they do not bind in this panel. Reweighting controls can only produce a weighted average of the control trends, so a treated trend steeper than every control's is out of reach; that is the same overlap problem seen earlier, now in the design, and the simplex is what makes it a hard boundary rather than a silent extrapolation. And if we are confident the trends really are unit-specific and linear, the direct response is to put unit-specific linear trends in the regression, not to reweight at all. Reweighting is the tool for when we will not commit to a functional form, which is the case Equation 14.18 stands in for.

14.5.5 Implied versus chosen

One distinction cuts across everything above. OLS, AIPW, DML, TWFE, and the imputation estimator all *imply* weights — they fall out of the estimation and are diagnosed afterward. Synthetic controls and synthetic difference-in-differences *choose* weights as part of the design, selecting ω to balance pre-treatment outcomes or covariates. For those estimators $\hat{\tau} = \omega' Y$ is the definition, not a decomposition.

That difference changes how the weights should be read, in three ways. First, chosen weights are fitted on Y itself, so $\hat{\tau}$ is not linear in Y and Equation 14.16 describes the estimate at the fitted weights rather than decomposing it the way Equation 14.2 decomposes OLS. Inference has to account for the weights having been estimated, which is why `synthdid` reports standard errors by jackknife or bootstrap rather than treating ω as fixed.

Second, a clean-looking balance diagnostic is weaker evidence here than it looks. When the weights were chosen to make the pre-period match, the match is not a test the design passed — it is what the design optimised. Fitting weights on a noisy pre-period statistic is a close cousin of a pretest. The analogy is with Roth (2022), who shows that conditioning an analysis on passing a parallel-trends test distorts the estimate and undercovers. Weight-fitting is not literally that procedure, since nothing is accepted or rejected, but it shares the mechanism: noise in the pre-period is allowed to select the analysis. Matching on pre-period *levels* is the sharp case: it induces regression to the mean, with a bias that grows as serial correlation weakens (Daw and Hatfield 2018; Chabé-Ferret 2015, 2017). Synthetic difference-in-differences avoids that particular trap by fitting ω with an intercept, so it never needs to match levels, only trajectories up to a constant.

Third, the effective count of Equation 14.19 is the number that reports the price. A design that matches well by concentrating weight on a few controls has bought that match with effective sample size, and Equation 14.20 shows the estimator charging itself for exactly that. Read the two together rather than either alone.

A closing caution. These weights diagnose estimation. They show which comparisons a difference-in-differences rests on, and whether it uses non-overlapping or forbidden ones. They do not test identification. Parallel trends is an assumption about the treated group's unobserved untreated path, and a negative weight and a failure of parallel trends stay independent: either can occur without the other. Chosen weights can buy robustness to one class of violations, those the design's trend model can express, and the weights let us see what that cost. No weighting of observed outcomes tests the assumption. To do that we bound the violation instead of reweighting, using the observed pre-period differences to limit how large the post-period departure can be (Rambachan and Roth 2023).

14.6 Summary

1. **A signed weighting of the outcomes.** FWL writes $\hat{\beta} = \sum_i \omega_i y_i$ with $\omega_i = \tilde{V}_i / \sum_j \tilde{V}_j^2$ (Equation 14.2): a signed, computable weighting of the observed outcomes, different from the effect weights $w_i = \tilde{V}_i V_i$.
2. **Two constraints.** The weights sum to zero and the weighted regressor sums to one: $\sum_i \omega_i = 0$ and $\sum_i \omega_i V_i = 1$.
3. **The sign shows direction.** Because ω_i is proportional to $\tilde{V}_i$, not to $\tilde{V}_i^2$, it carries a sign: it shows which rows push $\hat{\beta}$ up and which pull it down, and, under controls with thin overlap, which enter with the sign their raw position would not predict.
4. **The same weights beyond OLS.** Any estimator that can be written $\hat{\tau} = \omega' Y$ has outcome weights, so the same object and its checks reach AIPW, double machine learning, and causal forests (Knaus 2024). PLR's denominator $\tilde{D}'\tilde{D}$ reveals its estimand as the VWATE (Equation 14.11), not the ATE; and the sum-to-zero of OLS need not become a sum to ± 1 within the treated and control groups.
5. **From difference in means to difference-in-differences.** Regressed on the treatment alone, $\hat{\beta}$ is the difference in group means; a 2×2 difference-in-differences is a signed contrast of four cell means. With covariates, a doubly-robust DiD weights the outcome changes, $\widehat{\text{ATT}} = \omega' \Delta Y$ (Sant'Anna and Zhao 2020), open to the same checks.
6. **Staggered designs.** In staggered TWFE the outcome weights flag both treated cells used as controls and untreated cells used as treated — the observation-level version of the Bacon (2021) forbidden comparisons. Estimators that fit counterfactuals on untreated observations alone (Borusyak, Jaravel, and Spiess 2024; Callaway and Sant'Anna 2021) cannot assign negative weight to any treated cell, so the forbidden-comparison diagnostic is clean by construction.
7. **Rank counts the cohorts.** As an $N \times T$ matrix, the outcome weights of a block design have rank one and factor into a unit part times a time part (Equation 14.16). The plain 2×2 and synthetic difference-in-differences differ only in whether those margins are uniform or fitted. Staggered adoption raises the rank — to at most the number of treated cohorts, generically to exactly that — and only above rank one can a treated cell enter negatively.
8. **Both margins carry correction, and the effective count reports its price.** Weighting can offset differing trends only under a model in which trends vary by unit; then either margin helps alone and both together help most. What it costs is effective sample size: by Equation 14.19 the count reaches N_0 only under uniform weights, so a closer match always means fewer effective controls, and Equation 14.20 shows synthetic difference-in-differences charging itself for exactly that.
9. **Chosen weights are read differently from implied ones.** Weights fitted on Y make $\hat{\tau}$ non-linear in Y, so inference must account for their estimation, and a good pre-period match is what the design optimised rather than a test it passed. Fitting on a noisy pre-period statistic is a pretest (Roth 2022), and matching on pre-period *levels* invites regression to the mean (Daw and Hatfield 2018); fitting with an intercept, so only trajectories need to match, avoids that case.

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